SPARSE WAVELET ESTIMATION OF ADDITIVE FUNCTIONAL COEFFICIENTS

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Supplementary Material

This supplement contains the proofs of the results stated in the paper and the auxiliary lemmas they use. Section S1 fixes the notation and the objects. Section S2 proves the identification results of Section 2 of the paper, Section S3 the approximation lemma, Section S4 the bounds for the Gram matrix of the product design, Section S5 the oracle inequality, ~~and~~ Section S6 the rates and a minimax lower bound, ~~and~~ Section S7 the recovery of the block structure by thresholding and the risk of the thresholded estimator, and Section S8 gives the theory of the coordinatewise lasso, the option of the software that replaces the penalty of equation (2.5) by the ℓ_1 norm. Section S9 gives the details and the further results of the simulation study of Section 5 of the paper, and Section S10 further results of the application of Section 6 and a second application. References to numbered results without a section prefix are to the paper; results numbered with the prefix S are stated and proved here.

S1 Notation and objects

The model, the basis and the estimator are those of Sections 2.1 to 2.3 of the paper. We write $\mathbf{Z} = [\mathbf{A}\ \mathbf{B}]$ for the design matrix of equation (2.4), with $\mathbf{A} \in \mathbb{R}^{n\times p}$ the unpenalized block and $\mathbf{B} \in \mathbb{R}^{n\times d}$, $d = pqN_J$, the block of products; $\mathbf{Y}$, f and $\boldsymbol{\varepsilon}$ for the vectors with entries Y_i, $f(\mathbf{X}_i, \mathbf{U}_i)$ and ε_i; $\widehat{\Sigma} = \mathbf{Z}^\top\mathbf{Z}/n$ and $\Sigma = \mathrm{E}(\mathbf{Z}\mathbf{Z}^\top)$; and, for $a, h \in \mathbb{R}^n$, $\|a\|_n^2 = n^{-1}\sum_i a_i^2$ and $\langle a, h\rangle_n = n^{-1}\sum_i a_i h_i$.

The oracle of the approximation space is $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$, with $\theta^*_{\ell m,jk} = \langle g_{\ell m}, \psi_{jk}\rangle$ for $j < J$, so that $f_J = \mathbf{Z}\boldsymbol{\beta}^*$ is the best approximation of f in the approximation space in the sense of Lemma 2; the deterministic residual is $\mathbf{b} = f - f_J$, the oracle support is $S_0 = \mathrm{supp}(\boldsymbol{\theta}^*)$ and $s_0 = |S_0|$, and $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$. When $\mathbf{A}$ has rank p we write $\Pi_{\mathbf{A}} = \mathbf{A}(\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$, $M_{\mathbf{A}} = I_n - \Pi_{\mathbf{A}}$, $\widetilde{\mathbf{B}} = M_{\mathbf{A}}\mathbf{B}$, $\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} = \widetilde{\mathbf{B}}^\top\widetilde{\mathbf{B}}/n$ and $\widetilde{\gamma} = \lambda_{\min}(\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}})$, and finally $\widehat{\sigma}^2_{\max} = \max_a(\widehat{\Sigma})_{aa}$, the maximum being over the d penalized columns.

For the penalty of equation (2.5), $\mathcal{G}$ is the partition of the d penalized coordinates into the chunks of Section 2.3 of the paper, $|\mathcal{G}|$ the number of chunks, $\boldsymbol{\theta}_G$ and $|G|$ the subvector and the size of a chunk G, $w_G > 0$ its weight and $\varrho = \max_G w_G / \min_G w_G$ the ratio of the weights; $\|\boldsymbol{\theta}\|_{\mathcal{G},w} = \sum_G w_G \|\boldsymbol{\theta}_G\|_2$, $\|\boldsymbol{\theta}\|_{\mathcal{G},1}$ is the same norm with unit weights, and, for $\mathcal{H} \subseteq \mathcal{G}$, $\|\boldsymbol{\theta}_{\mathcal{H}}\|_{\mathcal{G},w} = \sum_{G\in\mathcal{H}} w_G \|\boldsymbol{\theta}_G\|_2$ and $\mathcal{W}(\mathcal{H}) = \sum_{G\in\mathcal{H}} w_G^2$; $\mathcal{G}_0(\boldsymbol{\theta}) = \{G : \boldsymbol{\theta}_G \neq 0\}$

are the active chunks of $\boldsymbol{\theta}$. With unit weights $\mathcal{W}(\mathcal{G}_0(\boldsymbol{\theta}))$ is the number of active chunks, and with $w_G = |G|^{1/2}$ it is the number of coordinates in them. A comparator is any $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}) \in \mathbb{R}^p \times \mathbb{R}^d$, with deterministic residual $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$, active chunks $\bar{\mathcal{G}} = \mathcal{G}_0(\bar{\boldsymbol{\theta}})$, $\bar{\mathcal{W}} = \mathcal{W}(\bar{\mathcal{G}})$ and $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$. For a chunk, $\mathbf{B}_G$ and $\widetilde{\mathbf{B}}_G$ are the columns of $\mathbf{B}$ and $\widetilde{\mathbf{B}}$ in G, $\widetilde{\Psi}_G = \widetilde{\mathbf{B}}_G^\top \widetilde{\mathbf{B}}_G / n$ and $\widehat{\Sigma}_{GG} = \mathbf{B}_G^\top \mathbf{B}_G / n$. The ideal risk over chunks is $R_{\mathcal{G}}(\boldsymbol{\theta}; \eta) = \sum_G \min(\|\boldsymbol{\theta}_G\|_2^2, \eta)$, for $\eta > 0$; it is nondecreasing in η, and, since $\min(u, \kappa\eta) \le \kappa \min(u, \eta)$ for $\kappa \ge 1$, $R_{\mathcal{G}}(\boldsymbol{\theta}; \kappa\eta) \le \kappa R_{\mathcal{G}}(\boldsymbol{\theta}; \eta)$. Finally $\gamma = \kappa_1 c_U$, $\Lambda = \kappa_2 C_U + \gamma/2$ is the constant of equation (3.1), and

$$\mathcal{E}_\Sigma \;=\; \Big\{ \Big\| \widehat{\Sigma} - \Sigma \Big\|_{\mathrm{op}} < \gamma/2 \Big\};$$

on $\mathcal{E}_\Sigma$, Weyl's inequality and Proposition 2(i) give $\lambda_{\min}(\widehat{\Sigma}) > \gamma/2$ and $\lambda_{\max}(\widehat{\Sigma}) < \Lambda$, and the matrix Bernstein bound in the proof of Proposition 2(ii), Section S4, gives $\mathrm{P}(\mathcal{E}_\Sigma) \to 1$ whenever $pq2^J \log(pq2^J)/n \to 0$.

For a support S and a symmetric matrix A, the compatibility constant is

$$\phi_0^2(S; A) = \min\big\{ |S|\, \delta^\top A \delta : \; \|\delta_S\|_1 = 1, \; \|\delta_{S^c}\|_1 \le 3 \big\},$$

and the elementary bound $\delta^\top A \delta \ge \lambda_{\min}(A)\, \|\delta\|_2^2 \ge \lambda_{\min}(A)\, \|\delta_S\|_1^2 / |S|$ gives

$$\phi_0^2(S; A) \;\ge\; \lambda_{\min}(A) \qquad \text{for every } S, \tag{S1.1}$$

so that a lower bound on the smallest eigenvalue is a compatibility bound on every support, with the same constant and no cone restriction.

S2 Identification and the periodized basis

Proof of Proposition 1

Levels and components are determined by the coefficients. Let β and $\tilde{\beta}$ be measurable maps from $[0,1]^q$ to $\mathbb{R}^p$ with $\sum_\ell \beta_\ell(\mathbf{U})X_\ell = \sum_\ell \tilde{\beta}_\ell(\mathbf{U})X_\ell$ almost surely, and put $\delta(u) = \beta(u) - \tilde{\beta}(u)$. Then $\{\delta(\mathbf{U})^\top \mathbf{X}\}^2 = 0$ almost surely, and since $\delta(\mathbf{U})$ is measurable with respect to $\mathbf{U}$ and $\mathrm{E}\,\|\mathbf{X}\|_2^2 < \infty$,

$$0 \;=\; \mathrm{E}\big[\{\delta(\mathbf{U})^\top \mathbf{X}\}^2 \mid \mathbf{U}\big] \;=\; \delta(\mathbf{U})^\top \mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\, \delta(\mathbf{U}) \qquad \text{almost surely.}$$

The conditional second moment matrix is positive definite almost surely, and a positive definite quadratic form vanishes only at the origin, so $\delta(\mathbf{U}) = 0$ almost surely.

Levels and components are determined within a coefficient. Because the density of $\mathbf{U}$ is positive Lebesgue almost everywhere, a set is null for the law of $\mathbf{U}$ if and only if it is Lebesgue null, so the identity $c_\ell + \sum_m g_{\ell m}(u_m) = \tilde{c}_\ell + \sum_m \tilde{g}_{\ell m}(u_m)$ holds Lebesgue almost everywhere on $[0,1]^q$. Write $\Delta = c_\ell - \tilde{c}_\ell$ and $\delta_m = g_{\ell m} - \tilde{g}_{\ell m} \in L_2[0,1] \subset L_1[0,1]$, so that $\Delta + \sum_m \delta_m(u_m) = 0$ almost everywhere. Each summand is integrable on the cube by Tonelli's theorem.

Integrating in all coordinates and using $\int_0^1 \delta_m = 0$, which holds because both parametrizations satisfy (2.2), gives $\Delta = 0$; fixing m and integrating in the remaining coordinates gives $\delta_m(u_m) = 0$ for almost every u_m. □

Remark S2.1. If the centering $\int_0^1 g_{\ell m} = 0$ is replaced by $\mathrm{E}\{g_{\ell m}(U_m)\} = 0$, the same proof applies under the two sided density bound of Assumption 3, since then $L_2[0,1]$ and $L_2(P_{U_m})$ coincide as sets: the step that uses Fubini's theorem yields δ_m constant almost everywhere, and the constant is its mean, which is zero. The two parametrizations are related by $c_\ell \mapsto c_\ell + \sum_m \mathrm{E}\{g_{\ell m}(U_m)\}$, as stated in Section 2.1 of the paper. The difference is not cosmetic for the estimator: the design of Section 2.3, which omits the constant scaling function, estimates the version centered with respect to Lebesgue measure.

Proof of Lemma 1

The basis. The periodization of the scaling function is $\phi_{00}^{\mathrm{per}}(u) = \sum_\iota \phi(u+\iota)$, a finite sum at each u because the support is compact, and it equals the constant function one, since the integer translates of an orthonormal scaling function with $\int \phi = 1$ form a partition of unity (Mallat, 2009, Section 7.5.1). For a wavelet, exchanging the finite sum with the integral, $\int_0^1 \psi_{jk} = \int_{\mathbb{R}} 2^{j/2}\psi(2^j y - k)\, dy = 2^{-j/2} \int_{\mathbb{R}} \psi = 0$ by the vanishing moment of

order zero. That $\{\mathbf{1}\} \cup \{\psi_{jk}\}$ is an orthonormal basis of $L_2[0,1]$ is the standard construction of the periodized multiresolution analysis (Daubechies, 1992, Section 9.3). Since $\dim V_J = 2^J$ and the $2^J - 1$ wavelets of level below J are orthogonal to the constant, W_J is exactly the set of elements of V_J with zero integral.

Injectivity. Suppose two parameter vectors give the same regression function almost surely. The functions $\beta_\ell^J(u) = c_\ell + \sum_m g_{\ell m}^J(u_m)$, with $g_{\ell m}^J = \sum_{jk} \theta_{\ell m,jk}\psi_{jk}$, are measurable and bounded, so the first part of Proposition 1 gives $\beta_\ell^J = \tilde{\beta}_\ell^J$ almost surely; each $g_{\ell m}^J$ has zero integral by the previous paragraph, so the second part gives $c_\ell = \tilde{c}_\ell$ and $\sum_{jk}(\boldsymbol{\theta} - \tilde{\boldsymbol{\theta}})_{\ell m,jk}\psi_{jk} = 0$ almost everywhere, whence $\boldsymbol{\theta} = \tilde{\boldsymbol{\theta}}$ by orthonormality. For the Gram matrix, $\mathrm{E}\,\|\mathbf{Z}\|_2^2 < \infty$ because $\mathrm{E}\,\|\mathbf{X}\|_2^2 < \infty$ and each wavelet is bounded, and for $a \neq 0$, $a^\top \Sigma a = \mathrm{E}\{(a^\top \mathbf{Z})^2\} = \|f_a\|_{L_2(P)}^2 > 0$ by injectivity, where f_a is the regression function with parameter a.

Nullity when the scaling column is kept. With $\phi_{00} \equiv 1$ the column $X_\ell \phi_{00}(U_m)$ coincides with the unpenalized column X_ℓ, for each of the pq pairs, so the pq vectors that oppose one to the other lie in the kernel and are linearly independent. There are no others: a kernel vector defines $\sum_\ell X_\ell\{a_\ell + \sum_m(\alpha_{\ell m} + \sum_{jk} a_{\ell m,jk}\psi_{jk}(U_m))\} = 0$ almost surely; by Proposition 1 applied to the part with zero integral, $a_{\ell m,jk} = 0$ and $a_\ell + \sum_m \alpha_{\ell m} = 0$,

which is the space of dimension pq spanned by those vectors. □

Remark S2.2 (The interval basis)**.** With the basis of Cohen, Daubechies, and Vial (1993) the wavelets of level $j \geq j_0$ are orthogonal to V_{j_0}, which contains the constants, so they still have zero integral and the penalized part of each block satisfies (2.2) with no restriction. The 2^{j_0} scaling functions do not: the constant is the combination $1 = \sum_k \mu_k \phi^{\mathrm{int}}_{j_0 k}$ with $\mu_k = \int_0^1 \phi^{\mathrm{int}}_{j_0 k}$, and the subspace with zero integral is $\{\mu^\top \alpha = 0\}$. Keeping the 2^{j_0} scaling columns of each block unrestricted repeats the constant and makes Σ singular with nullity pq, as in the last part of Lemma 1; reparametrizing $\alpha_{\ell m} = Q\gamma_{\ell m}$ with Q an orthonormal basis of $\{\mu^\top \alpha = 0\}$ restores injectivity, at the price of $pq(2^{j_0} - 1)$ unpenalized parameters.

S3 Approximation by the wavelet spaces W_J

Proof of Lemma 2

Fix a pair (ℓ, m) and write $g = g_{\ell m}$ and $\theta_{jk} = \theta^*_{\ell m, jk}$.

Tail sum. By Step 2 below the coefficients are square summable, so $g \in L_2[0,1]$ and, the system being complete and g orthogonal to the constant, $g = \sum_{j \geq 0} \sum_k \theta_{jk} \psi_{jk}$ in L_2. The orthogonal projection onto W_J retains the levels below J, and Parseval's identity gives $\|g - \Pi_J g\|^2_{L_2[0,1]} = \sum_{j \geq J} \|\theta_{j\cdot}\|_2^2$.

From ℓ_π to ℓ_2 inside a level. In $\mathbb{R}^{2^j}$ one has $\|a\|_2 \le \|a\|_\pi$ when $\pi \le 2$, and $\|a\|_2 \le (2^j)^{1/2-1/\pi} \|a\|_\pi$ by Hölder's inequality when $\pi > 2$. In both cases $\|\theta_{j\cdot}\|_2 \le 2^{j(1/2-1/\pi)_+} \|\theta_{j\cdot}\|_\pi$, and Assumption 5 gives

$$\|\theta_{j\cdot}\|_2 \;\le\; C_g\, 2^{-j\{s+1/2-1/\pi-(1/2-1/\pi)_+\}} \;=\; C_g\, 2^{-js'},$$

since the exponent equals s when $\pi \ge 2$ and $s-(1/\pi-1/2)$ when $\pi < 2$. This is where the loss $(1/\pi-1/2)_+$ appears.

Geometric sum and density. $\sum_{j\ge J} C_g^2 2^{-2js'} = C_g^2 2^{-2Js'}/(1-2^{-2s'})$, finite because $s' > 0$, which is the first display of Lemma 2. For the second, write $f - f_J = \sum_{\ell,m} X_\ell h_{\ell m}(U_m)$ with $h_{\ell m} = g_{\ell m} - \Pi_J g_{\ell m}$, the levels cancelling. By the Cauchy-Schwarz inequality over the pq terms and $|X_\ell| \le B_X$,

$$\mathrm{E}\{(f-f_J)^2\} \;\le\; pq \sum_{\ell,m} \mathrm{E}\{X_\ell^2 h_{\ell m}^2(U_m)\} \;\le\; pq\, B_X^2 \sum_{\ell,m} \mathrm{E}\{h_{\ell m}^2(U_m)\},$$

and $\mathrm{E}\{h^2(U_m)\} = \int_0^1 h^2 p_{U_m} \le C_U \|h\|^2_{L_2[0,1]}$, the marginal density being bounded by C_U because integrating the joint density over the other $q-1$ coordinates of $[0,1]^{q-1}$ does not increase the bound. One factor pq comes from the Cauchy-Schwarz inequality and the other from the sum. □

The uniform version is used in Section S5 to bound the empirical norm of the bias without an appeal to Markov's inequality.

Lemma S3.1 (Uniform approximation error)**.** *Under Assumptions 4 and 5 with $s > 1/\pi$, each component has a continuous version and, for every*

$J \geq 0$,

$$\|g_{\ell m} - \Pi_J g_{\ell m}\|_\infty \;\leq\; \frac{(L-1)\,\|\psi\|_\infty\, C_g}{1 - 2^{-(s-1/\pi)}}\, 2^{-J(s-1/\pi)},$$

and therefore $\|\mathbf{b}\|_n \leq \|f - f_J\|_\infty \leq pqB_X(L-1)\,\|\psi\|_\infty\, C_g\, 2^{-J(s-1/\pi)}/\{1 - 2^{-(s-1/\pi)}\}$ *for every sample.*

Proof. At a point u and a level j at most $L-1$ periodized translates are nonzero, because $\psi(2^j u - k) \neq 0$ requires $2^j u - k \in$ ~~$(0, L-1)$~~$(1 - L/2,\, L/2)$, so $\sum_k |\psi_{jk}(u)| \leq (L-1)2^{j/2}\,\|\psi\|_\infty$. With $\max_k |\theta_{jk}| \leq \|\theta_{j\cdot}\|_\pi \leq C_g 2^{-j(s+1/2-1/\pi)}$ the level contributes at most $(L-1)\,\|\psi\|_\infty\, C_g 2^{-j(s-1/\pi)}$ in the supremum norm. Since $s > 1/\pi$ the series converges absolutely and uniformly, so its sum is continuous and agrees almost everywhere with g; the tail from level J on is the geometric sum of the displayed bounds, and the statement for $f - f_J$ is the triangle inequality over the pq terms. □

The next proposition is the one referred to in Section 2.2 of the paper: it quantifies what the periodized basis charges for a component that does not match at the two ends of the interval.

Proposition S3.1 (Cost of periodization)**.** *Let the basis be as in Assumption 4 and let* $g \in B^s_{\pi,r}[0,1]$*, the Besov space of the interval and not of the circle, with* $s > 1/\pi$*,* $N > s$ *and* $\int_0^1 g = 0$*. Then there is a constant* C

depending on g, on ψ and on (s,π) such that, for every J with $2^J \geq L-1$,

$$\|g - \Pi_J g\|^2_{L_2[0,1]} \;\leq\; C\, 2^{-2Js'} \;+\; 2(L-1)\,\|\psi\|_1^2\,\|g\|_\infty^2\; 2^{-J},$$

so that the L_2 rate is $2^{-J\min(s',1/2)}$. If g is continuous with $g(0) \neq g(1)$, the second term is of the exact order 2^{-J} and $\|g - \Pi_J g\|_\infty$ does not converge to zero.

Proof. Let $\tilde{g}$ be the one periodic extension of g, so that $\theta_{jk} = \int_{\mathbb{R}} \tilde{g}(y) 2^{j/2}\psi(2^j y - k)\,dy$. At a level with $2^j \geq L-1$, separate the interior translates, whose support lies inside $[0,1]$, from the at most $L-1$ boundary translates that cross the identification of 0 with 1. For an interior translate, $\theta_{jk} = \langle Eg, \psi_{jk}\rangle_{\mathbb{R}}$ for any extension Eg of g to the line; taking E a bounded extension operator from $B^s_{\pi,r}[0,1]$ to $B^s_{\pi,r}(\mathbb{R})$ (Triebel, 1983, Chapter 3) and using the wavelet characterization of the latter, which for the implication from the function to the sequence requires only $N > s$ vanishing moments, gives $\|(\theta_{jk})_{\text{int}}\|_\pi \leq C_1 2^{-j(s+1/2-1/\pi)}$, whose tail sum is bounded as in Lemma 2. For a boundary translate, $s > 1/\pi$ makes g bounded and $|\theta_{jk}| \leq \|g\|_\infty \|\psi_{jk}\|_1 = \|g\|_\infty \|\psi\|_1 2^{-j/2}$, and there are at most $L-1$ of them per level, so their tail sum is at most $2(L-1)\,\|\psi\|_1^2\,\|g\|_\infty^2\, 2^{-J}$. For the exact order, if g is continuous with $g(0) \neq g(1)$ then on the support of a boundary translate $\tilde{g}$ equals $g(1) + o(1)$ to the left of the identification point and $g(0) + o(1)$ to the right, so $\theta_{jk} = 2^{-j/2}\{g(0) - g(1)\}\int_{y > a_k} \psi(y)\,dy + o(2^{-j/2})$;

since $\int \psi = 0$ and ψ vanishes on no subinterval of its support, that partial integral is nonzero for some translate, and the boundary contribution of the level is of order 2^{-j}. The failure of uniform convergence is that of any wavelet series at a jump. □

Under Assumption 5 nothing changes, since that assumption is already a periodic one; what Proposition S3.1 gives is the price of its failure. With the basis of Cohen, Daubechies, and Vial (1993) the space V_J reproduces polynomials of degree below N and its basis characterizes $B^s_{\pi,r}[0,1]$ without periodicity (Cohen, 2003, Section 3.9), so Lemma 2 holds with that space and the same proof.

S4 The Gram matrix of the product design

Recall $\mathbf{\Psi}(u) \in \mathbb{R}^K$, $K = 1 + qN_J$, the vector that collects the constant and the wavelets of the q modulators, and the identity $\mathbf{Z} = \Pi\{\mathbf{X} \otimes \mathbf{\Psi}(\mathbf{U})\}$, where Π is the fixed permutation that moves the p entries $X_\ell \cdot 1$ to the front. Since Π is orthogonal, every statement about eigenvalues is invariant to the column order. Assumption 4 also gives a constant C_ψ, depending only on the family, with

$$\sup_{u\in[0,1]} \sum_{j<J} \sum_{k} \psi_{jk}(u)^2 \;\le\; C_\psi\, 2^J \qquad \text{for every } J \ge 1 : \tag{S4.1}$$

at each level at most L periodized wavelets are nonzero at a given u and $\|\psi_{jk}\|_\infty = 2^{j/2}\|\psi\|_\infty$, so the left side is at most $L\|\psi\|_\infty^2 \sum_{j<J} 2^j \le L\|\psi\|_\infty^2 2^J$.

Proof of Proposition 2

Part (i), the modulators. For $a \in \mathbb{R}^K$ write $h_a(u) = a^\top \mathbf{\Psi}(u) = a_0 + \sum_m h_{a,m}(u_m)$ with $h_{a,m} = \sum_{jk} a_{m,jk}\psi_{jk}$. The functions 1 and $\psi_{jk}(u_m)$, $m = 1,\dots,q$, are orthonormal in $L_2([0,1]^q, du)$: within one coordinate this is the orthonormality of the periodized system, and across coordinates $\int \psi_{jk}(u_m)\psi_{j'k'}(u_{m'})\,du = \int_0^1 \psi_{jk} \int_0^1 \psi_{j'k'} = 0$ for $m \neq m'$, because every wavelet integrates to zero. Hence $\int_{[0,1]^q} h_a^2 = \|a\|_2^2$ and, by Assumption 3, $c_U\|a\|_2^2 \le \mathrm{E}\{h_a(\mathbf{U})^2\} \le C_U\|a\|_2^2$, which is the claim since $a^\top \Sigma_{\mathbf{\Psi}} a = \mathrm{E}\{h_a(\mathbf{U})^2\}$.

Part (i), the full matrix. Let $\delta \in \mathbb{R}^{pK}$ and reshape it as $\Delta \in \mathbb{R}^{p\times K}$, row ℓ holding the coefficients attached to X_ℓ, so that $\{\mathbf{X}\otimes\mathbf{\Psi}(\mathbf{U})\}^\top\delta = \mathbf{X}^\top\Delta\mathbf{\Psi}(\mathbf{U})$. By the tower property,

$$\delta^\top \Pi^\top \Sigma \Pi \delta = \mathrm{E}\big[\{\mathbf{X}^\top\Delta\mathbf{\Psi}(\mathbf{U})\}^2\big] = \mathrm{E}\big[\mathbf{\Psi}(\mathbf{U})^\top\Delta^\top \mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\,\Delta\mathbf{\Psi}(\mathbf{U})\big].$$

Assumption 2 bounds the inner quadratic form between $\kappa_1\|\Delta\mathbf{\Psi}(\mathbf{U})\|_2^2$ and $\kappa_2\|\Delta\mathbf{\Psi}(\mathbf{U})\|_2^2$ almost surely, and $\mathrm{E}\,\|\Delta\mathbf{\Psi}(\mathbf{U})\|_2^2 = \delta^\top(I_p\otimes\Sigma_{\mathbf{\Psi}})\delta$ lies between $\lambda_{\min}(\Sigma_{\mathbf{\Psi}})\|\delta\|_2^2$ and $\lambda_{\max}(\Sigma_{\mathbf{\Psi}})\|\delta\|_2^2$. Combining the two and using the first

part gives $\gamma = \kappa_1 c_U \le \lambda_{\min}(\Sigma)$ and $\lambda_{\max}(\Sigma) \le \kappa_2 C_U$. If $\mathbf{X}$ and $\mathbf{U}$ are independent, the conditional second moment matrix is constant, the same computation gives $\delta^\top \Pi^\top \Sigma \Pi \delta = \delta^\top \{\mathrm{E}(\mathbf{X}\mathbf{X}^\top) \otimes \Sigma_{\mathbf{\Psi}}\}\delta$ for every δ, and two symmetric matrices with the same quadratic form are equal; the eigenvalues of a Kronecker product of symmetric matrices are the products of the eigenvalues, and $\lambda_{\min}\{\mathrm{E}(\mathbf{X}\mathbf{X}^\top)\} \ge \kappa_1$ by Jensen's inequality, since $\lambda_{\min}$ is concave and $\mathrm{E}(\mathbf{X}\mathbf{X}^\top) = \mathrm{E}\{\mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U})\}$.

Part (ii), the empirical matrix. First the range: $\|\mathbf{Z}_i\|_2^2 = \|\mathbf{X}_i\|_2^2 \, \|\mathbf{\Psi}(\mathbf{U}_i)\|_2^2 \le pB_X^2(1 + qC_\psi 2^J) = R_J$ by (S4.1) applied to each of the q coordinates. Set $A_i = (\mathbf{Z}_i\mathbf{Z}_i^\top - \Sigma)/n$, independent, centered and symmetric, with $\widehat{\Sigma} - \Sigma = \sum_i A_i$. Then $\|\mathbf{Z}_i\mathbf{Z}_i^\top\|_{\mathrm{op}} = \|\mathbf{Z}_i\|_2^2 \le R_J$ and $\|\Sigma\|_{\mathrm{op}} \le R_J$, so $\|A_i\|_{\mathrm{op}} \le 2R_J/n$; and $\mathrm{E}\{(\mathbf{Z}\mathbf{Z}^\top - \Sigma)^2\} = \mathrm{E}(\|\mathbf{Z}\|_2^2 \, \mathbf{Z}\mathbf{Z}^\top) - \Sigma^2 \preceq R_J\Sigma$, so that $\|\sum_i \mathrm{E}A_i^2\|_{\mathrm{op}} \le R_J\lambda_{\max}(\Sigma)/n \le R_J\kappa_2 C_U/n$. The matrix Bernstein inequality (Tropp, 2012, Theorem 1.4) applied to $\pm\sum_i A_i$ gives, for every $t > 0$,

$$\mathrm{P}\Big(\Big\|\widehat{\Sigma} - \Sigma\Big\|_{\mathrm{op}} \ge t\Big) \;\le\; 2D \exp\Big\{-\frac{nt^2}{2R_J(\kappa_2 C_U + 2t/3)}\Big\},$$

and $t = \gamma/2$ with Weyl's inequality gives $\lambda_{\min}(\widehat{\Sigma}) \ge \lambda_{\min}(\Sigma) - \gamma/2 \ge \gamma/2$ on the complementary event, which is the display of part (ii). The exponent is $-cn/R_J$ with c fixed and $R_J \asymp pq2^J$, while $\log(2D) \asymp \log(pq2^J)$, so the bound tends to zero exactly when $n/(pq2^J) - \log(pq2^J) \to \infty$, which is implied by $pq2^J \log(pq2^J)/n \to 0$. On that event $\phi_0^2(S; \widehat{\Sigma}) \ge \gamma/2$ for every

support, by (S1.1). □

Remark S4.1 (What the two assumptions exclude)**.** Assumption 2 fails exactly when a nontrivial linear combination of the linear covariates is almost surely a function of the modulators on a set of positive probability; in the canonical case $X_1 \equiv 1$ and $X_2 = h(U_1)$ the conditional second moment matrix has rank one and the population Gram matrix degenerates as J grows, because $h(u_1)\psi_{jk}(u_1)$ is approached by the span of the basis. In Assumption 3 it is the joint density that is bounded below: if $U_2 = U_1$ almost surely, both marginals are uniform but $\psi_{jk}(U_1) - \psi_{jk}(U_2) \equiv 0$ and $\Sigma_{\mathbf{\Psi}}$ is singular. Independence between $\mathbf{X}$ and $\mathbf{U}$ buys the exact factorization of part (i) and nothing else.

S5 The oracle inequality

[k = 4: the calibration lemma of the lasso, the proof of Theorem 2 for the lasso, the remark on the cone and the corollary that instantiates the design condition, which stood in this section, are moved to Section S8, unchanged except for the numbering of the results they cite; Lemmas S5.1, S5.3 and S5.4 stay, the first extended to the penalty of equation (2.5).]

~~Four lemmas precede the proof of Theorem 2.~~ A proposition on the partition into chunks and four lemmas precede the proof of Theorem 2.

Proposition S5.1 (The partition into chunks)**.** *Let $\mathcal{G}$ be the partition of Section 2.3 of the paper with $b \geq 2$ and $J \geq 1$, and let $j^* = \max\{j \geq 0 : 2^j < b\}$.*

(i) Every chunk of a fine level, $j > j^*$, is contained in that level and has between b and $2b-1$ coefficients. If $J > j^*+1$, the coarse chunk has $2^{j^*+1}-1$ coefficients, and $b-1 \leq 2^{j^*+1}-1 \leq 2b-3$; if $J \leq j^*+1$, each block is a single chunk of 2^J-1 coefficients.

(ii) $|\mathcal{G}| \leq pq(1+2^J/b)$.

(iii) With $w_G = |G|^{1/2}$, $\varrho^2 = \max_G |G| / \min_G |G| \leq (2b-1)/(b-1) \leq 3$.

Proof. All blocks have the same structure, so it suffices to look at one. The coarse chunk has $\sum_{j=0}^{\min(j^*,J-1)} 2^j = 2^{\min(j^*,J-1)+1} - 1$ coefficients. If $J \leq j^*+1$, this is 2^J-1 and there is no fine level. Otherwise $2^{j^*} < b \leq 2^{j^*+1}$ and, b being an integer, $2^{j^*} \leq b-1$, so that $b-1 \leq 2^{j^*+1}-1 \leq 2b-3$. A fine level j has $k_j = \lfloor 2^j/b \rfloor \geq 1$ chunks, $k_j - 1$ of them with b coefficients and the last with $b + r_j$, where $r_j = 2^j - k_j b \in \{0, \ldots, b-1\}$; all lie between b and $2b-1$, which is (i). For (ii), a block has $1 + \sum_{j=j^*+1}^{J-1} \lfloor 2^j/b \rfloor \leq 1 + (2^J - 2^{j^*+1})/b \leq 1 + 2^J/b$ chunks. For (iii), with $w_G = |G|^{1/2}$, $\varrho^2 = \max|G| / \min|G|$; with no fine level every chunk has 2^J-1 coefficients and $\varrho = 1$, and otherwise (i) gives $\min|G| \geq b-1$ and $\max|G| \leq 2b-1$, and

$(2b-1)/(b-1) = 2 + 1/(b-1) \le 3.$ □

Lemma S5.1 (Profiling and the Schur complement)**.** *Suppose* $\mathbf{A}$ *has rank* p*. Then* $\widehat{\boldsymbol{\theta}}$ *minimizes* ~~$\left\|M_{\mathbf{A}}\mathbf{Y} - \widetilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2 + 2\lambda\|\boldsymbol{\theta}\|_1$~~ $\left\|M_{\mathbf{A}}\mathbf{Y} - \widetilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2 + 2\lambda\|\boldsymbol{\theta}\|_{\mathcal{G},w}$*, or the same with* $\|\boldsymbol{\theta}\|_1$ *for the lasso, and* $\widehat{\mathbf{c}} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top(\mathbf{Y} - \mathbf{B}\widehat{\boldsymbol{\theta}})$*; with* $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$*,*

$$\widehat{f} - f_J \;=\; \widetilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}), \tag{S5.1}$$

and the same identity holds with any comparator $\bar{\boldsymbol{\beta}}$*,* $\bar{\mathbf{b}}$ *and* $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$ *in place of* $\boldsymbol{\beta}^*$*,* $\mathbf{b}$ *and* v*;* $\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}}$ *is the Schur complement of the block* $\mathbf{A}$ *in* $\widehat{\Sigma}$ *and*

$$\widetilde{\gamma} \;=\; \lambda_{\min}(\widehat{\Sigma}_{\mathbf{B}|\mathbf{A}}) \;\ge\; \lambda_{\min}(\widehat{\Sigma}); \tag{S5.2}$$

and if $\lambda_{\min}(\widehat{\Sigma}) > 0$ *the minimizer in (2.5) is unique.*

Proof. The argument uses only that the penalty does not depend on $\mathbf{c}$ and is convex, so it holds for both penalties. The objective is convex and, for fixed $\boldsymbol{\theta}$, quadratic and coercive in $\mathbf{c}$, with minimum at the stated value and minimized value $\|M_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\boldsymbol{\theta})\|_n^2 = \left\|M_{\mathbf{A}}\mathbf{Y} - \widetilde{\mathbf{B}}\boldsymbol{\theta}\right\|_n^2$, since $M_{\mathbf{A}}$ is a symmetric idempotent; minimizing first in $\mathbf{c}$ and then in $\boldsymbol{\theta}$ gives the same set of minimizers as minimizing jointly. For (S5.1), $\mathbf{A}\widehat{\mathbf{c}} = \Pi_{\mathbf{A}}(\mathbf{Y} - \mathbf{B}\widehat{\boldsymbol{\theta}})$ and $\mathbf{A}\mathbf{c} = \Pi_{\mathbf{A}}\mathbf{A}\mathbf{c}$, and $\mathbf{Y} = \mathbf{A}\mathbf{c} + \mathbf{B}\boldsymbol{\theta}^* + \mathbf{b} + \boldsymbol{\varepsilon}$, so $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$ and $\widehat{f} - f_J = M_{\mathbf{A}}\mathbf{B}v + \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$; the same computation with $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} +$

$\mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \boldsymbol{\varepsilon}$ gives the identity for a comparator. For (S5.2), the variational characterization of the Schur complement gives, for every $v \in \mathbb{R}^d$,

$$v^\top \widehat{\Sigma}_{\mathbf{B}|\mathbf{A}} v \;=\; \min_{u \in \mathbb{R}^p} (u^\top, v^\top)\, \widehat{\Sigma}\, (u^\top, v^\top)^\top \;\geq\; \min_{u} \lambda_{\min}(\widehat{\Sigma})(\|u\|_2^2 + \|v\|_2^2) \;=\; \lambda_{\min}(\widehat{\Sigma})\, \|v\|_2^2 .$$

Finally $\lambda_{\min}(\widehat{\Sigma}) > 0$ makes the quadratic term strictly convex, and adding a convex function preserves strict convexity. □

Lemma S5.2 (Calibration of the penalty over chunks)**.** *Under Assumption 1, let $\mathcal{G}$ be any partition of the penalized coordinates and $\alpha \in (0,1)$.*

(i) With $\lambda_{0,G}$ as in Theorem 2, the event $\mathcal{E} = \{\left\|(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G\right\|_2 \leq \lambda_{0,G}$ for every $G\}$ has $\mathrm{P}(\mathcal{E} \mid \mathbf{X}, \mathbf{U}) \geq 1 - \alpha$. Consequently, for any weights w and every $\lambda \geq \lambda_w$, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$ and $\mathrm{P}(\mathcal{T}_{\mathcal{G},w}) \geq 1 - \alpha$.

(ii) (Closed form.) $\mathrm{tr}(\widetilde{\Psi}_G) \leq |G| \widehat{\sigma}^2_{\max}$ and $\left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}} \leq \lambda_{\max}(\widehat{\Sigma})$; hence, on the event $\{\widehat{\sigma}^2_{\max} \leq 2B_X^2 C_U\} \cap \{\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\}$, $\lambda_{0,G} \leq \bar{\lambda}_{0,G}$ for every G, with

$$\bar{\lambda}_{0,G} \;=\; \frac{\sigma}{\sqrt{n}} \Big\{ B_X \sqrt{2 C_U |G|} + \sqrt{2\Lambda \log(|\mathcal{G}|/\alpha)} \Big\},$$

which is deterministic and is the quantity of equation (3.1) of the paper.

(iii) (The price of the weights.) With $\lambda_n^{\mathcal{G}} = 2 \max_G \bar{\lambda}_{0,G}/w_G$ and $\lambda_{n,1}^{\mathcal{G}} = 2 \max_G \bar{\lambda}_{0,G}$, $\lambda_n^{\mathcal{G}} w_G \leq \varrho \lambda_{n,1}^{\mathcal{G}}$ for every G, and therefore $(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{H}) \leq \varrho^2 (\lambda_{n,1}^{\mathcal{G}})^2 |\mathcal{H}|$ for every $\mathcal{H} \subseteq \mathcal{G}$. The same holds with λ_w and $2 \max_G \lambda_{0,G}$ in place of $\lambda_n^{\mathcal{G}}$ and $\lambda_{n,1}^{\mathcal{G}}$.

The event $\mathcal{E}$ does not depend on the weights: they only choose the chunk that dictates the penalty, the one with the largest $\lambda_{0,G}/w_G$, and the price of unequal weights is the factor ϱ of (iii), paid in the estimation term. With unit weights and chunks of at most $b_{\max}$ coefficients, $\lambda^{\mathcal{G}}_{n,1} \asymp \sigma\{(b_{\max} + \log|\mathcal{G}|)/n\}^{1/2}$, which is the penalty level of Klopp and Pensky (2015, eq. 3.14) up to constants and the sub-Gaussian version of the level of Lounici et al. (2011, Lemma 3.1).

Proof. (i). Condition on $(\mathbf{X}, \mathbf{U})$. Then $\widetilde{\mathbf{B}}$ is fixed and, by Assumption 1, the errors are independent with $\mathrm{E}\{\exp(a_i\varepsilon_i) \mid \mathbf{X}, \mathbf{U}\} \le \exp(\sigma^2 a_i^2/2)$, so that $\mathrm{E}\{\exp(a^\top\boldsymbol{\varepsilon}) \mid \mathbf{X}, \mathbf{U}\} \le \exp(\sigma^2 \|a\|_2^2/2)$ for every $a \in \mathbb{R}^n$, which is the condition of Hsu et al. (2012, Theorem 2.1) with mean zero. Fix G and write $M_G = \widetilde{\mathbf{B}}_G^\top/n$, so that $(\widetilde{\mathbf{B}}^\top\boldsymbol{\varepsilon}/n)_G = M_G\boldsymbol{\varepsilon}$, and $\Sigma_G = M_G^\top M_G \in \mathbb{R}^{n\times n}$. The theorem is stated for a square matrix A with $\Sigma = A^\top A$; take $A = \Sigma_G^{1/2}$, which has the same Σ_G, and note that $\left\|\Sigma_G^{1/2}\boldsymbol{\varepsilon}\right\|_2^2 = \boldsymbol{\varepsilon}^\top\Sigma_G\boldsymbol{\varepsilon} = \|M_G\boldsymbol{\varepsilon}\|_2^2$. Since $\mathrm{tr}(\Sigma_G) = \mathrm{tr}(\widetilde{\Psi}_G)/n$, $\|\Sigma_G\|_{\mathrm{op}} = \left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}}/n$ and $\mathrm{tr}(\Sigma_G^2) \le \|\Sigma_G\|_{\mathrm{op}}\,\mathrm{tr}(\Sigma_G)$, the theorem gives, for every $t > 0$, with conditional probability at least $1 - e^{-t}$,

$$\left\|(\widetilde{\mathbf{B}}^\top\boldsymbol{\varepsilon}/n)_G\right\|_2^2 \;\le\; \sigma^2\left[\mathrm{tr}(\Sigma_G) + 2\{\mathrm{tr}(\Sigma_G^2)t\}^{1/2} + 2\|\Sigma_G\|_{\mathrm{op}}\,t\right] \;\le\; \frac{\sigma^2}{n}\left[\{\mathrm{tr}(\widetilde{\Psi}_G)\}^{1/2} + \{2\left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}}t\}^{1/2}\right]^2.$$

With $t = \log(|\mathcal{G}|/\alpha)$ each chunk fails with conditional probability at most $\alpha/|\mathcal{G}|$, and a union bound over the chunks and the tower property give

$\mathrm{P}(\mathcal{E}) \geq 1-\alpha$. On $\mathcal{E}$, $\left\|(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G\right\|_2 / w_G \leq \lambda_{0,G}/w_G \leq \lambda_w/2 \leq \lambda/2$ for every G, so $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$.

(ii). Since $M_{\mathbf{A}} \preceq I_n$, $\widetilde{\Psi}_G = \mathbf{B}_G^\top M_{\mathbf{A}} \mathbf{B}_G/n \preceq \widehat{\Sigma}_{GG}$, the principal submatrix of $\widehat{\Sigma}$ on G. Hence $\mathrm{tr}(\widetilde{\Psi}_G) \leq \sum_{a\in G}(\widehat{\Sigma})_{aa} \leq |G|\widehat{\sigma}^2_{\max}$, and $\left\|\widetilde{\Psi}_G\right\|_{\mathrm{op}} \leq \lambda_{\max}(\widehat{\Sigma}_{GG}) \leq \lambda_{\max}(\widehat{\Sigma})$ by Cauchy interlacing. On the event of the statement, $\widehat{\sigma}_{\max} \leq B_X(2C_U)^{1/2}$ and $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$.

(iii). For every G, $\lambda_n^{\mathcal{G}} w_G = 2\max_{G'}(\bar{\lambda}_{0,G'} w_G/w_{G'}) \leq 2\max_{G'}\bar{\lambda}_{0,G'} \cdot \max_{G'}(w_G/w_{G'}) \leq \varrho\lambda_{n,1}^{\mathcal{G}}$, and $(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{H}) = \sum_{G\in\mathcal{H}}(\lambda_n^{\mathcal{G}} w_G)^2 \leq \varrho^2(\lambda_{n,1}^{\mathcal{G}})^2|\mathcal{H}|$. The same computation holds with the $\lambda_{0,G}$. □

Lemma S5.3 (Empirical column norms)**.** *Under Assumptions 2 to 4, every penalized column has* $\mathrm{E}\{(\widehat{\Sigma})_{aa}\} \leq B_X^2 C_U$ *and, for every* $t > 0$*, with* $R'_J = B_X^2 \|\psi\|_\infty^2 2^{J-1}$*,*

$$\mathrm{P}(\widehat{\sigma}^2_{\max} > B_X^2 C_U + t) \;\leq\; d\exp\Big\{-\frac{nt^2}{2(R'_J B_X^2 C_U + R'_J t/3)}\Big\}.$$

In particular $\mathrm{P}(\widehat{\sigma}^2_{\max} \leq 2B_X^2 C_U) \to 1$ *whenever* $2^J \log d/n \to 0$.

Proof. For the column $a = (\ell m, jk)$ the summand is $W_i = X_{i\ell}^2 \psi_{jk}(U_{im})^2 \geq 0$, with $\mathrm{E}(W) \leq B_X^2 \int_0^1 \psi_{jk}^2 p_{U_m} \leq B_X^2 C_U$ by orthonormality and the bounded marginal density, and $0 \leq W_i \leq B_X^2 \|\psi\|_\infty^2 2^{J-1} = R'_J$ because $j \leq J-1$. Then $\mathrm{Var}(W) \leq \mathrm{E}(W^2) \leq R'_J \mathrm{E}(W) \leq R'_J B_X^2 C_U$, and the one sided Bernstein inequality with a union bound over the d columns gives the display.

With $t = B_X^2 C_U$ the exponent is $-3nB_X^2 C_U/(8R'_J)$ and $R'_J \asymp 2^J$. □

Lemma S5.4 (The ℓ_1 norm of the oracle)**.** *Under Assumption 5, for every pair (ℓ, m) and every level, $\sum_k |\theta^*_{\ell m,jk}| \le 2^{j(1-1/\pi)} \left\|\theta^*_{\ell m,j\cdot}\right\|_\pi \le C_g 2^{j(1/2-s)}$, with no dependence on π, and therefore $\|\boldsymbol{\theta}^*\|_1 \le pqC_g \sum_{j<J} 2^{j(1/2-s)}$, which is $O(1)$ if $s > 1/2$, of order J if $s = 1/2$ and of order $2^{J(1/2-s)}$ if $s < 1/2$.*

Proof. Hölder's inequality in $\mathbb{R}^{2^j}$ gives $\|a\|_1 \le (2^j)^{1-1/\pi} \|a\|_\pi$, and Assumption 5 bounds the ℓ_π norm of the level by $C_g 2^{-j(s+1/2-1/\pi)}$; the resulting exponent is $j(1-1/\pi) - j(s+1/2-1/\pi) = j(1/2-s)$, in which π has disappeared. Summing over the levels below J and over the pq blocks gives the three regimes of the geometric sum. □

Proof of Theorem 2

The event $\mathcal{T}_{\mathcal{G},w}$ has probability at least $1-\alpha$ whenever $\lambda \ge \lambda_w$, by Lemma S5.2(i). Fix a comparator $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}})$ and write $\|\cdot\|$ for $\|\cdot\|_{\mathcal{G},w}$ in this proof.

Step 1, the basic inequality. By Lemma S5.1, $\widehat{\boldsymbol{\theta}}$ minimizes the residualized objective; comparing its value at $\widehat{\boldsymbol{\theta}}$ with its value at $\bar{\boldsymbol{\theta}}$, and using $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \boldsymbol{\varepsilon}$ and $M_{\mathbf{A}}\mathbf{A} = 0$, so that $M_{\mathbf{A}}\mathbf{Y} = \widetilde{\mathbf{B}}\bar{\boldsymbol{\theta}} + r$ with $r = M_{\mathbf{A}}(\bar{\mathbf{b}} + \boldsymbol{\varepsilon})$,

$$\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + 2\lambda \left\|\widehat{\boldsymbol{\theta}}\right\| \;\le\; 2\langle r, \widetilde{\mathbf{B}}\bar{v}\rangle_n + 2\lambda \left\|\bar{\boldsymbol{\theta}}\right\|. \tag{S5.3}$$

Step 2, the two noises. Since $M_{\mathbf{A}}$ is symmetric and $M_{\mathbf{A}}\widetilde{\mathbf{B}} = \widetilde{\mathbf{B}}$, $\langle r, \widetilde{\mathbf{B}}\bar{v}\rangle_n = \langle \bar{\mathbf{b}} + \boldsymbol{\varepsilon}, \widetilde{\mathbf{B}}\bar{v}\rangle_n$. The stochastic part is handled by the dual norm of $\|\cdot\|$, which is $z \mapsto \max_G \|z_G\|_2 / w_G$: on $\mathcal{T}_{\mathcal{G},w}$,

$$2\left|\langle \boldsymbol{\varepsilon}, \widetilde{\mathbf{B}}\bar{v}\rangle_n\right| = 2\left|\sum_G \langle (\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G, \bar{v}_G\rangle\right| \leq 2\max_G \frac{\left\|(\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G\right\|_2}{w_G} \sum_G w_G \|\bar{v}_G\|_2 \leq \lambda \|\bar{v}\|.$$

The deterministic part is handled by the Cauchy-Schwarz and Young inequalities: $2|\langle \bar{\mathbf{b}}, \widetilde{\mathbf{B}}\bar{v}\rangle_n| = 2|\langle M_{\mathbf{A}}\bar{\mathbf{b}}, \widetilde{\mathbf{B}}\bar{v}\rangle_n| \leq 2\left\|\bar{\mathbf{b}}\right\|_n \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n \leq \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2/2 + 2\left\|\bar{\mathbf{b}}\right\|_n^2$. Substituting both in (S5.3),

$$\tfrac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + 2\lambda\left\|\widehat{\boldsymbol{\theta}}\right\| \leq \lambda\|\bar{v}\| + 2\lambda\left\|\bar{\boldsymbol{\theta}}\right\| + 2\left\|\bar{\mathbf{b}}\right\|_n^2. \tag{S5.4}$$

Step 3, the slow rate. Bounding $\|\bar{v}\| \leq \left\|\widehat{\boldsymbol{\theta}}\right\| + \left\|\bar{\boldsymbol{\theta}}\right\|$ in (S5.4) gives part (i). No property of $\widehat{\Sigma}$ was used.

Step 4, decomposability. For $G \notin \bar{\mathcal{G}}$, $\bar{\boldsymbol{\theta}}_G = 0$ and $\widehat{\boldsymbol{\theta}}_G = \bar{v}_G$; for $G \in \bar{\mathcal{G}}$, $\left\|\widehat{\boldsymbol{\theta}}_G\right\|_2 \geq \left\|\bar{\boldsymbol{\theta}}_G\right\|_2 - \|\bar{v}_G\|_2$. Hence $\left\|\widehat{\boldsymbol{\theta}}\right\| \geq \left\|\bar{\boldsymbol{\theta}}\right\| - \|\bar{v}_{\bar{\mathcal{G}}}\| + \|\bar{v}_{\bar{\mathcal{G}}^c}\|$, while $\|\bar{v}\| = \|\bar{v}_{\bar{\mathcal{G}}}\| + \|\bar{v}_{\bar{\mathcal{G}}^c}\|$; substituting in (S5.4) and cancelling $2\lambda\left\|\bar{\boldsymbol{\theta}}\right\|$,

$$\tfrac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + \lambda\|\bar{v}_{\bar{\mathcal{G}}^c}\| \leq 3\lambda\|\bar{v}_{\bar{\mathcal{G}}}\| + 2\left\|\bar{\mathbf{b}}\right\|_n^2. \tag{S5.5}$$

Step 5, two cases, with no cone. By the Cauchy-Schwarz inequality and the definition of $\widetilde{\gamma}$,

$$\|\bar{v}_{\bar{\mathcal{G}}}\| = \sum_{G\in\bar{\mathcal{G}}} w_G \|\bar{v}_G\|_2 \leq \bar{\mathcal{W}}^{1/2}\|\bar{v}\|_2 \leq (\bar{\mathcal{W}}/\widetilde{\gamma})^{1/2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n, \tag{S5.6}$$

because $\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 = \bar{v}^\top \widehat{\Sigma}_{\mathbf{B}|\mathbf{A}}\bar{v} \geq \widetilde{\gamma}\,\|\bar{v}\|_2^2$. If $2\left\|\bar{\mathbf{b}}\right\|_n^2 \leq \lambda\,\|\bar{v}_{\bar{\mathcal{G}}}\|$, then (S5.5) gives $\frac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + \lambda\,\|\bar{v}_{\bar{\mathcal{G}}^c}\| \leq 4\lambda\,\|\bar{v}_{\bar{\mathcal{G}}}\|$, and with (S5.6), $\frac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \leq 4\lambda(\bar{\mathcal{W}}/\widetilde{\gamma})^{1/2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n$, that is $\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \leq 64\lambda^2\bar{\mathcal{W}}/\widetilde{\gamma}$ and $\|\bar{v}\|_2^2 \leq \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2/\widetilde{\gamma} \leq 64\lambda^2\bar{\mathcal{W}}/\widetilde{\gamma}^2$. If instead $2\left\|\bar{\mathbf{b}}\right\|_n^2 > \lambda\,\|\bar{v}_{\bar{\mathcal{G}}}\|$, then (S5.5) gives $\frac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 < 8\left\|\bar{\mathbf{b}}\right\|_n^2$, that is $\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 < 16\left\|\bar{\mathbf{b}}\right\|_n^2$, and $\|\bar{v}\|_2^2 < 16\left\|\bar{\mathbf{b}}\right\|_n^2/\widetilde{\gamma}$. Each quantity is bounded by the maximum of the two cases, hence by their sum, which is part (ii).

Step 6, prediction. By Lemma S5.1 with the comparator, $\widehat{f} - f = \widetilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}}+\boldsymbol{\varepsilon}) - \bar{\mathbf{b}}$, and $\left\|\Pi_{\mathbf{A}}\bar{\mathbf{b}}\right\|_n \leq \left\|\bar{\mathbf{b}}\right\|_n$, which gives the triangle inequality of part (iii). Conditionally on the covariates the errors are independent, centered and of variance at most σ^2, so $\mathrm{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2\mathrm{tr}(\Pi_{\mathbf{A}})/n \leq \sigma^2 p/n$. Since $\mathcal{T}_{\mathcal{G},w}$, $\widetilde{\gamma}$ and λ do not depend on the comparator, the bounds hold simultaneously for all comparators, and with $(x+y+z)^2 \leq 3(x^2+y^2+z^2)$ and part (ii) they give, on $\mathcal{T}_{\mathcal{G},w} \cap \{\widetilde{\gamma} > 0\}$,

$$\left\|\widehat{f} - f\right\|_n^2 \;\leq\; 3\min_{\bar{\boldsymbol{\beta}}}\Big\{\frac{64\lambda^2\bar{\mathcal{W}}}{\widetilde{\gamma}} + 20\left\|\bar{\mathbf{b}}\right\|_n^2\Big\} + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2\,. \qquad \square \qquad \text{(S5.7)}$$

Remark S5.1 (Why there is no cone condition, also over chunks)**.** The step that dispenses with the cone is the one of the lasso (Remark S8.1): profiling removes the levels from the problem, and the full smallest eigenvalue of Proposition 2 controls $\|\bar{v}_{\bar{\mathcal{G}}}\| \leq (\bar{\mathcal{W}}/\widetilde{\gamma})^{1/2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n$ over the whole space, by (S5.6). Lounici et al. (2011) need a restricted eigenvalue condition for groups on a weighted cone (their Assumption 3.1); here it is dominated by

the full smallest eigenvalue, and the constant is the same for every $\bar{\mathcal{G}}$.

Remark S5.2 (The estimator that the software computes)**.** The software standardizes the columns, orthonormalizes each chunk and penalizes the orthonormalized coordinates with the weights $|G|^{1/2}$. On the original scale its penalty is $\sum_G w_G \left\| Q_G^{1/2} \boldsymbol{\theta}_G \right\|_2$, with $Q_G = \mathbf{B}_G^\top M_1 \mathbf{B}_G / n$ the centered Gram matrix of the chunk, $M_1 = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$. Since $\|M_1 \mathbf{B}_G \boldsymbol{\theta}_G\|_2 = n^{1/2} \left\| Q_G^{1/2} \boldsymbol{\theta}_G \right\|_2$, this is the standardized group lasso of Simon and Tibshirani (2012, eq. 1.4), which penalizes the fit of each chunk rather than its coefficients and which they show to be equivalent to orthonormalizing within groups (their Section 2). Theorem 2 holds with any Q_G positive definite and measurable with respect to the covariates, with three changes: the event becomes $\{\max_G \left\| Q_G^{-1/2} (\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n)_G \right\|_2 / w_G \le \lambda/2\}$, by the dual norm; $\bar{\mathcal{W}}$ becomes $q_{\max}\bar{\mathcal{W}}$ in parts (ii) and (iii), with $q_{\max} = \max_G \lambda_{\max}(Q_G)$, because Step 5 reads $\sum_{G\in\bar{\mathcal{G}}} w_G \left\| Q_G^{1/2} \bar{v}_G \right\|_2 \le (q_{\max}\bar{\mathcal{W}})^{1/2} \|\bar{v}\|_2$; and Lemma S5.2(i) holds with $Q_G^{-1/2} \widetilde{\Psi}_G Q_G^{-1/2}$ in place of $\widetilde{\Psi}_G$, by the same proof with $Q_G^{-1/2} \widetilde{\mathbf{B}}_G^\top / n$ in place of $\widetilde{\mathbf{B}}_G^\top / n$. When the constant covariate $X_1 \equiv 1$ is among the columns of $\mathbf{A}$, $\Pi_{\mathbf{A}} \succeq I_n - M_1$, so that $M_{\mathbf{A}} \preceq M_1$ and $\widetilde{\Psi}_G \preceq Q_G$: the trace of $Q_G^{-1/2} \widetilde{\Psi}_G Q_G^{-1/2}$ is at most $|G|$ and its norm at most one, and

$$\lambda_{0,G} \;\le\; \frac{\sigma}{\sqrt{n}} \Big\{ |G|^{1/2} + \sqrt{2\log(|\mathcal{G}|/\alpha)} \Big\},$$

which is pivotal, with neither $\widehat{\sigma}_{\max}$ nor Λ; and $q_{\max} \le \lambda_{\max}(\widehat{\Sigma}) \le \Lambda$ on $\mathcal{E}_\Sigma$,

since $Q_G \preceq \widehat{\Sigma}_{GG}$. The rates of Section S6 hold for this estimator with the constant multiplied by Λ.

Corollary S5.1 (The ideal risk form)**.** *In the conditions of Theorem 2, on the event $\mathcal{T}_{\mathcal{G},w} \cap \{\widetilde{\gamma} > 0\}$, let $\Lambda' \geq \lambda_{\max}(\widehat{\Sigma})$, for instance $\Lambda' = \Lambda$ on $\mathcal{E}_\Sigma$, and*

$$\eta \;=\; \frac{1.6 \max_G (\lambda w_G)^2}{\Lambda' \widetilde{\gamma}}.$$

Then

$$\left\| \widehat{f} - f \right\|_n^2 \;\leq\; 120\Lambda' R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) + 120 \left\| \mathbf{b} \right\|_n^2 + 3 \left\| \Pi_{\mathbf{A}} \boldsymbol{\varepsilon} \right\|_n^2, \qquad \left\| \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^* \right\|_2^2 \;\leq\; \frac{82\Lambda' R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) + 64 \left\| \mathbf{b} \right\|_n^2}{\widetilde{\gamma}}.$$

With $\lambda = \lambda_n^{\mathcal{G}}$, or $\lambda = \lambda_w$, Lemma S5.2(iii) gives $\eta \leq \varrho^2 \eta_1$, with $\eta_1 = 1.6(\lambda_{n,1}^{\mathcal{G}})^2/(\Lambda'\widetilde{\gamma})$ the level of unit weights, and $R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta) \leq \varrho^2 R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta_1)$.

Proof. Let $T = \{G : \|\boldsymbol{\theta}_G^*\|_2^2 > \eta\}$ and take the comparator $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \bar{\boldsymbol{\theta}})$ with $\bar{\boldsymbol{\theta}}_G = \boldsymbol{\theta}_G^* \mathbf{1}\{G \in T\}$. Then $\bar{\mathcal{G}} \subseteq T$ and, by the definition of η, $\lambda^2 \bar{\mathcal{W}} \leq \sum_{G \in T} (\lambda w_G)^2 \leq |T| \max_G (\lambda w_G)^2 = |T| \eta \Lambda' \widetilde{\gamma}/1.6$. The residual is $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})$, and since $\mathbf{B}^\top \mathbf{B}/n$ is a principal submatrix of $\widehat{\Sigma}$,

$$\left\| \bar{\mathbf{b}} \right\|_n^2 \;\leq\; 2 \left\| \mathbf{b} \right\|_n^2 + 2\lambda_{\max}(\widehat{\Sigma}) \left\| \boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}} \right\|_2^2 \;\leq\; 2 \left\| \mathbf{b} \right\|_n^2 + 2\Lambda' \sum_{G \notin T} \left\| \boldsymbol{\theta}_G^* \right\|_2^2.$$

By (S5.7), $\left\| \widehat{f} - f \right\|_n^2 \leq 192\lambda^2 \bar{\mathcal{W}}/\widetilde{\gamma} + 60 \left\| \bar{\mathbf{b}} \right\|_n^2 + 3 \left\| \Pi_{\mathbf{A}} \boldsymbol{\varepsilon} \right\|_n^2 \leq 120\Lambda' \{\eta |T| + \sum_{G \notin T} \left\| \boldsymbol{\theta}_G^* \right\|_2^2\} + 120 \left\| \mathbf{b} \right\|_n^2 + 3 \left\| \Pi_{\mathbf{A}} \boldsymbol{\varepsilon} \right\|_n^2$, and the term in braces is exactly $R_{\mathcal{G}}(\boldsymbol{\theta}^*; \eta)$. For the coefficients, Theorem 2(ii) gives $\|\bar{v}\|_2^2 \leq 64\lambda^2 \bar{\mathcal{W}}/\widetilde{\gamma}^2 + 16 \left\| \bar{\mathbf{b}} \right\|_n^2 / \widetilde{\gamma} \;\leq\; (40\Lambda' \eta |T| + 32 \left\| \mathbf{b} \right\|_n^2 + 32\Lambda' \sum_{G \notin T} \left\| \boldsymbol{\theta}_G^* \right\|_2^2)/\widetilde{\gamma} \;\leq\; (40\Lambda' R_{\mathcal{G}} +$

$32\,\|\mathbf{b}\|_n^2)/\widetilde{\gamma}$, and $\left\|\widehat{\boldsymbol{\theta}}-\boldsymbol{\theta}^*\right\|_2^2 \le 2\,\|\bar{v}\|_2^2+2\left\|\bar{\boldsymbol{\theta}}-\boldsymbol{\theta}^*\right\|_2^2 \le (80\Lambda' R_{\mathcal{G}}+64\,\|\mathbf{b}\|_n^2)/\widetilde{\gamma}+2R_{\mathcal{G}}$; since $\widetilde{\gamma} \le \lambda_{\min}(\mathbf{B}^\top\mathbf{B}/n) \le \lambda_{\max}(\widehat{\Sigma}) \le \Lambda'$, the Schur complement being dominated by the block, $2R_{\mathcal{G}} \le 2\Lambda' R_{\mathcal{G}}/\widetilde{\gamma}$, which gives the second bound. The last statement is Lemma S5.2(iii) and the two properties of $R_{\mathcal{G}}$ in Section S1. □

S6 Rates

[k = 4: the lemma that read Theorem 2 at an arbitrary comparator for the lasso, and the proofs of Lemma 3, Proposition 3, Theorem 3, Corollary 1, Theorem 1 and the remark on compressibility for the lasso, which stood in this section, are moved to Section S8, unchanged except for the numbering of the results they cite.]

Throughout this section p, q, σ, B_X, c_U, C_U, κ_1, κ_2, C_g, s, π, r and α are fixed as n grows, $J = J_n \to \infty$, $d_n = pq(2^{J_n}-1)$, ~~$L_n = \log(2d_n/\alpha) \asymp J_n$,~~ ~~and λ_n is as in (3.1)~~$2^{J_n} \le 2n$, $b_n = \lceil \log n \rceil$ with $n \ge 3$, so that $b_n \ge 2$, $\mathcal{G}_n$ is the partition of Section 2.3 of the paper with $b = b_n$, the weights have ratio $\varrho \le \bar{\varrho}$ for a fixed $\bar{\varrho}$, and $\lambda = \lambda_n^{\mathcal{G}}$ is as in (3.1). ~~Since the constants of Theorem 2 depend on the confidence levels only through $1/\alpha'$ and $1/\alpha''$, taking $\alpha' = \alpha'' = \epsilon/3$ gives, for each $\epsilon > 0$, a constant M_ϵ with~~

~~$\mathrm{P}(\|\widehat{f} - f\|_n^2 > M_\epsilon r_n) < \epsilon$, which is the meaning of the statements in O_p below. We also write $\Lambda = \kappa_2 C_U + \gamma/2$.~~ The statements in O_p are proved by exhibiting, for each $\epsilon > 0$, an event of probability at least $1 - \epsilon$ for n large, built with the confidence levels $\alpha = \alpha' = \alpha'' = \epsilon/4$ for the noise, the bias and the levels, on which the error is at most a constant M_ϵ times the rate. The constant $\Lambda = \kappa_2 C_U + \gamma/2$ and the event $\mathcal{E}_\Sigma$ are those of Section S1.

Lemma S6.1 (The penalty level and the regime)**.** *In the setting above,*

$$(\lambda_{n,1}^{\mathcal{G}})^2 \;\le\; C_\lambda \,\frac{\sigma^2 b_n}{n}, \qquad C_\lambda = 16\big\{2B_X^2 C_U + \Lambda\big(1 + \tfrac{1}{2}\log(2pq/\alpha)\big)\big\}, \quad \text{(S6.1)}$$

and, on the event $\mathcal{A}_n = \{\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U\} \cap \mathcal{E}_\Sigma$, $\lambda_n^{\mathcal{G}} \ge \lambda_w$, *so that* $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$ *for the event* $\mathcal{E}$ *of Lemma S5.2(i),* $\widetilde{\gamma} > \gamma/2$, $\mathbf{A}$ *has rank* p *and* $\lambda_{\max}(\widehat{\Sigma}) < \Lambda$. *Moreover* $\mathrm{P}(\mathcal{A}_n) \to 1$ *whenever* $pq2^{J_n}\log(pq2^{J_n})/n \to 0$.

Proof. By $(x+y)^2 \le 2x^2+2y^2$, $(\lambda_{n,1}^{\mathcal{G}})^2 = 4\max_G \bar{\lambda}_{0,G}^2 \le (8\sigma^2/n)\{2B_X^2 C_U b_{\max} + 2\Lambda\log(|\mathcal{G}_n|/\alpha)\}$, with $b_{\max} = \max_G |G|$. By Proposition S5.1, $b_{\max} \le 2b_n - 1 < 2b_n$ and $|\mathcal{G}_n| \le pq(1 + 2^{J_n}/b_n) \le pq(1+n) \le 2pqn$, because $2^{J_n} \le 2n$ and $b_n \ge 2$; hence $\log(|\mathcal{G}_n|/\alpha) \le \log(2pq/\alpha) + \log n \le \log(2pq/\alpha) + b_n$, and $(\lambda_{n,1}^{\mathcal{G}})^2 \le (16\sigma^2/n)\{(2B_X^2 C_U + \Lambda)b_n + \Lambda\log(2pq/\alpha)\} \le C_\lambda \sigma^2 b_n/n$, using $b_n \ge 2$. On $\mathcal{A}_n$, Lemma S5.2(ii) gives $\lambda_{0,G} \le \bar{\lambda}_{0,G}$ for every G, so that $\lambda_n^{\mathcal{G}} \ge \lambda_w$ and, by the same lemma, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$. On $\mathcal{E}_\Sigma$, $\lambda_{\min}(\widehat{\Sigma}) > \gamma/2$ and $\lambda_{\max}(\widehat{\Sigma}) < \Lambda$; (S5.2) gives $\widetilde{\gamma} > \gamma/2$, and $\lambda_{\min}(\mathbf{A}^\top\mathbf{A}/n) \ge \lambda_{\min}(\widehat{\Sigma}) > 0$

gives the rank. Finally $\mathrm{P}(\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U) \to 1$ by Lemma S5.3, which asks $2^J \log d/n \to 0$, and $\mathrm{P}(\mathcal{E}_\Sigma) \to 1$ when $pq2^{J_n} \log(pq2^{J_n})/n \to 0$ (Section S1); the second condition implies the first. □

Proof of Lemma 3

Fix a block and write $\theta_j = \theta^*_{\ell m,j\cdot} \in \mathbb{R}^{2^j}$, set to zero for $j \ge J$, which only decreases the risk; write $\mathcal{G}_j$ for the chunks of a fine level j, those with $2^j \ge b$, and $r_j = \sum_{G\in\mathcal{G}_j} \min(\|\theta_G\|_2^2, \eta)$. The coarse chunk, if there is one, contributes at most η, so the block contributes at most $\eta + \sum_j r_j$, the sum running over the fine levels. The first bound on r_j is the count: each chunk of $\mathcal{G}_j$ has at least b of the 2^j coefficients of the level, so that $r_j \le |\mathcal{G}_j|\eta \le 2^j\eta/b$.

The case $\pi \le 2$. Write $a = \pi(s + 1/2) > 1$. For $u \ge 0$, $\min(u, \eta) \le \eta^{1-\pi/2}u^{\pi/2}$, and $\|x\|_2 \le \|x\|_\pi$ for $\pi \le 2$; hence the second bound,

$$r_j \;\le\; \eta^{1-\pi/2} \sum_{G\in\mathcal{G}_j} \|\theta_G\|_\pi^\pi \;\le\; \eta^{1-\pi/2}\, \|\theta_j\|_\pi^\pi \;\le\; \eta^{1-\pi/2} C_g^\pi 2^{-j(a-1)},$$

by Assumption 5, because the chunks of $\mathcal{G}_j$ are disjoint and contained in the level. Define $x \in \mathbb{R}$ by $2^{xa} = bC_g^\pi \eta^{-\pi/2}$, so that the two bounds agree at $j = x$, with common value $h = 2^x\eta/b = \eta^{1-\pi/2}C_g^\pi 2^{-x(a-1)}$. Summing the first bound over the levels $j \le x$ gives at most $2^{\lfloor x\rfloor+1}\eta/b \le 2h$ (an empty sum if $x < 0$), and summing the second over the levels $j > x$ gives at most

$h\sum_{i\geq 0} 2^{-i(a-1)} = h/(1-2^{1-a})$, since the first integer $j > x$ has $2^{-j(a-1)} \leq 2^{-x(a-1)}$. The block therefore contributes at most $\eta + \{2 + (1-2^{1-a})^{-1}\}h$. Since $1/a = \tau/\pi$, $2^x = b^{\tau/\pi} C_g^\tau \eta^{-\tau/2}$ and $h = b^{\tau/\pi - 1} C_g^\tau \eta^{1-\tau/2}$, and, with $1 - \tau/2 = 2s/(2s+1)$,

$$h = C_g^\tau \Big(\frac{\eta}{b}\Big)^{2s/(2s+1)} b^{\tau/\pi - 1 + 2s/(2s+1)} = C_g^\tau \Big(\frac{\eta}{b}\Big)^{2s/(2s+1)} b^{\tau(1/\pi - 1/2)}, \qquad \tau\Big(\frac{1}{\pi} - \frac{1}{2}\Big) = \frac{2/\pi - 1}{2s+1}.$$

The case $\pi \geq 2$. Hölder's inequality in $\mathbb{R}^{2^j}$ gives $\|\theta_j\|_2 \leq 2^{j(1/2-1/\pi)} \|\theta_j\|_\pi \leq C_g 2^{-js}$, with $1/\pi = 0$ if $\pi = \infty$, and the second bound is $r_j \leq \|\theta_j\|_2^2 \leq C_g^2 2^{-2js}$. With $2^{x(2s+1)} = C_g^2 b/\eta$ and $h = 2^x \eta/b = C_g^2 2^{-2xs}$, the same computation gives the contribution $\eta + \{2 + (1-2^{-2s})^{-1}\}h$, and $h = (C_g^2 b/\eta)^{1/(2s+1)} \eta/b = C_g^\tau (\eta/b)^{2s/(2s+1)}$, because $2/(2s+1) = \tau$.

Summing over the pq blocks gives the statement; at $\pi = 2$, $1 - 2^{1-a} = 1 - 2^{-2s}$ and the two constants agree. □

We also record that, on the event of Proposition 2(ii) with $t = \gamma/2$,

$$\|\mathbf{B}\delta\|_n^2 \leq \lambda_{\max}(\widehat{\Sigma})\, \|\delta\|_2^2 \quad \text{and} \quad \lambda_{\max}(\widehat{\Sigma}) \leq \Lambda, \tag{S6.2}$$

the first because $\mathbf{B}^\top \mathbf{B}/n$ is a principal submatrix of $\widehat{\Sigma}$ and the second by Weyl's inequality and Proposition 2(i).

Lemma S6.2 (The norm of the oracle over chunks)**.** *Under Assumptions 4 and 5, let $\mathcal{G}$ be the partition of Section 2.3 of the paper with $b \geq 2$, and let*

$\vartheta_\pi = \min(1 - 1/\pi,\, 1/2)$, *with* $\vartheta_\infty = 1/2$. *Then, for every* J,

$$\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \;\le\; pqC_g\Big\{\frac{1}{1-2^{-s'}} + b^{-\vartheta_\pi}\sum_{j=j^*+1}^{J-1} 2^{j(1/2-s)}\Big\}. \tag{S6.3}$$

The bound for each fine level is attained, by a level whose energy is spread evenly when $\pi \ge 2$, *and by a level with one large coefficient in each chunk when* $\pi \le 2$.

Proof. Fix a block and write θ_j for its coefficients of level j. By Assumption 5, $\|\theta_j\|_2 \le 2^{j(1/2-1/\pi)_+}\|\theta_j\|_\pi \le C_g 2^{-js'}$, by Hölder's inequality in $\mathbb{R}^{2^j}$ if $\pi \ge 2$ and by $\|\cdot\|_2 \le \|\cdot\|_\pi$ if $\pi \le 2$. The coarse chunk contributes at most $\sum_{j\le j^*}\|\theta_j\|_2 \le C_g(1-2^{-s'})^{-1}$. At a fine level j the $k_j \le 2^j/b$ chunks $G \in \mathcal{G}_j$ partition the level. If $\pi \ge 2$, by the Cauchy-Schwarz inequality,

$$\sum_{G\in\mathcal{G}_j}\|\theta_G\|_2 \;\le\; k_j^{1/2}\,\|\theta_j\|_2 \;\le\; \Big(\frac{2^j}{b}\Big)^{1/2} C_g 2^{-js} \;=\; C_g\, b^{-1/2}\, 2^{j(1/2-s)};$$

if $\pi \le 2$, by $\|\theta_G\|_2 \le \|\theta_G\|_\pi$ and Hölder's inequality over the chunks,

$$\sum_{G\in\mathcal{G}_j}\|\theta_G\|_2 \;\le\; k_j^{1-1/\pi}\Big(\sum_{G\in\mathcal{G}_j}\|\theta_G\|_\pi^\pi\Big)^{1/\pi} = k_j^{1-1/\pi}\,\|\theta_j\|_\pi \;\le\; \Big(\frac{2^j}{b}\Big)^{1-1/\pi} C_g 2^{-j(s+1/2-1/\pi)} = C_g\, b^{-(1-1/\pi)}\, 2^{j(1/2-}$$

Both are $C_g b^{-\vartheta_\pi} 2^{j(1/2-s)}$, and the equality cases of the two inequalities are the configurations of the statement. Summing over the fine levels $j^* < j < J$ and over the pq blocks gives (S6.3). □

Proof of Proposition 3

The nonasymptotic bound. On $\mathcal{T}_{\mathcal{G},w}$, Theorem 2(i) with the comparator $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$, so that $\bar{\mathbf{b}} = \mathbf{b}$ and $\bar{v} = v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$, gives $\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \le 6\lambda\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 4\,\|\mathbf{b}\|_n^2$, and the first part of Theorem 2(iii), with $(x+y+z)^2 \le 3(x^2+y^2+z^2)$, gives

$$\left\|\widehat{f} - f\right\|_n^2 \;\le\; 18\lambda\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 24\,\|\mathbf{b}\|_n^2 + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \qquad \text{on } \mathcal{T}_{\mathcal{G},w}. \tag{S6.4}$$

Steps 1 to 3 and 6 of the proof of Theorem 2, the only ones used, do not involve $\widehat{\Sigma}$. With $\lambda \ge \lambda_w$, calibrated on the covariates by Lemma S5.2(i), the bound holds with probability at least $1-\alpha$ in any design in which $\mathbf{A}$ has rank p, including designs with $d > n$, where $\widetilde{\gamma} = 0$.

The penalty level of the theory. Let $\mathcal{A}_n^+ = \{\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U\} \cap \{\lambda_{\max}(\widehat{\Sigma}) \le \Lambda\}$. On it Lemma S5.2(ii) gives $\lambda_{0,G} \le \bar{\lambda}_{0,G}$ for every G, hence $\lambda_n^{\mathcal{G}} \ge \lambda_w$ and $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$. The first event has probability tending to one by Lemma S5.3, whose proof uses only $|X_\ell| \le B_X$ and $p_{\mathbf{U}} \le C_U$. For the second, $\mathrm{P}\{\lambda_{\max}(\widehat{\Sigma}) \le \Lambda\} \ge \mathrm{P}\{\left\|\widehat{\Sigma} - \Sigma\right\|_{\mathrm{op}} < \Lambda - \kappa_2 C_U\}$ by Weyl's inequality, since $\lambda_{\max}(\Sigma) \le \kappa_2 C_U$, and the matrix Bernstein bound in the proof of Proposition 2(ii), with $t = \Lambda - \kappa_2 C_U$, sends it to one; that bound uses only $\|\mathbf{Z}_i\|_2^2 \le R_J$ and $\lambda_{\max}(\Sigma) \le \kappa_2 C_U$, the upper half of Proposition 2(i), and neither c_U nor κ_1, so that any $\Lambda > \kappa_2 C_U$ serves. Both probabilities tend

to one when $pq2^{J_n}\log(pq2^{J_n})/n \to 0$. Fix $\epsilon > 0$, take $\alpha = \alpha' = \alpha'' = \epsilon/4$ and intersect $\mathcal{A}_n^+$, $\mathcal{E}$, $\{\|\mathbf{b}\|_n^2 \le \mathcal{B}_{J_n}^2/\alpha'\}$ and $\{\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \le \sigma^2 p/(n\alpha'')\}$, the last two by Markov's inequality, Lemma 2 and Theorem 2(iii); for n large the intersection has probability at least $1-\epsilon$, and on it $\left\|\widehat{f}-f\right\|_n^2 \le 18\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + 24\mathcal{B}_{J_n}^2/\alpha' + 3\sigma^2 p/(n\alpha'')$, which is the first display of Proposition 3. For the chain, Lemma S5.2(iii) gives $\lambda_n^{\mathcal{G}} w_G \le \varrho\lambda_{n,1}^{\mathcal{G}}$ for every G, and summing with the weights $\|\boldsymbol{\theta}_G^*\|_2$ gives $\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \le \varrho\lambda_{n,1}^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$, with $\varrho \le \bar{\varrho}$; and $\|x\|_2 \le \|x\|_1$ in each chunk gives $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \le \|\boldsymbol{\theta}^*\|_1$. The order of $\lambda_{n,1}^{\mathcal{G}}$ is (S6.1).

Part (i). With $s > 1/2$ and $2^{j^*+1} \ge b$, $\sum_{j>j^*} 2^{j(1/2-s)} \le 2^{(j^*+1)(1/2-s)}/(1-2^{1/2-s}) \le b_n^{1/2-s}/(1-2^{1/2-s})$, and Lemma S6.2 gives $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \le pqC_g\{(1-2^{-s'})^{-1} + b_n^{1/2-s-\vartheta_\pi}(1-2^{1/2-s})^{-1}\} = O(1)$; hence $\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} = O\{(b_n/n)^{1/2}\} = O\{(\log n/n)^{1/2}\}$. With $J_n \ge \log_2 n/(4s')$, $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \le n^{-1/2}$, and p/n is of smaller order. With $s' > 1/4$ the choice $J_n = \lceil \log_2 n/(4s')\rceil$ has $2^{J_n} < 2n^{1/(4s')}$ and satisfies the conditions of the statement.

Part (ii). With $s < 1/2$, $\sum_{j^*<j<J_n} 2^{j(1/2-s)} \le 2^{J_n(1/2-s)}/(2^{1/2-s}-1)$, so that $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O\{1 + b_n^{-\vartheta_\pi}2^{J_n(1/2-s)}\}$ and, since $1/2 - \vartheta_\pi = (1/\pi - 1/2)_+$,

$$\lambda_n^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \lesssim \left(\frac{b_n}{n}\right)^{1/2} + n^{-1/2}\, b_n^{(1/\pi-1/2)_+}\, 2^{J_n(1/2-s)}.$$

Equating the second term with the bias $2^{-2J_n s'}$ gives $2^{J_n(1/2-s+2s')} \asymp n^{1/2} b_n^{-(1/\pi-1/2)_+}$, which is the choice of the statement, with common value $2^{-2J_n s'} \asymp \{b_n^{(2/\pi-1)_+}/n\}^{2s'/(1-2s+4s')}$.

The term $(b_n/n)^{1/2}$ and p/n are of smaller order, because $2s'/(1-2s+4s') < 1/2$ is equivalent to $s < 1/2$. The conditions of the statement hold because $s' > s/2$ gives $1/(1-2s+4s') < 1$, and $2^{J_n} \geq b_n$ for n large. When $\pi \geq 2$, $s' = s$, $(2/\pi - 1)_+ = 0$ and $1 - 2s + 4s = 2s + 1$. □

Proposition 3 above the concentration ceiling

The conditions $s' > 1/4$ in Proposition 3(i) and $s' > s/2$ in Proposition 3(ii) keep the resolution of the balance below the ceiling $2^{J_n} \asymp n/\log n$, under which $\widehat{\sigma}^2_{\max} \leq 2B_X^2 C_U$ and $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda$ hold with probability tending to one. The pairs that they exclude, all with $\pi < 2$, and the boundary $s = 1/2$ call for 2^{J_n} above that ceiling, where neither bound holds: the wavelets of the finest level take values up to $2^{(J-1)/2} \|\psi\|_\infty$ and have one or two sample points in their support, so that $\widehat{\sigma}^2_{\max}$ and $\lambda_{\max}(\widehat{\Sigma})$ grow with $2^J/n$, and the deterministic level (3.1) is no longer admissible. With $D = p + d$, $\xi = \Lambda - \kappa_2 C_U > 0$, C_ψ as in Proposition 2(ii) and $\|\psi\|_\infty$ as in Lemma S5.3, write

$$\mu_J^{\mathcal{G}} = 1 + c_\mu \frac{2^J \log(2D)}{n}, \qquad \lambda_n^{\mathcal{G},+} = (\mu_{J_n}^{\mathcal{G}})^{1/2} \lambda_n^{\mathcal{G}}, \tag{S6.5}$$

with

$$c_\psi = \frac{7\|\psi\|_\infty^2}{12 C_U}, \qquad c_\mu = \max\Big\{c_\psi,\ \frac{2pB_X^2(1+qC_\psi)}{\Lambda}\Big(\frac{\kappa_2 C_U}{2\xi} + \frac{4}{3}\Big)\Big\}.$$

When $pq2^{J_n}\log(pq2^{J_n})/n \to 0$, $\lambda_n^{\mathcal{G},+} = \lambda_n^{\mathcal{G}}\{1 + o(1)\}$.

Proposition S6.1 (Proposition 3 above the concentration ceiling)**.** *Let the assumptions of Proposition 3 hold, with the conditions* $2^{J_n} \le 2n$ *and* $pq2^{J_n}\log(pq2^{J_n})/n \to 0$ *replaced by* $2^{J_n} \le n^2$*, and with* $\lambda_n^{\mathcal{G},+}$ *of* (S6.5) *in place of* $\lambda_n^{\mathcal{G}}$ *in the estimator. Then*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{\lambda_n^{\mathcal{G},+}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} + \mathcal{B}_{J_n}^2 + p/n\},$$

where $\lambda_n^{\mathcal{G},+}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \le \bar{\varrho}\,(\mu_{J_n}^{\mathcal{G}})^{1/2}\lambda_{n,1}^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$*, with* $(\lambda_{n,1}^{\mathcal{G}})^2 \le (C_\lambda + 16\Lambda)\sigma^2 b_n/n$ *and* C_λ *as in* (S6.1)*, and:*

(iii) (the pairs that Proposition 3(i) and (ii) exclude, all with $\pi < 2$*) (a) if* $s > 1/2$ *and* $s' \le 1/4$*, the choice* $2^{J_n} \asymp (n/\log n)^{2/(1+4s')}$ *gives*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{(\log n/n)^{4s'/(1+4s')}\};$$

(b) if $s < 1/2$ *and* $s' \le s/2$*, the choice* $2^{J_n} \asymp (n/b_n^{1/\pi})^{1/(1-s+2s')}$ *gives*

$$\left\|\widehat{f} - f\right\|_n^2 = O_p\{n^{-2s'/(1-s+2s')}(\log n)^{(1/\pi)2s'/(1-s+2s')}\};$$

in both, $2^{J_n} \gtrsim n/\log n$*, and the powers of* n *join those of Proposition 3(i) and (ii) at* $s' = 1/4$ *and* $s' = s/2$*;*

(iv) (the boundary $s = 1/2$*, with* $\pi > 1$*)* $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \le pqC_g\{(1-2^{-s'})^{-1} + b_n^{-\vartheta_\pi}J_n\}$*, and: if* $s' \ge 1/4$*, that is* $\pi \ge 4/3$*, the choice* $2^{J_n} \asymp (n/b_n)^{1/(4s')}$ *gives* $\left\|\widehat{f} - f\right\|_n^2 = O_p\{n^{-1/2}(\log n)^{3/2-\vartheta_\pi}\}$*, which is* $n^{-1/2}\log n$ *when* $\pi \ge 2$*; if* $s' < 1/4$*, that is* $1 < \pi < 4/3$*, the choice* $2^{J_n} \asymp \{n/b_n^{1+1/\pi}\}^{2/(1+4s')}$ *gives* $\left\|\widehat{f} - f\right\|_n^2 = O_p[\{(\log n)^{1+1/\pi}/n\}^{4s'/(1+4s')}]$.

The proof uses no lower bound on $\widehat{\Sigma}$, and no property of $\widehat{\Sigma}$ other than the upper bound $\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\mu_{J_n}^{\mathcal{G}}$, which holds with probability tending to one under the upper bounds of Assumptions 2 and 3 alone, with no condition on J_n.

Stopping J_n at the ceiling instead, with $2^{J_n} = n/(K \log n)$ and K fixed and large enough, keeps $\lambda_n^{\mathcal{G}}$ admissible and gives $O_p\{(\log n/n)^{2s'}\}$ in the pairs of (iii); above the ceiling the estimation term grows more slowly than the bias decreases, and the exponents of (iii) are strictly larger than $2s'$ away from $s' = 1/4$ and $s' = s/2$ (Remark S8.2 in Section S8, for the lasso). Compared with Proposition S8.1(iii) and (iv) for the lasso, the chunks remove nothing in (iii)(a), where the coarse chunk dominates both norms, as in Proposition 3(i); the factor $(\log n)^{(1-1/\pi)2s'/(1-s+2s')}$ in (iii)(b); and, in (iv), $(\log n)^{\vartheta_\pi}$ when $s' \geq 1/4$, half a logarithm when $\pi \geq 2$, and $(\log n)^{(1-1/\pi)4s'/(1+4s')}$ when $s' < 1/4$. This is the mechanism of Proposition 3(ii): the logarithm comes from the size of the chunks, and the count of Lemma S6.2 returns it to the extent that the signal is spread over them.

Proof of Proposition S6.1

The two events above the ceiling. Let $\mathcal{A}_n^{++} = \{\widehat{\sigma}^2_{\max} \leq 2B_X^2 C_U \mu_J^{\mathcal{G}}\} \cap \{\lambda_{\max}(\widehat{\Sigma}) \leq \Lambda\mu_J^{\mathcal{G}}\}$. For the first, the Bernstein bound of Lemma S5.3 holds

for every J and every $t > 0$. With $x = 2\log d$ and $t = (2xR'_J B_X^2 C_U/n)^{1/2} + 2xR'_J/(3n)$, $nt^2 \geq 2xR'_J B_X^2 C_U + (2/3)xR'_J t$, so that the exponent is at most $-x$; and $(2ab)^{1/2} \leq a + b/2$ with $a = B_X^2 C_U$ and $b = xR'_J/n$ gives $t \leq B_X^2 C_U + (7/6)xR'_J/n$. Since $2B_X^2 C_U + (7/3)R'_J \log d/n = 2B_X^2 C_U(1 + c_\psi 2^J \log d/n)$,

$$\mathrm{P}\big\{\widehat{\sigma}^2_{\max} > 2B_X^2 C_U(1 + c_\psi 2^J \log d/n)\big\} \;\leq\; \frac{1}{d} \qquad \text{for every } J, \tag{S6.6}$$

and $c_\psi 2^J \log d \leq c_\mu 2^J \log(2D)$. For the second, the matrix Bernstein bound in the proof of Proposition 2(ii), $\mathrm{P}(\big\|\widehat{\Sigma} - \Sigma\big\|_{\mathrm{op}} \geq t) \leq 2D \exp\{-nt^2/(2R_J(\kappa_2 C_U + 2t/3))\}$, holds for every J and every $t > 0$. With $x = 2\log(2D)$ and $t = (2xR_J\kappa_2 C_U/n)^{1/2} + (4/3)xR_J/n$, $nt^2 \geq 2xR_J\kappa_2 C_U + (4/3)xR_J t$, and the probability is at most $2De^{-x} = 1/(2D)$; and $(2ab)^{1/2} \leq \xi + ab/(2\xi)$ with $a = \kappa_2 C_U$ and $b = xR_J/n$ gives $t \leq \xi + \{\kappa_2 C_U/(2\xi) + 4/3\}xR_J/n$. By Weyl's inequality and $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$, off that event $\lambda_{\max}(\widehat{\Sigma}) < \Lambda + \{\kappa_2 C_U/(2\xi) + 4/3\}2\log(2D)R_J/n \leq \Lambda\mu_J^{\mathcal{G}}$, because $R_J = pB_X^2(1 + qC_\psi 2^J) \leq pB_X^2(1 + qC_\psi)2^J$. Hence $\mathrm{P}(\mathcal{A}_n^{++}) \geq 1 - 1/d - 1/(2D) \to 1$, with no condition on J_n. Both bounds use only $|X_\ell| \leq B_X$, $p_{\mathbf{U}} \leq C_U$ and $\lambda_{\max}(\Sigma) \leq \kappa_2 C_U$.

The general bound. On $\mathcal{A}_n^{++}$ the two bounds in the proof of Lemma S5.2(ii), $\mathrm{tr}(\widetilde{\Psi}_G) \leq |G|\widehat{\sigma}^2_{\max}$ and $\big\|\widetilde{\Psi}_G\big\|_{\mathrm{op}} \leq \lambda_{\max}(\widehat{\Sigma})$, give $\lambda_{0,G} \leq (\mu_J^{\mathcal{G}})^{1/2}\bar{\lambda}_{0,G}$ for every G, so that $\lambda_n^{\mathcal{G},+} \geq \lambda_w$ and $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$, and the proof of Proposition 3 goes through with $\lambda_n^{\mathcal{G},+}$ in place of $\lambda_n^{\mathcal{G}}$ and $\mathcal{A}_n^{++}$ in place of $\mathcal{A}_n^{+}$; the chain gives

$\lambda_n^{\mathcal{G},+}\|\boldsymbol{\theta}^*\|_{\mathcal{G},w} \le \bar{\varrho}(\mu_{J_n}^{\mathcal{G}})^{1/2}\lambda_{n,1}^{\mathcal{G}}\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$. With $2^{J_n} \le n^2$, $|\mathcal{G}_n| \le pq(1+n^2/b_n) \le pqn^2$ and $\log(|\mathcal{G}_n|/\alpha) \le \log(2pq/\alpha)+2b_n$, and the computation in the proof of Lemma S6.1 gives $(\lambda_{n,1}^{\mathcal{G}})^2 \le (16\sigma^2/n)\{(2B_X^2C_U+2\Lambda)b_n+\Lambda\log(2pq/\alpha)\} \le (C_\lambda+16\Lambda)\sigma^2 b_n/n$. Since $\log(2D_n) \asymp \log n \asymp b_n$,

$$(\mu_{J_n}^{\mathcal{G}})^{1/2}\lambda_{n,1}^{\mathcal{G}} \;\asymp\; \Big(\frac{\log n}{n}\Big)^{1/2}\Big(1+\frac{2^{J_n}\log n}{n}\Big)^{1/2}, \tag{S6.7}$$

which is of order $2^{J_n/2}\log n/n$ when $2^{J_n} \gtrsim n/\log n$.

Part (iii)(a). With $s > 1/2$, $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O(1)$ as in the proof of Proposition 3(i). With $2^{J_n} \asymp (n/\log n)^{2/(1+4s')} \gtrsim n/\log n$, because $s' \le 1/4$, (S6.7) gives the estimation term $(\log n/n)(n/\log n)^{1/(1+4s')} = (\log n/n)^{4s'/(1+4s')}$, the order of the bias $2^{-2J_n s'}$; p/n is of smaller order.

Part (iii)(b). With $s < 1/2$ and $\pi < 2$, $\vartheta_\pi = 1-1/\pi$ and $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} = O\{1+b_n^{-\vartheta_\pi}2^{J_n(1/2-s)}\}$ as in the proof of Proposition 3(ii). With $2^{J_n} \asymp (n/b_n^{1/\pi})^{1/(1-s+2s')}$, which exceeds n/b_n for n large, because the exponent is at least 1 and $1/\pi \le 1$, and which is at most n^2, because $1-s+2s' > 1/2$, (S6.7) gives the estimation term

$$2^{J_n/2}\frac{\log n}{n}\big\{1+b_n^{-(1-1/\pi)}2^{J_n(1/2-s)}\big\} \;\asymp\; 2^{J_n/2}\frac{\log n}{n}+2^{J_n(1-s)}\frac{(\log n)^{1/\pi}}{n},$$

whose second term dominates and, at the choice of the statement, equals $(b_n^{1/\pi}/n)^{1-(1-s)/(1-s+2s')} = (b_n^{1/\pi}/n)^{2s'/(1-s+2s')}$, the order of the bias $2^{-2J_n s'}$.

Part (iv). With $s = 1/2$, the sum in (S6.3) is at most J_n, which gives the

bound on $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$ of the statement, and $J_n \asymp \log n \asymp b_n$ in both choices, so that $\|\boldsymbol{\theta}^*\|_{\mathcal{G},1} \asymp (\log n)^{1-\vartheta_\pi}$, since $\vartheta_\pi \leq 1/2$. If $s' \geq 1/4$, $2^{J_n} \asymp (n/b_n)^{1/(4s')} \lesssim n/\log n$, $\mu^{\mathcal{G}}_{J_n}$ is bounded, the estimation term is $(\log n/n)^{1/2}(\log n)^{1-\vartheta_\pi}$, and the bias, of order $(b_n/n)^{1/2}$, is of smaller order. If $s' < 1/4$, then $\pi < 4/3$ and $\vartheta_\pi = 1 - 1/\pi$; with $2^{J_n} \asymp (n/b_n^{1+1/\pi})^{2/(1+4s')}$, which exceeds $n/\log n$, the estimation term is $2^{J_n/2}(\log n)^{1+1/\pi}/n \asymp \{(\log n)^{1+1/\pi}/n\}^{4s'/(1+4s')}$, the order of the bias. □

Proof of Theorem 3

By the definition of J_n, $n^{1/(2s'+1)} \leq 2^{J_n} < 2n^{1/(2s'+1)} \leq 2n$, so the setting of this section holds, and $pq2^{J_n}\log(pq2^{J_n})/n \asymp n^{1/(2s'+1)-1}\log n = n^{-2s'/(2s'+1)}\log n \to 0$, which is the requirement of Proposition 2(ii); by Lemma S6.1, $\mathrm{P}(\mathcal{A}_n) \to 1$. Fix $\epsilon > 0$, take $\alpha = \alpha' = \alpha'' = \epsilon/4$, and intersect $\mathcal{A}_n$, the event $\mathcal{E}$ of Lemma S5.2(i), $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}^2_{J_n}/\alpha'\}$ (Markov's inequality, since $\mathrm{E}\,\|\mathbf{b}\|_n^2 = \|f - f_{J_n}\|^2_{L_2(P)} \leq \mathcal{B}^2_{J_n}$ by Lemma 2) and $\{\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \leq \sigma^2 p/(n\alpha'')\}$ (Markov's inequality and Theorem 2(iii)); for n large the intersection has probability at least $1-\epsilon$. On it $\mathcal{T}_{\mathcal{G},w}$ holds, and (S5.7) with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives

$$\left\|\widehat{f} - f\right\|_n^2 \leq \frac{192(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{G}_0(\boldsymbol{\theta}^*))}{\widetilde{\gamma}} + 60\,\|\mathbf{b}\|_n^2 + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2\,.$$

By Lemma S5.2(iii), Lemma S6.1 and Proposition S5.1(ii), with $\widetilde{\gamma} > \gamma/2$,

$$\frac{192(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\widetilde{\gamma}} \leq \frac{384\bar{\varrho}^2(\lambda_{n,1}^{\mathcal{G}})^2|\mathcal{G}_n|}{\gamma} \leq \frac{384\bar{\varrho}^2 C_\lambda \sigma^2 pq(b_n + 2^{J_n})}{\gamma n},$$

which is of order $n^{-2s'/(2s'+1)}$, because $2^{J_n} \asymp n^{1/(2s'+1)}$ dominates $b_n \asymp \log n$. The bias term is $60\mathcal{B}_{J_n}^2/\alpha' \asymp 2^{-2J_n s'} \leq n^{-2s'/(2s'+1)}$, by the lower bound on 2^{J_n}, and the last term is $O(1/n)$, of strictly smaller order. Hence $\mathrm{P}\{\left\|\widehat{f} - f\right\|_n^2 > M_\epsilon n^{-2s'/(2s'+1)}\} < \epsilon$ for a constant M_ϵ. □

Proof of Corollary 1

By the orthonormality of the wavelets of level below J_n in $L_2[0,1]$ and Lemma 2, $\sum_{\ell,m}\|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 \leq 2\left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2^2 + 2pqC_g^2(1-2^{-2s'})^{-1}2^{-2J_n s'}$. On the event of the proof of Theorem 3, Theorem 2(ii) with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives $\left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2^2 \leq 64(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{G}_n)/\widetilde{\gamma}^2 + 16\left\|\mathbf{b}\right\|_n^2/\widetilde{\gamma}$, with $\widetilde{\gamma} > \gamma/2$: the two terms have the order of those of Theorem 3, with γ^{-2} in place of γ^{-1}. For the levels, Lemma S5.1 with $\bar{\boldsymbol{\beta}} = \boldsymbol{\beta}^*$ gives $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$, so that $\left\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\right\|_n \leq \left\|\mathbf{b}\right\|_n + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n + \Lambda^{1/2}\left\|v\right\|_2$, and $\lambda_{\min}(\mathbf{A}^\top\mathbf{A}/n) > \gamma/2$ gives $\left\|\widehat{\mathbf{c}} - \mathbf{c}\right\|_2^2 \leq 2\left\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\right\|_n^2/\gamma$. Finally the components of a functional coefficient depend on distinct coordinates of u and the basis is orthonormal, so that $\|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} \leq |\widehat{c}_\ell - c_\ell| + \sum_m \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, with q fixed. □

Proof of Theorem 1

The window. Since $s' \le s$, $s/\{(2s+1)s'\} \ge 1/(2s+1)$, and the window is nonempty if and only if $s/\{(2s+1)s'\} < 1$, that is $s' > s/(2s+1)$. With $c < 1$, $2^{J_n} \le 2n^c \le 2n$, so the setting of this section holds; the requirement of Proposition 2(ii) is $pq2^{J_n}\log(pq2^{J_n})/n \asymp n^{c-1}\log n \to 0$, and $\mathrm{P}(\mathcal{A}_n) \to 1$ by Lemma S6.1.

The rate. Fix $\epsilon > 0$ and the event of the proof of Theorem 3. On it, Corollary S5.1 with $\Lambda' = \Lambda$ gives $\left\|\widehat{f} - f\right\|_n^2 \le 120\Lambda R_{\mathcal{G}}(\boldsymbol{\theta}^*;\eta) + 120\,\|\mathbf{b}\|_n^2 + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$, and, by Lemma S5.2(iii), (S6.1) and $\widetilde{\gamma} > \gamma/2$,

$$\eta \;=\; \frac{1.6\max_G(\lambda_n^{\mathcal{G}} w_G)^2}{\Lambda\widetilde{\gamma}} \;\le\; \frac{3.2\bar{\varrho}^2(\lambda_{n,1}^{\mathcal{G}})^2}{\Lambda\gamma} \;\le\; \bar{\eta}_n := \kappa\,\frac{b_n}{n}, \qquad \kappa = \frac{3.2\bar{\varrho}^2 C_\lambda \sigma^2}{\Lambda\gamma}.$$

Since $R_{\mathcal{G}}$ is nondecreasing in η, Lemma 3 with $b = b_n$ gives

$$R_{\mathcal{G}}(\boldsymbol{\theta}^*;\eta) \;\le\; R_{\mathcal{G}}(\boldsymbol{\theta}^*;\bar{\eta}_n) \;\le\; pq\Big\{A_{s,\pi}^{\mathcal{G}} C_g^{\tau}\kappa^{2s/(2s+1)}\, n^{-2s/(2s+1)} b_n^{(2/\pi-1)_+/(2s+1)} + \kappa\frac{b_n}{n}\Big\},$$

with $b_n \asymp \log n$; the second term is of smaller order because $2s/(2s+1) < 1$. The bias is $O(2^{-2J_n s'}) = O(n^{-2cs'})$, and $2cs' \ge 2s/(2s+1)$ by the window; the term $\sigma^2 p/n$ is of smaller order. For the components, the second bound of Corollary S5.1 and Lemma 2 give the same rate, as in the proof of Corollary 1.

Uniformity over the class. The events $\mathcal{A}_n$ and $\mathcal{E}$ and the event for $\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$ do not depend on f; the event for $\|\mathbf{b}\|_n^2$ does, but its probability is

at least $1-\alpha'$ for every f whose components satisfy Assumption 5, because the bound $\mathcal{B}_{J_n}^2$ on $\mathrm{E}\,\|\mathbf{b}\|_n^2$ of Lemma 2 depends on f only through C_g. On the event, the bound depends on f only through C_g, by Lemmas 2 and 3, and not on the levels $\mathbf{c}$, since $\mathbf{b} = f - f_{J_n}$ does not involve them. Hence $\sup_f \mathrm{P}_f\{\left\|\widehat{f} - f\right\|_n^2 > M_\epsilon r_n\} < \epsilon$ for n large, with P_f the law of the sample when the regression function is f and the supremum over the functions of Assumption 5 with any levels, and the same holds for the components: the bound of Theorem 1 is uniform over the class, which is the form that Theorem S6.1 below is compared with.

Two readings. The constant grows with the weights at most as $\bar{\varrho}^{4s/(2s+1)}$ in the leading term, through $\kappa^{2s/(2s+1)}$. The gain over the rate $(\log n/n)^{2s/(2s+1)}$ of the lasso (Theorem S8.3) is the factor $(\log n)^{2s'/(2s+1)}$, because $2s/(2s+1) - (2/\pi - 1)_+/(2s+1) = 2s'/(2s+1)$: when $\pi < 2$, $2s - 2/\pi + 1 = 2(s - 1/\pi + 1/2) = 2s'$, and when $\pi \geq 2$, $s' = s$. The logarithm of the union bound over the columns is absorbed by the size of the chunks, which is why they must have at least of the order of $\log|\mathcal{G}_n|$ coefficients. □

[removed in k = 4]

Remark (The lower bound of Klopp and Pensky with one modulator)**.** For $q = 1$, $\mathbf{X}$ independent of U_1 and Gaussian errors, Klopp and Pensky (2015, Theorem 1, eqs. 3.6 and 3.7), with the number of covariates fixed,

give the minimax lower bound $n^{-2r/(2r+1)}$ over their class (A3) with the periodized wavelet basis, which their condition (A1)(b) admits. That class, a weighted ℓ_ν ball $\sum_k |a_k|^\nu (k+1)^{\nu(r+1/2-1/\nu)} \le C_a^\nu$ over all the coefficients, implies Assumption 5 with $(s,\pi) = (r,\nu)$ and $C_g = C_a$, level by level, since $k+1 \ge 2^j$ at level j; so the lower bound holds over the class of Assumption 5, and when $\pi \ge 2$ Theorem 1 is minimax optimal up to a constant in that case. When $\pi < 2$ the factor $(\log n)^{(2/\pi-1)/(2s+1)}$ remains, the same as in eqs. (3.19) and (3.20) of Klopp and Pensky with the number of covariates fixed, and the same as in the rate of block thresholding in the sequence model (Cai, 1999, Theorem 4, eq. 5.4), which holds for $s \ge 1/\pi$; the condition $s' > s/(2s+1)$ of Theorem 1 is implied by $s \ge 1/\pi$, since then $s' \ge 1/2 > s/(2s+1)$, and is strictly weaker. That this factor is necessary for the estimator is proved neither here nor by Klopp and Pensky, and for $q \ge 2$ no lower bound is available.

A minimax lower bound

The rates of this section are upper bounds. For a lower bound, fix any law P of $(\mathbf{X}, \mathbf{U})$ that satisfies Assumptions 2 and 3, with $\mathbf{X}$ allowed to depend on $\mathbf{U}$, and let the errors be Gaussian,

$$\varepsilon_i \sim N(0, \sigma^2), \text{ independent of } (\mathbf{X}_i, \mathbf{U}_i), \tag{S6.8}$$

which satisfy Assumption 1. Write P_f and E_f for the law and the expectation of the sample when the regression function is f, and $\mathcal{F} = \mathcal{F}(s, \pi, C_g)$ for the set of the functions $f(x,u) = \sum_\ell x_\ell\{c_\ell + \sum_m g_{\ell m}(u_m)\}$ with $\mathbf{c} \in \mathbb{R}^p$ arbitrary and every component satisfying the centering and the bound of Assumption 5; by Proposition 1, f determines its levels and its components. An estimator is any measurable function of the sample: for prediction, $\widehat{f}$ with values in $L_2(P)$, or only its values $\widehat{f}(\mathbf{X}_i, \mathbf{U}_i)$, and for the components, $\widehat{g} = (\widehat{g}_{\ell m})$ with $\widehat{g}_{\ell m} \in L_2[0,1]$. For a nonempty set $\mathcal{K}$ of blocks (ℓ, m), the loss for the components in $\mathcal{K}$ is

$$L_{\mathcal{K}}(\widehat{g}, f) \;=\; \sum_{(\ell,m)\in\mathcal{K}} \|\widehat{g}_{\ell m} - g_{\ell m}\|^2_{L_2[0,1]}.$$

With $\gamma = \kappa_1 c_U$ as in Section S1 and $c_{\mathrm{A}} = (1 - 2^{-1/2})/2 \approx 0.146$, put, for $n \ge n_0 = \lceil 2\sigma^2/(C_g^2 B_X^2 C_U)\rceil$,

$$\begin{gathered}\zeta_n^2 = \frac{2\sigma^2}{nB_X^2 C_U}, \qquad j_n = \max\Big\{j \ge 0 : 2^{j(2s+1)} \le \frac{C_g^2}{\zeta_n^2}\Big\}, \\ \underline{\rho}_n^2 = C_g^{2/(2s+1)}\Big(\frac{\sigma^2}{nB_X^2 C_U}\Big)^{2s/(2s+1)};\end{gathered} \tag{S6.9}$$

j_n exists because $C_g^2/\zeta_n^2 \ge 1$ when $n \ge n_0$, and $\underline{\rho}_n^2$, of order $n^{-2s/(2s+1)}$, is the rate of the lower bound.

Theorem S6.1 (Minimax lower bound)**.** *Let P be any law of $(\mathbf{X}, \mathbf{U})$ that satisfies Assumptions 2 and 3, with $q \ge 1$ and $\mathbf{X}$ possibly dependent on $\mathbf{U}$, let the basis satisfy Assumption 4, and let the errors be Gaussian, as in*

(S6.8). *For every $s > 0$, every $1 \le \pi \le \infty$, every nonempty set $\mathcal{K}$ of blocks and every $n \ge n_0$, with $\underline{\rho}_n$ as in* (S6.9)*:*

(i) (components)

$$\inf_{\widehat{g}} \sup_{f \in \mathcal{F}} \mathrm{E}_f L_{\mathcal{K}}(\widehat{g}, f) \;\ge\; \frac{c_{\mathrm{A}}}{8} |\mathcal{K}| \, \underline{\rho}_n^2,$$

$$\inf_{\widehat{g}} \sup_{f \in \mathcal{F}} \mathrm{P}_f \Big\{ L_{\mathcal{K}}(\widehat{g}, f) \ge \frac{c_{\mathrm{A}}}{16} |\mathcal{K}| \, \underline{\rho}_n^2 \Big\} \;\ge\; \frac{c_{\mathrm{A}}}{2};$$

(ii) (prediction, population norm) the same holds for $\left\| \widehat{f} - f \right\|_{L_2(P)}^2$, with $\gamma |\mathcal{K}| \underline{\rho}_n^2$ in place of $|\mathcal{K}| \underline{\rho}_n^2$;

(iii) (prediction, empirical norm) if moreover $n \ge n_1$, the smallest integer from which on $2D \exp[-n\gamma^2/\{8R_J(\kappa_2 C_U + \gamma/3)\}] \le c_{\mathrm{A}}/4$, with $J = j_n + 1$ and $D = p + pqN_J$ and R_J as in Proposition 2(ii), then

$$\inf_{\widehat{f}} \sup_{f \in \mathcal{F}} \mathrm{E}_f \left\| \widehat{f} - f \right\|_n^2 \;\ge\; \frac{c_{\mathrm{A}}}{32} \gamma |\mathcal{K}| \, \underline{\rho}_n^2,$$

$$\inf_{\widehat{f}} \sup_{f \in \mathcal{F}} \mathrm{P}_f \Big\{ \left\| \widehat{f} - f \right\|_n^2 \ge \frac{c_{\mathrm{A}}}{32} \gamma |\mathcal{K}| \, \underline{\rho}_n^2 \Big\} \;\ge\; \frac{c_{\mathrm{A}}}{4}.$$

The constants depend neither on π, r, p or q, nor on P beyond (B_X, C_U) in $\underline{\rho}_n$ and $\gamma = \kappa_1 c_U$; n_0 depends only on (σ, C_g, B_X, C_U), and n_1, which is finite, also on $(p, q, s, \kappa_1, \kappa_2, c_U)$ and the wavelet family. In particular, with $\mathcal{K}$ a single block, every component and the prediction have minimax risk of order at least $n^{-2s/(2s+1)}$, and with $\mathcal{K}$ all the blocks, so that $|\mathcal{K}| = pq$, the bound is proportional to pq.

The bound is not asymptotic, and it holds for every pair (s, π), in particular for every pair that Assumption 5 allows. It has the form of the lower bound of Donoho and Johnstone (1998, Theorem 5) for Besov bodies in the Gaussian sequence model, with radius C_g and noise level $\sigma(nB_X^2 C_U)^{-1/2}$ per coordinate. Since Gaussian errors satisfy Assumption 1, it holds a fortiori over the models of Assumptions 1 to 5, and the form in probability is the one to compare with the upper bounds, which are stated in O_p and are uniform over the class (proof of Theorem 1 above). Hence, when $\pi \geq 2$, $n^{-2s/(2s+1)}$ is the minimax rate in the prediction norm and for the components, for every q and with covariates that depend on the modulators, and the estimator of Theorem 1 attains it uniformly over the class; when $\pi < 2$ and $s' > s/(2s+1)$ it attains it up to the factor $(\log n)^{(2/\pi-1)/(2s+1)}$, which the lower bound does not carry (the same factor as in the rate of block thresholding in the sequence model, Cai, 1999, Theorem 4, which holds for $s \geq 1/\pi$, a condition strictly stronger than $s' > s/(2s+1)$), and the coordinatewise lasso, at the rate $(\log n/n)^{2s/(2s+1)}$ of Theorem S8.3, stays above it by the factor $(\log n)^{2s/(2s+1)}$ for every π. When $q = 1$ and $\mathbf{X}$ is independent of U_1, part (i) with $\mathcal{K}$ all the blocks is, up to the constants, the nonparametric term of the lower bound of Klopp and Pensky (2015, Theorem 1, eqs. 3.6 and 3.7) with a fixed number of covariates.

The proof applies Assouad's lemma to a hypercube of wavelets at a single level. Fix $\mathcal{K}$, a level $j \geq 0$ and a height $\zeta > 0$; let

$$\mathcal{C} = \big\{(\ell, m, k) : (\ell, m) \in \mathcal{K},\ 0 \leq k < 2^j\big\}, \qquad |\mathcal{C}| = |\mathcal{K}|\, 2^j,$$

and $\Omega = \{0,1\}^{\mathcal{C}}$. For $\omega \in \Omega$ the components are $g^{\omega}_{\ell m} = \zeta \sum_k \omega_{\ell m,k} \psi_{jk}$ for $(\ell, m) \in \mathcal{K}$ and zero otherwise, the levels are zero, and

$$f_\omega(x,u) \;=\; \zeta\, z_{\mathcal{C}}(x,u)^{\top}\omega, \qquad z_{\mathcal{C}}(x,u) = \big(x_\ell \psi_{jk}(u_m)\big)_{(\ell,m,k)\in\mathcal{C}},$$

whose entries are columns of the design at resolution $J = j + 1$. Write $\Sigma_{\mathcal{C}\mathcal{C}}$ and $\widehat{\Sigma}_{\mathcal{C}\mathcal{C}}$ for the principal submatrices of Σ and $\widehat{\Sigma}$ at that resolution on the columns in $\mathcal{C}$, $\mathbf{Z}_{\mathcal{C}}$ for those columns of $\mathbf{Z}$, $d_H(\omega, \omega') = \sum_{(\ell,m,k)\in\mathcal{C}} \mathbf{1}\{\omega_{\ell m,k} \neq \omega'_{\ell m,k}\}$ for the Hamming distance, and P_ω and E_ω for P_{f_ω} and E_{f_ω}. The form of Assouad's lemma that we use is Tsybakov (2009, Theorem 2.12(iv)) with the Kullback–Leibler divergence bounded by one: if $\mathrm{KL}(\mathrm{P}_\omega, \mathrm{P}_{\omega'}) \leq 1$ whenever $d_H(\omega, \omega') = 1$, then

$$\inf_{\widehat{\omega}} \max_{\omega\in\Omega} \mathrm{E}_\omega d_H(\widehat{\omega}, \omega) \;\geq\; \frac{|\mathcal{C}|}{2} \max\Big(\frac{e^{-1}}{2},\, 1 - 2^{-1/2}\Big) \;=\; c_{\mathrm{A}}|\mathcal{C}|, \qquad \text{(S6.10)}$$

the infimum being over all the estimators with values in Ω.

Lemma S6.3 (A hypercube of wavelets)**.** *In the setting of Theorem S6.1, for every nonempty $\mathcal{K}$, every $j \geq 0$ and every $\zeta > 0$:*

(i) $f_\omega \in \mathcal{F}$ for every $\omega \in \Omega$ if and only if $\zeta 2^{j(s+1/2)} \leq C_g$, for every $s > 0$ and every $1 \leq \pi \leq \infty$;

(ii) for every pair, $\mathrm{KL}(\mathrm{P}_\omega, \mathrm{P}_{\omega'}) = \{n\zeta^2/(2\sigma^2)\}(\omega - \omega')^\top \Sigma_{\mathcal{C}\mathcal{C}}(\omega - \omega')$*, and, if* ω *and* ω' *differ only in the coordinate* (ℓ, m, k)*,*

$$\mathrm{KL}(\mathrm{P}_\omega, \mathrm{P}_{\omega'}) \;=\; \frac{n\zeta^2}{2\sigma^2}\,\mathrm{E}\big\{X_\ell^2 \psi_{jk}(U_m)^2\big\} \;\le\; \frac{n\zeta^2}{2\sigma^2}\,\min(B_X^2, \kappa_2)\, C_U;$$

(iii) $\lambda_{\min}(\Sigma_{\mathcal{C}\mathcal{C}}) \ge \gamma$*, so that*

$$\|f_\omega - f_{\omega'}\|_{L_2(P)}^2 \ge \gamma\zeta^2 d_H(\omega, \omega'),$$

$$\sum_{(\ell,m)\in\mathcal{K}} \|g_{\ell m}^{\omega} - g_{\ell m}^{\omega'}\|_{L_2[0,1]}^2 = \zeta^2 d_H(\omega, \omega');$$

with a single block (ℓ, m)*, the first statement uses only* $\mathrm{E}(X_\ell^2 \mid \mathbf{U}) \ge \kappa_1$ *and the lower bound* c_U *on the density of* U_m*;*

(iv) for each of the three losses and every estimator there is a measurable function $\widehat{\omega}$ *of the sample with values in* Ω *such that, for every* $\omega \in \Omega$*,*

(a) $L_{\mathcal{K}}(\widehat{g}, f_\omega) \ge (\zeta^2/4)\, d_H(\widehat{\omega}, \omega)$*,*

(b) $\left\|\widehat{f} - f_\omega\right\|_{L_2(P)}^2 \ge (\gamma\zeta^2/4)\, d_H(\widehat{\omega}, \omega)$*,*

(c) $\left\|\widehat{f} - f_\omega\right\|_{n}^2 \ge (\gamma\zeta^2/8)\, d_H(\widehat{\omega}, \omega)\, \mathbf{1}_{\mathcal{E}_\mathcal{C}}$*, where* $\mathcal{E}_\mathcal{C} = \{\lambda_{\min}(\widehat{\Sigma}_{\mathcal{C}\mathcal{C}}) \ge \gamma/2\}$ *depends only on the* $(\mathbf{X}_i, \mathbf{U}_i)$ *and has, with* $J = j + 1$ *and* D *and* R_J *as in Theorem S6.1(iii),*

$$\mathrm{P}(\mathcal{E}_\mathcal{C}^c) \;\le\; 2D \exp\Big\{-\frac{n\gamma^2}{8R_J(\kappa_2 C_U + \gamma/3)}\Big\}.$$

The Kullback–Leibler divergence uses only upper bounds on the design, and through the conditional second moment $\mathrm{E}(X_\ell^2 \mid \mathbf{U})$, which is why the

covariates may depend on the modulators at no cost. The separation of the components uses no property of the design; that of the prediction uses the smallest eigenvalue, and it is there that the cross terms between modulators and between covariates enter, through Proposition 2(i).

Proof of Lemma S6.3

Part (i). By the orthonormality of $\{\mathbf{1}\} \cup \{\psi_{jk}\}$ in $L_2[0,1]$ (Section S2), the coefficients of $g^{\omega}_{\ell m}$ are $\theta_{\ell m,j'k} = \zeta\omega_{\ell m,k}\mathbf{1}\{j' = j\}$, and $\int_0^1 g^{\omega}_{\ell m} = 0$ because $\int_0^1 \psi_{jk} = 0$. Only the level j is nonzero, so the bound of Assumption 5 reduces to that level, whatever the index r. For $\pi < \infty$, $\|\theta_{\ell m,j\cdot}\|_\pi = \zeta(\sum_k \omega_{\ell m,k})^{1/\pi} \leq \zeta 2^{j/\pi}$, with equality at $\omega \equiv 1$; for $\pi = \infty$ it is ζ if $\omega_{\ell m,\cdot} \neq 0$. In both cases the maximum over Ω of $2^{j(s+1/2-1/\pi)}\|\theta_{\ell m,j\cdot}\|_\pi$ is $\zeta 2^{j(s+1/2)}$. The components outside $\mathcal{K}$ and the levels are zero.

Part (ii). Under P_ω the pairs $(\mathbf{X}_i, \mathbf{U}_i)$ have the law P, the same for every ω, and, given them, the Y_i are independent with $Y_i \sim N\{f_\omega(\mathbf{X}_i, \mathbf{U}_i), \sigma^2\}$. The divergence between product laws is the sum of the divergences, and that between two laws with the same marginal is the mean of the divergence between the conditionals, so that $\mathrm{KL}(\mathrm{P}_\omega, \mathrm{P}_{\omega'}) = n\mathrm{E}\{(f_\omega - f_{\omega'})^2(\mathbf{X}, \mathbf{U})\}/(2\sigma^2)$, and $f_\omega - f_{\omega'} = \zeta z_{\mathcal{C}}^\top(\omega - \omega')$ gives the quadratic form. For a neighbor, $\mathrm{E}\{(f_\omega - f_{\omega'})^2\} = \zeta^2\mathrm{E}\{\mathrm{E}(X_\ell^2 \mid \mathbf{U})\psi_{jk}(U_m)^2\}$, with $\mathrm{E}(X_\ell^2 \mid \mathbf{U}) \leq B_X^2$ and

$\mathrm{E}(X_\ell^2 \mid \mathbf{U}) = e_\ell^\top \mathrm{E}(\mathbf{X}\mathbf{X}^\top \mid \mathbf{U}) e_\ell \le \kappa_2$ by Assumption 2, and $\mathrm{E}\{\psi_{jk}(U_m)^2\} \le C_U \int_0^1 \psi_{jk}^2 = C_U$, because the density of U_m inherits the bound C_U of Assumption 3.

Part (iii). For $v \in \mathbb{R}^{\mathcal{C}}$ and $\tilde{v}$ its completion with zeros, $v^\top \Sigma_{\mathcal{C}\mathcal{C}} v = \tilde{v}^\top \Sigma \tilde{v} \ge \lambda_{\min}(\Sigma) \|v\|_2^2 \ge \gamma \|v\|_2^2$, by Proposition 2(i). Since $\omega - \omega' \in \{-1, 0, 1\}^{\mathcal{C}}$, $\|\omega - \omega'\|_2^2 = d_H(\omega, \omega')$, and $\|f_\omega - f_{\omega'}\|_{L_2(P)}^2 = \zeta^2 (\omega - \omega')^\top \Sigma_{\mathcal{C}\mathcal{C}} (\omega - \omega') \ge \gamma \zeta^2 d_H(\omega, \omega')$. The identity for the components is Parseval's. With a single block,

$$
\begin{aligned}
v^\top \Sigma_{\mathcal{C}\mathcal{C}} v &= \mathrm{E}\Big[\mathrm{E}(X_\ell^2 \mid \mathbf{U}) \Big\{ \sum_k v_k \psi_{jk}(U_m) \Big\}^2 \Big] \\
&\ge \kappa_1 c_U \int_0^1 \Big(\sum_k v_k \psi_{jk} \Big)^2 = \gamma \|v\|_2^2 .
\end{aligned}
$$

Part (iv). Let $\varpi(y) = \mathbf{1}\{y \ge \zeta/2\}$ for real y. For $\omega_{\ell m,k} \in \{0, 1\}$,

$$
|y - \zeta \omega_{\ell m,k}| \;\ge\; \frac{\zeta}{2} |\varpi(y) - \omega_{\ell m,k}|, \tag{S6.11}
$$

because if $\varpi(y) \ne \omega_{\ell m,k}$ then y and $\zeta \omega_{\ell m,k}$ lie on opposite sides of $\zeta/2$. Applied coordinatewise to a vector $a \in \mathbb{R}^{\mathcal{C}}$, it gives

$$
\|a - \zeta \omega\|_2^2 \;\ge\; \frac{\zeta^2}{4} d_H(\varpi(a), \omega).
$$

For (a), take $a_{\ell m,k} = \langle \widehat{g}_{\ell m}, \psi_{jk} \rangle$ and $\widehat{\omega} = \varpi(a)$: by Bessel's inequality, $\|\widehat{g}_{\ell m} - g_{\ell m}^{\omega}\|_{L_2[0,1]}^2 \ge \sum_k (a_{\ell m,k} - \zeta \omega_{\ell m,k})^2$ for each $(\ell, m) \in \mathcal{K}$, and it remains to sum over $\mathcal{K}$. For (b), let $\Pi_{\mathcal{C}}$ be the orthogonal projection of $L_2(P)$ onto the span

of the entries of $z_{\mathcal{C}}$, of dimension $|\mathcal{C}|$ because $\Sigma_{\mathcal{C}\mathcal{C}}$ is positive definite by (iii): $\Pi_{\mathcal{C}}\widehat{f} = z_{\mathcal{C}}^{\top}a$ with $a = \Sigma_{\mathcal{C}\mathcal{C}}^{-1}\mathrm{E}\{z_{\mathcal{C}}(\mathbf{X},\mathbf{U})\widehat{f}(\mathbf{X},\mathbf{U})\}$, the expectation being over a new $(\mathbf{X},\mathbf{U})$ with $\widehat{f}$ fixed, so that a is a measurable function of the sample, P being fixed. Since f_ω lies in the span, Pythagoras' theorem gives $\left\|\widehat{f} - f_\omega\right\|_{L_2(P)}^2 \geq \left\|\Pi_{\mathcal{C}}\widehat{f} - f_\omega\right\|_{L_2(P)}^2 = (a-\zeta\omega)^{\top}\Sigma_{\mathcal{C}\mathcal{C}}(a-\zeta\omega) \geq \gamma\left\|a-\zeta\omega\right\|_2^2$; take $\widehat{\omega} = \varpi(a)$. For (c), write $\widehat{\mathbf{F}} = (\widehat{f}(\mathbf{X}_i,\mathbf{U}_i))_{i\leq n}$. On $\mathcal{E}_{\mathcal{C}}$, $\mathbf{Z}_{\mathcal{C}}$ has rank $|\mathcal{C}|$; take $a = (\mathbf{Z}_{\mathcal{C}}^{\top}\mathbf{Z}_{\mathcal{C}})^{-1}\mathbf{Z}_{\mathcal{C}}^{\top}\widehat{\mathbf{F}}$ and $\widehat{\omega} = \varpi(a)$ there, and $\widehat{\omega} = 0$ off it. On $\mathcal{E}_{\mathcal{C}}$, since $\zeta\mathbf{Z}_{\mathcal{C}}\omega$ is the vector of the values of f_ω and lies in the column space of $\mathbf{Z}_{\mathcal{C}}$, $n\left\|\widehat{f} - f_\omega\right\|_n^2 = \left\|\widehat{\mathbf{F}} - \zeta\mathbf{Z}_{\mathcal{C}}\omega\right\|_2^2 \geq \left\|\mathbf{Z}_{\mathcal{C}}(a-\zeta\omega)\right\|_2^2 = n(a-\zeta\omega)^{\top}\widehat{\Sigma}_{\mathcal{C}\mathcal{C}}(a-\zeta\omega) \geq n(\gamma/2)\left\|a-\zeta\omega\right\|_2^2$; off it the right side of (c) is zero. Finally $\widehat{\Sigma}_{\mathcal{C}\mathcal{C}}$ is a principal submatrix of $\widehat{\Sigma}$ at resolution $J = j+1$, so that $\lambda_{\min}(\widehat{\Sigma}_{\mathcal{C}\mathcal{C}}) \geq \lambda_{\min}(\widehat{\Sigma})$, and $\mathcal{E}_{\mathcal{C}}$ contains the event of Proposition 2(ii), which gives the bound on its complement. □

Proof of Theorem S6.1

The hypercube. Take $\zeta = \zeta_n$ and $j = j_n$ of (S6.9). Since $2^{j_n(2s+1)} \leq C_g^2/\zeta_n^2$, $\zeta_n 2^{j_n(s+1/2)} \leq C_g$, and Lemma S6.3(i) gives $f_\omega \in \mathcal{F}$ for every ω. By Lemma S6.3(ii), the divergence between neighbors is at most one, because $n\zeta_n^2 B_X^2 C_U = 2\sigma^2$, and (S6.10) gives, for every $\widehat{\omega}$ with values in Ω,

$$\max_{\omega\in\Omega}\mathrm{E}_\omega d_H(\widehat{\omega},\omega) \;\geq\; c_{\mathrm{A}}|\mathcal{C}|, \qquad |\mathcal{C}| = |\mathcal{K}|\,2^{j_n}. \tag{S6.12}$$

The scale. By the maximality of j_n, $2^{(j_n+1)(2s+1)} > C_g^2/\zeta_n^2$, so that $2^{j_n} > (C_g/\zeta_n)^{2/(2s+1)}/2$ and

$$2^{j_n}\zeta_n^2 \;>\; \frac{1}{2}\,C_g^{2/(2s+1)}(\zeta_n^2)^{2s/(2s+1)} = \frac{1}{2}\,C_g^{2/(2s+1)}\Big(\frac{2\sigma^2}{nB_X^2C_U}\Big)^{2s/(2s+1)}$$
$$= 2^{-1/(2s+1)}\underline{\rho}_n^2 \;\geq\; \frac{1}{2}\,\underline{\rho}_n^2,$$

that is, $\zeta_n^2|\mathcal{C}| \geq |\mathcal{K}|\underline{\rho}_n^2/2$.

Expectation. The supremum over $\mathcal{F}$ is at least the maximum over the hypercube. With the $\widehat{\omega}$ of Lemma S6.3(iv)(a) and (S6.12),

$$\max_{\omega} \mathrm{E}_\omega L_{\mathcal{K}}(\widehat{g}, f_\omega) \;\geq\; \frac{\zeta_n^2}{4}\,c_{\mathrm{A}}|\mathcal{C}| \;\geq\; \frac{c_{\mathrm{A}}}{8}\,|\mathcal{K}|\,\underline{\rho}_n^2;$$

with that of (iv)(b), the same with the factor γ. For (iii), since $d_H \leq |\mathcal{C}|$ and the law of the $(\mathbf{X}_i, \mathbf{U}_i)$ does not depend on ω, $\mathrm{E}_\omega\{d_H(\widehat{\omega},\omega)\mathbf{1}_{\mathcal{E}_\mathcal{C}}\} \geq \mathrm{E}_\omega d_H(\widehat{\omega},\omega) - |\mathcal{C}|\mathrm{P}(\mathcal{E}_\mathcal{C}^c)$, and for $n \geq n_1$

$$\max_{\omega} \mathrm{E}_\omega \Big\|\widehat{f} - f_\omega\Big\|_n^2 \;\geq\; \frac{\gamma\zeta_n^2}{8}\Big(c_{\mathrm{A}}|\mathcal{C}| - \frac{c_{\mathrm{A}}}{4}|\mathcal{C}|\Big) = \frac{3c_{\mathrm{A}}}{32}\,\gamma\zeta_n^2|\mathcal{C}| \;\geq\; \frac{c_{\mathrm{A}}}{32}\,\gamma|\mathcal{K}|\,\underline{\rho}_n^2.$$

Probability. Let ω^* attain the maximum in (S6.12), which exists because Ω is finite. Since $0 \leq d_H \leq |\mathcal{C}|$, $c_{\mathrm{A}}|\mathcal{C}| \leq \mathrm{E}_{\omega^*} d_H \leq c_{\mathrm{A}}|\mathcal{C}|/2 + |\mathcal{C}|\mathrm{P}_{\omega^*}(d_H \geq c_{\mathrm{A}}|\mathcal{C}|/2)$, so that $\mathrm{P}_{\omega^*}(d_H \geq c_{\mathrm{A}}|\mathcal{C}|/2) \geq c_{\mathrm{A}}/2$. On that event, (iv)(a) gives $L_{\mathcal{K}} \geq (\zeta_n^2/4)c_{\mathrm{A}}|\mathcal{C}|/2 \geq (c_{\mathrm{A}}/16)|\mathcal{K}|\underline{\rho}_n^2$, and (iv)(b) the same with the factor γ; on its intersection with $\mathcal{E}_\mathcal{C}$, of probability at least $c_{\mathrm{A}}/2 - c_{\mathrm{A}}/4$ when $n \geq n_1$, (iv)(c) gives $\Big\|\widehat{f} - f_{\omega^*}\Big\|_n^2 \geq (\gamma\zeta_n^2/8)c_{\mathrm{A}}|\mathcal{C}|/2 \geq (c_{\mathrm{A}}/32)\gamma|\mathcal{K}|\underline{\rho}_n^2$. The estimator being arbitrary, the infima follow.

The integer n_1 is finite. Since $2^{j_n+1} \le 2(C_g/\zeta_n)^{2/(2s+1)}$, of order $n^{1/(2s+1)}$, D and R_J with $J = j_n + 1$ are of order $n^{1/(2s+1)}$, and the bound of Lemma S6.3(iv)(c) is of order $n^{1/(2s+1)} \exp(-cn^{2s/(2s+1)})$ for a constant $c > 0$, which tends to zero. The two particular cases are $\mathcal{K}$ a single block and $\mathcal{K}$ all the blocks. □

The slow rate through a truncated comparator

Proposition 3 reads Theorem 2(i) at the comparator $\boldsymbol{\beta}^*$ only, and pays $\lambda w_G \|\boldsymbol{\theta}^*_G\|_2$ for every chunk, including the chunks whose energy would cost less as bias. Theorem 2 holds for every comparator on the same event, and the comparator truncated chunk by chunk,

$$\bar{\boldsymbol{\theta}}_G \;=\; \boldsymbol{\theta}^*_G \, \mathbf{1}\big\{\|\boldsymbol{\theta}^*_G\|_2 > \lambda w_G\big\}, \qquad G \in \mathcal{G}, \tag{S6.13}$$

charges the penalty only for the chunks above their own level and sends the others to the bias. What remains is

$$\mathcal{R}_{\mathcal{G},w}(\boldsymbol{\theta};\lambda) = \sum_{G\in\mathcal{G}} \min\big(\lambda w_G \|\boldsymbol{\theta}_G\|_2 \,,\, \|\boldsymbol{\theta}_G\|_2^2\big), \qquad \mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta};t) = \sum_{G\in\mathcal{G}} \min\big(t \|\boldsymbol{\theta}_G\|_2 \,,\, \|\boldsymbol{\theta}_G\|_2^2\big), \tag{S6.14}$$

the counterparts in the group norm of the ideal risk $R_{\mathcal{G}}$, with $\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta};t) \le t\|\boldsymbol{\theta}\|_{\mathcal{G},1}$.

Proposition S6.2 (Slow rate through the truncated comparator)**.** *Let the*

assumptions of Proposition 3 hold; in the statements with $\lambda_n^{\mathcal{G},+}$ in the estimator, the conditions on J_n are replaced by $2^{J_n} \le n^2$, as in Proposition S6.1.

(i) On the event $\mathcal{T}_{\mathcal{G},w}$, with $\bar{\boldsymbol{\theta}}$ of (S6.13),

$$\left\|\widehat{f} - f\right\|_n^2 \;\le\; 18\lambda\|\bar{\boldsymbol{\theta}}\|_{\mathcal{G},w} + 48\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2 + 48\left\|\mathbf{b}\right\|_n^2 + 3\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2.$$

If λ and w are deterministic, $\mathrm{E}\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2 \le \kappa_2 C_U \left\|\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}\right\|_2^2$,

$$18\lambda\|\bar{\boldsymbol{\theta}}\|_{\mathcal{G},w} + 48\kappa_2 C_U \left\|\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}\right\|_2^2 \;\le\; \max(18, 48\kappa_2 C_U)\,\mathcal{R}_{\mathcal{G},w}(\boldsymbol{\theta}^*;\lambda),$$

and $\mathcal{R}_{\mathcal{G},w}(\boldsymbol{\theta}^*;\lambda_n^{\mathcal{G}}) \le \bar{\varrho}\,\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;\lambda_{n,1}^{\mathcal{G}})$. Consequently, with deterministic weights,

$$\left\|\widehat{f} - f\right\|_n^2 \;=\; O_p\big\{\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;\lambda_{n,1}^{\mathcal{G}}) + \mathcal{B}_{J_n}^2 + p/n\big\},$$

and, with $\lambda_n^{\mathcal{G},+}$ in the estimator, the same holds with $(\mu_{J_n}^{\mathcal{G}})^{1/2}\lambda_{n,1}^{\mathcal{G}}$ in place of $\lambda_{n,1}^{\mathcal{G}}$, for every sequence J_n.

(ii) For every J, every $b \ge 2$ and every $t > 0$, with $\tau = (s+1/2)^{-1}$: if $s < 1/2$,

$$\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;t) \;\le\; pq\Big\{\frac{C_g\,t}{1-2^{-s'}} + C_g^{\tau}\,t^{4s/(2s+1)}\,b^{\{(2/\pi-1)_+-2s\}/(2s+1)}\Big(\frac{1}{1-2^{s-1/2}} + \frac{1}{1-2^{-\min(\pi,2)s'}}\Big)\Big\}; \tag{S6.15}$$

if $s \ge 1/2$, nothing is gained in order: $\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;t) \le t\|\boldsymbol{\theta}^*\|_{\mathcal{G},1}$, and for $t \le C_g$ the class of Assumption 5 contains a $\boldsymbol{\theta}^*$ with $\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;t) \ge tC_g$ if $s > 1/2$, and, if $s = 1/2$, one in which every fine level $j < J$ with

$2^j \le C_g b^{1-\vartheta_\pi}/t$ contributes at least $tC_g b^{-\vartheta_\pi}/2$ to $\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*; t)$, the order of Lemma S6.2.

(iii) If $s < 1/2$ and $s' > s/(2s+1)$, then with $J_n = \lceil c \log_2 n \rceil$ and c in the window (3.2) of Theorem 1,

$$\left\| \widehat{f} - f \right\|_n^2 \;=\; O_p\big\{ n^{-2s/(2s+1)} (\log n)^{(2/\pi - 1)_+/(2s+1)} \big\},$$

the prediction rate of Theorem 1.

(iv) If $s < 1/2$ and $s' \le s/(2s+1)$, which forces $\pi < 2$, then with $\lambda_n^{\mathcal{G},+}$ in the estimator and the choice $2^{J_n} \asymp \{ n^{2s} b_n^{-(2s+2/\pi-1)/2} \}^{1/\{s+s'(2s+1)\}}$, which exceeds n/b_n,

$$\left\| \widehat{f} - f \right\|_n^2 \;=\; O_p\big\{ n^{-e} (\log n)^{e(2s+2/\pi-1)/(4s)} \big\}, \qquad e = \frac{4ss'}{s + s'(2s+1)};$$

at $s' = s/(2s+1)$ this is $n^{-2s/(2s+1)} (\log n)^{(2s+2/\pi-1)/\{2(2s+1)\}}$.

Proposition S6.2 is a statement on prediction. Like Proposition 3, it uses no lower bound on $\widehat{\Sigma}$: besides the upper bound on $\lambda_{\max}(\widehat{\Sigma})$ that makes the level admissible, it uses only $\lambda_{\max}(\Sigma) \le \kappa_2 C_U$, in the new bias, and the components still need the design condition (Corollary 1). When $s < 1/2$ and $\pi < 2$ it improves on Proposition 3(ii) and on Proposition S6.1(iii)(b) in every pair: (iii) gives the rate of Theorem 1 in the window of that theorem, and (iv) replaces Proposition S6.1(iii)(b) and the part of Proposition 3(ii)

with $s/2 < s' \le s/(2s+1)$. When $\pi \ge 2$ the rate is that of Proposition 3(ii), $n^{-2s/(2s+1)}$, now for every $c \in [1/(2s+1), 1)$. When $s \ge 1/2$ nothing changes: a coarse chunk of norm C_g, which the class allows, costs λC_g in the penalty or C_g^2 in the bias, and the slow route does not go below $\lambda_{n,1}^{\mathcal{G}} \asymp (\log n/n)^{1/2}$; Proposition 3(i) and Proposition S6.1(iii)(a) and (iv) remain the statements with no design condition in those pairs. Against the lasso read at its own truncated comparator (Proposition S8.2), the chunks remove the factor $(\log n)^{2s'/(2s+1)}$ in (iii), the gain of Theorem 1 over Theorem S8.3, and $(\log n)^{es'/(2s)}$ in (iv). The proposition is stated for the estimator (2.5) and not for the one that the software computes (Remark S5.2); for the latter the comparator would truncate at $\lambda w_G \Lambda^{1/2}$ on the event $q_{\max} \le \Lambda$, and we do not pursue it.

Proof of Proposition S6.2

Part (i). Theorem 2(i) and (iii) with $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \bar{\boldsymbol{\theta}})$ give, as in the proof of (S6.4), $\left\|\widehat{f} - f\right\|_n^2 \le 18\lambda\|\bar{\boldsymbol{\theta}}\|_{\mathcal{G},w} + 24\left\|\bar{\mathbf{b}}\right\|_n^2 + 3\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2$, and $\bar{\mathbf{b}} = f - \mathbf{A}\mathbf{c} - \mathbf{B}\bar{\boldsymbol{\theta}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})$ gives $\left\|\bar{\mathbf{b}}\right\|_n^2 \le 2\left\|\mathbf{b}\right\|_n^2 + 2\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2$. If λ and w are deterministic, so is $u = \boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}$, and $\mathrm{E}\left\|\mathbf{B}u\right\|_n^2 = u^\top \mathrm{E}(\mathbf{B}^\top\mathbf{B}/n)u \le \lambda_{\max}(\Sigma)\left\|u\right\|_2^2 \le \kappa_2 C_U \left\|u\right\|_2^2$, because $\mathrm{E}(\mathbf{B}^\top\mathbf{B}/n)$ is a principal submatrix of Σ, by the upper half of Proposition 2(i). On the chunks with $\|\boldsymbol{\theta}_G^*\|_2 > \lambda w_G$ the minimum

in (S6.14) is $\lambda w_G \|\boldsymbol{\theta}^*_G\|_2$, and on the others it is $\|\boldsymbol{\theta}^*_G\|_2^2$; summing gives the bound by $\max(18, 48\kappa_2 C_U)\mathcal{R}_{\mathcal{G},w}(\boldsymbol{\theta}^*;\lambda)$. By Lemma S5.2(iii), $\lambda_n^{\mathcal{G}} w_G \le \varrho\lambda_{n,1}^{\mathcal{G}}$, and $\min(\varrho x, y) \le \varrho \min(x,y)$ for $\varrho \ge 1$ gives $\mathcal{R}_{\mathcal{G},w}(\boldsymbol{\theta}^*;\lambda_n^{\mathcal{G}}) \le \varrho\, \mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*;\lambda_{n,1}^{\mathcal{G}})$; with $\lambda_n^{\mathcal{G},+}$, the same with $(\mu_J^{\mathcal{G}})^{1/2}\lambda_{n,1}^{\mathcal{G}}$. The form in O_p is obtained as in the proof of Proposition 3, on the intersection of $\mathcal{A}_n^+$ below the ceiling, or of $\mathcal{A}_n^{++}$ of the proof of Proposition S6.1 for every J_n, with $\mathcal{E}$ and with the events given by Markov's inequality for $\|\mathbf{b}\|_n^2$, $\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2$ and $\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$.

Part (ii), $s < 1/2$. Fix a block. The coarse chunk contributes at most $t\|\theta_{\text{coarse}}\|_2 \le tC_g(1-2^{-s'})^{-1}$, by the proof of Lemma S6.2. At a fine level j, with $\mathcal{G}_j$ its chunks and $v_j = \sum_{G\in\mathcal{G}_j} \min(t\|\theta_G\|_2, \|\theta_G\|_2^2)$, there are two bounds. The first is in the group norm: $v_j \le t\sum_G \|\theta_G\|_2 \le tC_g b^{-\vartheta_\pi} 2^{j(1/2-s)} =: A_j$, by the proof of Lemma S6.2. The second is the tail. If $\pi \le 2$, $\min(ty, y^2) \le t^{2-\pi}y^\pi$ for $y \ge 0$ (if $y \le t$, $y^2 = y^\pi y^{2-\pi} \le y^\pi t^{2-\pi}$; if $y > t$, $ty = t^{2-\pi}t^{\pi-1}y \le t^{2-\pi}y^\pi$, because $\pi \ge 1$), and $\|\theta_G\|_2 \le \|\theta_G\|_\pi$ gives $v_j \le t^{2-\pi}\sum_G \|\theta_G\|_\pi^\pi = t^{2-\pi}\|\theta_j\|_\pi^\pi \le t^{2-\pi}C_g^\pi 2^{-j\pi s'}$, since $\pi(s+1/2-1/\pi) = \pi s'$; if $\pi \ge 2$, $v_j \le \|\theta_j\|_2^2 \le C_g^2 2^{-2js}$. Call this second bound B_j. With

$$2^x = \left\{(C_g/t)\, b^{1-\vartheta_\pi}\right\}^\tau, \qquad h = C_g^\tau\, t^{2-\tau}\, b^{\tau(1-\vartheta_\pi)-1},$$

one has $A_j = h2^{(j-x)(1/2-s)}$ and $B_j = h2^{-(j-x)\min(\pi,2)s'}$. Indeed, at $j = x$, $A_x = C_g^\tau t^{2-\tau} b^{-\vartheta_\pi + (1-\vartheta_\pi)(\tau-1)}$, because $\tau(1/2-s) = \tau - 1$, and the exponent of b is $\tau(1-\vartheta_\pi) - 1$; if $\pi \le 2$, $1 - \vartheta_\pi = 1/\pi$ and $B_x =$

$t^{2-\pi}C_g^{\pi}\{(C_g/t)b^{1/\pi}\}^{-(\pi-\tau)} = C_g^{\tau}t^{2-\tau}b^{\tau/\pi-1}$, because $\tau\pi s' = \pi - \tau$; if $\pi \geq 2$, $1 - \vartheta_\pi = 1/2$ and $B_x = C_g^2\{(C_g/t)b^{1/2}\}^{-(2-\tau)} = C_g^{\tau}t^{2-\tau}b^{\tau/2-1}$, because $2s\tau = 2 - \tau$. For every π, $\tau(1-\vartheta_\pi) - 1 = \{(2/\pi - 1)_+ - 2s\}/(2s+1)$ and $2-\tau = 4s/(2s+1)$. Summing A_j over the fine levels with $j \leq x$ and B_j over those with $j > x$, two geometric series of ratios $2^{-(1/2-s)}$ and $2^{-\min(\pi,2)s'}$ that start from at most h, gives at most $h\{(1-2^{s-1/2})^{-1} + (1-2^{-\min(\pi,2)s'})^{-1}\}$ per block. Truncating at $j < J$ only removes terms, and summing over the pq blocks gives (S6.15).

Part (ii), $s \geq 1/2$. The first statement is $\min(ty, y^2) \leq ty$. If $s > 1/2$, the block with $\theta^*_{\ell m,00} = C_g$ and all its other coefficients zero satisfies Assumption 5, and its coarse chunk, of norm C_g, contributes $\min(tC_g, C_g^2) = tC_g$. If $s = 1/2$, take a fine level j with $2^j \leq C_g b^{1-\vartheta_\pi}/t$, and let k_j be the number of its chunks, so that $2^j/(2b) \leq k_j \leq 2^j/b$. If $\pi \leq 2$, put a spike of height $y_j = C_g 2^{-j(1-1/\pi)}k_j^{-1/\pi}$ in each chunk, which saturates Assumption 5 at that level; since $k_j \leq 2^j/b$, $y_j \geq C_g 2^{-j}b^{1/\pi} \geq t$, and the level contributes $k_j t y_j = tC_g 2^{-j(1-1/\pi)}k_j^{1-1/\pi} \geq tC_g(2b)^{-(1-1/\pi)}$. If $\pi \geq 2$, set all the coefficients of the level equal to $C_g 2^{-j}$, which again saturates Assumption 5: each chunk has norm $|G|^{1/2}C_g 2^{-j} \geq b^{1/2}C_g 2^{-j} \geq t$, and the level contributes $tC_g 2^{-j}\sum_G |G|^{1/2} \geq tC_g 2^{-j}k_j b^{1/2} \geq tC_g b^{-1/2}/2$. In both cases the contribution is at least $tC_g b^{-\vartheta_\pi}/2$.

Part (iii). With $\lambda = \lambda_n^{\mathcal{G}}$ and $(\lambda_{n,1}^{\mathcal{G}})^2 \le C_\lambda \sigma^2 b_n/n$ by (S6.1), (S6.15) with $b = b_n$ gives $\mathcal{R}_{\mathcal{G},1}(\boldsymbol{\theta}^*; \lambda_{n,1}^{\mathcal{G}}) \lesssim (b_n/n)^{1/2} + (b_n/n)^{2s/(2s+1)} b_n^{\{(2/\pi-1)_+ - 2s\}/(2s+1)} = (b_n/n)^{1/2} + n^{-2s/(2s+1)} b_n^{(2/\pi-1)_+/(2s+1)}$, and the first term is of smaller order because $2s/(2s+1) < 1/2$. The bias and the conditions on J_n are those of the proof of Theorem 1: $c \ge s/\{(2s+1)s'\}$ gives $\mathcal{B}_{J_n}^2 \lesssim n^{-2s/(2s+1)}$, and $c < 1$ gives $2^{J_n} \le 2n$ and $pq 2^{J_n} \log(pq 2^{J_n})/n \to 0$.

Part (iv). Write $\eta = 2s/\{s + s'(2s+1)\} \in [1, 2)$ and $\beta = (2s + 2/\pi - 1)/(2s+1)$, which lies in $(0,\, 4s/(2s+1))$ because $\pi < 2$ and $2s' = 2s + 1 - 2/\pi > 0$. The choice of the statement is $2^{J_n} \asymp n^\eta b_n^{-\eta\beta(2s+1)/(4s)}$, which exceeds n/b_n because $\beta(2s+1)/(4s) < 1$, and is at most n^2. By (S6.7), $t = (\mu_{J_n}^{\mathcal{G}})^{1/2} \lambda_{n,1}^{\mathcal{G}} \asymp 2^{J_n/2} b_n/n$, and (S6.15) gives

$$t^{4s/(2s+1)} b_n^{\beta - 4s/(2s+1)} \;\asymp\; 2^{J_n \cdot 2s/(2s+1)}\, n^{-4s/(2s+1)}\, b_n^{\beta},$$

which, equated with the bias $2^{-2J_n s'}$, gives $2^{J_n\{2s/(2s+1)+2s'\}} = n^{4s/(2s+1)} b_n^{-\beta}$; this is the choice of the statement, because $\{2s/(2s+1)+2s'\}(2s+1) = 2\{s + s'(2s+1)\}$. The common value is $2^{-2J_n s'} = \{b_n^\beta / n^{4s/(2s+1)}\}^{2s'/\{2s/(2s+1)+2s'\}} = n^{-e} b_n^{e\beta(2s+1)/(4s)}$, and $\beta(2s+1)/(4s) = (2s + 2/\pi - 1)/(4s)$. The term of the coarse chunk, of order t, is polynomially smaller, because $t \to 0$ and $4s/(2s+1) < 1$. □

S7 Recovery of the block structure by thresholding

Throughout this section $\nu_{\ell m}$, $\widehat{\nu}_{\ell m}$, $\mathcal{S}$, $\widehat{\mathcal{S}}(t)$, ~~ρ_n~~ $\rho_n^{\mathcal{G}}$ and $\nu_{\min,n}$ are as in Section 3.6 of the paper, $\nu_{\min} = \min_{(\ell,m)\in\mathcal{S}} \nu_{\ell m}$ for a fixed sample size, and

$$\Delta \;=\; \max_{\ell,m} \big\|\widehat{g}_{\ell m} - g_{\ell m}\big\|_{L_2[0,1]}$$

is the largest error over the blocks. The first statement is deterministic and knows nothing about the ~~lasso~~ estimator.

Lemma S7.1 (Thresholding under a uniform block error)**.** *Let $\{g_{\ell m}\}$ and $\{\widehat{g}_{\ell m}\}$ be any two families of functions in $L_2[0,1]$ indexed by the blocks, with $\nu_{\ell m}$, $\widehat{\nu}_{\ell m}$, $\mathcal{S}$, $\widehat{\mathcal{S}}(t)$, Δ and $\nu_{\min}$ defined from them as above, and let $\bar{\Delta} \geq \Delta$. Then, for every $t \geq 0$,*

$$\big\{(\ell,m):\ \nu_{\ell m} > t + \bar{\Delta}\big\} \;\subseteq\; \widehat{\mathcal{S}}(t) \;\subseteq\; \big\{(\ell,m):\ \nu_{\ell m} > t - \bar{\Delta}\big\}. \qquad \text{(S7.1)}$$

In particular: (i) if $t \geq \bar{\Delta}$, then $\widehat{\mathcal{S}}(t) \subseteq \mathcal{S}$, with no use of $\nu_{\min}$; (ii) if $t < \nu_{\min} - \bar{\Delta}$, then $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$; and (iii) if $\bar{\Delta} \leq t < \nu_{\min} - \bar{\Delta}$, then $\widehat{\mathcal{S}}(t) = \mathcal{S}$, and such a t exists if and only if $\nu_{\min} > 2\bar{\Delta}$.

Proof. By the triangle inequality in $L_2[0,1]$, for every block, $|\widehat{\nu}_{\ell m} - \nu_{\ell m}| \leq \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]} \leq \Delta \leq \bar{\Delta}$. If $\nu_{\ell m} > t + \bar{\Delta}$, then $\widehat{\nu}_{\ell m} \geq \nu_{\ell m} - \bar{\Delta} > t$, which is the left inclusion of (S7.1); if $\widehat{\nu}_{\ell m} > t$, then $\nu_{\ell m} \geq \widehat{\nu}_{\ell m} - \bar{\Delta} > t - \bar{\Delta}$, which is the right one. For (i), $t - \bar{\Delta} \geq 0$ and the right inclusion give

$\widehat{\mathcal{S}}(t) \subseteq \{\nu_{\ell m} > 0\} = \mathcal{S}$. For (ii), every block of $\mathcal{S}$ has $\nu_{\ell m} \geq \nu_{\min} > t + \bar{\Delta}$, and the left inclusion applies. Part (iii) is the conjunction of the two, and $[\bar{\Delta}, \nu_{\min} - \bar{\Delta})$ is nonempty exactly when $\bar{\Delta} < \nu_{\min} - \bar{\Delta}$. □

Remark S7.1 (The window is exact)**.** The condition $\nu_{\min} > 2\bar{\Delta}$ is not an artifact of the proof. Take two blocks, one active with $\nu_{11} = 2\bar{\Delta}$ and one null with $\nu_{21} = 0$, and fitted norms $\widehat{\nu}_{11} = \widehat{\nu}_{21} = \bar{\Delta}$, which is compatible with $\Delta \leq \bar{\Delta}$: the two statistics tie and no threshold separates them. This says that the argument does not extend below $\nu_{\min} = 2\bar{\Delta}$, not that no procedure does, which would be a lower bound and is not proved here.

Lemma S7.2 (The risk of the thresholded estimator)**.** *Let* $\{g_{\ell m}\}$ *and* $\{\widehat{g}_{\ell m}\}$ *be any two families of functions in* $L_2[0,1]$ *indexed by the blocks and* $t \geq 0$*, with* $\nu_{\ell m}$*,* $\widehat{\nu}_{\ell m}$*,* $\mathcal{S}$ *and* $\widehat{\mathcal{S}}(t)$ *as above,* $\Delta_{\ell m} = \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$*, and* $\widehat{g}^{(t)}_{\ell m} = \widehat{g}_{\ell m}\mathbf{1}\{(\ell, m) \in \widehat{\mathcal{S}}(t)\}$.

(i) With no assumption,

$$\sum_{\ell,m} \big\|\widehat{g}^{(t)}_{\ell m} - g_{\ell m}\big\|^2_{L_2[0,1]} = \sum_{(\ell,m)\in\widehat{\mathcal{S}}(t)} \Delta^2_{\ell m} + \sum_{(\ell,m)\in\mathcal{S}\setminus\widehat{\mathcal{S}}(t)} \nu^2_{\ell m},$$

and every $(\ell, m) \in \mathcal{S} \setminus \widehat{\mathcal{S}}(t)$ *has* $\nu_{\ell m} \leq t + \Delta_{\ell m}$*; hence the left side is at most* $2\sum_{\ell,m} \Delta^2_{\ell m} + 2t^2|\mathcal{S} \setminus \widehat{\mathcal{S}}(t)| \leq 2\sum_{\ell,m} \Delta^2_{\ell m} + 2t^2|\mathcal{S}|$.

(ii) If $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$*, the left side of (i) is* $\sum_{\widehat{\mathcal{S}}(t)} \Delta^2_{\ell m} \leq \sum_{\ell,m} \Delta^2_{\ell m}$*; if* $\widehat{\mathcal{S}}(t) = \mathcal{S}$*, it is* $\sum_{\mathcal{S}} \Delta^2_{\ell m}$*, and the null blocks no longer contribute.*

(iii) Let $\lambda_{\min}(\widehat{\Sigma}) > 0$, $\boldsymbol{\beta}^* = (\mathbf{c}, \boldsymbol{\theta}^*)$, $f_J = \mathbf{Z}\boldsymbol{\beta}^*$, $\widehat{\boldsymbol{\beta}} = (\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}})$ *any fit,* $\widehat{\mathcal{S}}(t) = \{(\ell, m) : \left\|\widehat{\boldsymbol{\theta}}_{\ell m,\cdot}\right\|_2 > t\}$, $\mathcal{S}_J = \{(\ell, m) : \boldsymbol{\theta}^*_{\ell m,\cdot} \neq 0\}$, $\widehat{\boldsymbol{\theta}}^{(t)}$ *the vector* $\widehat{\boldsymbol{\theta}}$ *with the blocks outside* $\widehat{\mathcal{S}}(t)$ *set to zero, and* $\widehat{f}^{(t)} = \mathbf{A}\widehat{\mathbf{c}} + \mathbf{B}\widehat{\boldsymbol{\theta}}^{(t)}$. *Then*

$$\left\|\widehat{f}^{(t)} - f_J\right\|_n^2 \leq \frac{2\lambda_{\max}(\widehat{\Sigma})}{\lambda_{\min}(\widehat{\Sigma})} \left\|\widehat{f} - f_J\right\|_n^2 + 2\lambda_{\max}(\widehat{\Sigma})\, t^2\, |\mathcal{S}_J \setminus \widehat{\mathcal{S}}(t)|,$$

and, if $\mathcal{S}_J \subseteq \widehat{\mathcal{S}}(t)$, $\left\|\widehat{f}^{(t)} - f_J\right\|_n^2 \leq \{\lambda_{\max}(\widehat{\Sigma})/\lambda_{\min}(\widehat{\Sigma})\} \left\|\widehat{f} - f_J\right\|_n^2$.

Proof. (i). If $(\ell, m) \in \widehat{\mathcal{S}}(t)$, then $\widehat{g}^{(t)}_{\ell m} = \widehat{g}_{\ell m}$ and the error of the block is $\Delta^2_{\ell m}$; otherwise $\widehat{g}^{(t)}_{\ell m} = 0$ and the error is $\nu^2_{\ell m}$, which vanishes outside $\mathcal{S}$. Summing gives the identity. For $(\ell, m) \notin \widehat{\mathcal{S}}(t)$, $\widehat{\nu}_{\ell m} \leq t$, and the triangle inequality gives $\nu_{\ell m} \leq \widehat{\nu}_{\ell m} + \Delta_{\ell m} \leq t + \Delta_{\ell m}$, hence $\nu^2_{\ell m} \leq 2t^2 + 2\Delta^2_{\ell m}$; summing over $\mathcal{S} \setminus \widehat{\mathcal{S}}(t)$ and adding the sum over $\widehat{\mathcal{S}}(t)$, a disjoint set, gives the bound. *(ii).* If $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t)$, the second sum of (i) is empty. *(iii).* The proof of (i) and (ii) uses only the triangle inequality in a norm; applied to the vectors $\widehat{\boldsymbol{\theta}}_{\ell m,\cdot}$ and $\boldsymbol{\theta}^*_{\ell m,\cdot}$ of $\mathbb{R}^{N_J}$, with $\mathcal{S}_J$ in place of $\mathcal{S}$, it gives $\left\|\widehat{\boldsymbol{\theta}}^{(t)} - \boldsymbol{\theta}^*\right\|_2^2 \leq 2\left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2^2 + 2t^2|\mathcal{S}_J \setminus \widehat{\mathcal{S}}(t)|$, and $\left\|\widehat{\boldsymbol{\theta}}^{(t)} - \boldsymbol{\theta}^*\right\|_2^2 \leq \left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2^2$ when $\mathcal{S}_J \subseteq \widehat{\mathcal{S}}(t)$. Since $\widehat{\mathbf{c}}$ does not change, the same bounds hold for $\widehat{\boldsymbol{\beta}}^{(t)} = (\widehat{\mathbf{c}}, \widehat{\boldsymbol{\theta}}^{(t)})$ against $\boldsymbol{\beta}^*$. Finally $\left\|\widehat{f}^{(t)} - f_J\right\|_n^2 = (\widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^*)^\top \widehat{\Sigma} (\widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^*) \leq \lambda_{\max}(\widehat{\Sigma}) \left\|\widehat{\boldsymbol{\beta}}^{(t)} - \boldsymbol{\beta}^*\right\|_2^2$ and $\left\|\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*\right\|_2^2 \leq \left\|\mathbf{Z}(\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta}^*)\right\|_n^2 / \lambda_{\min}(\widehat{\Sigma}) = \left\|\widehat{f} - f_J\right\|_n^2 / \lambda_{\min}(\widehat{\Sigma})$. By the orthonormality of the basis, $\left\|\widehat{\boldsymbol{\theta}}_{\ell m,\cdot}\right\|_2 = \widehat{\nu}_{\ell m}$, so that the $\widehat{\mathcal{S}}(t)$ of (iii) is that of (i). □

The factor 2 in (i) is attained: an active block with $\widehat{g}_{\ell m} = t\, g_{\ell m}/\nu_{\ell m}$ and $\nu_{\ell m} = 2t$ has $\widehat{\nu}_{\ell m} = t$, is set to zero, and pays $\nu_{\ell m}^2 = 4t^2 = 2\Delta_{\ell m}^2 + 2t^2$. In (iii), the passage between the norm of the coefficients and the prediction norm uses the smallest eigenvalue of $\widehat{\Sigma}$ to go from prediction to coefficients and the largest to come back, and the factor is the condition number of $\widehat{\Sigma}$, which on $\mathcal{E}_\Sigma$ is smaller than $2\Lambda/\gamma$.

[k = 4: the proof of Corollary 2 for the lasso, which stood here, is moved to Section S8, unchanged except for the numbering of the results it cites.]

Proof of Corollary 2

A nonasymptotic bound on Δ. Work at $J = J_n$ and $\lambda = \lambda_n^{\mathcal{G}}$, and intersect $\mathcal{A}_n$, the event $\mathcal{E}$ of Lemma S5.2(i) and $\{\|\mathbf{b}\|_n^2 \le \mathcal{B}_{J_n}^2/\alpha'\}$; the intersection has probability at least $1 - \alpha - \alpha' - \mathrm{P}(\mathcal{A}_n^c)$, by Lemma S5.2, Markov's inequality and Lemma 2. On $\mathcal{A}_n$, $\mathcal{E} \subseteq \mathcal{T}_{\mathcal{G},w}$ and $\widetilde{\gamma} > \gamma/2$ (Lemma S6.1), and Theorem 2(ii) with the comparator $\boldsymbol{\beta}^*$ and $\mathcal{W}(\mathcal{G}_0(\boldsymbol{\theta}^*)) \le \mathcal{W}(\mathcal{G}_n)$ gives, with $v = \widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*$,

$$\|v\|_2^2 \;\le\; \frac{64(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\widetilde{\gamma}^2} + \frac{16\,\|\mathbf{b}\|_n^2}{\widetilde{\gamma}} \;\le\; \frac{256(\lambda_n^{\mathcal{G}})^2 \mathcal{W}(\mathcal{G}_n)}{\gamma^2} + \frac{32\mathcal{B}_{J_n}^2}{\alpha'\gamma}.$$

For a fixed block, the orthonormality of the wavelets of level below J_n gives $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]} = \|v_{\ell m,\cdot}\|_2 \le \|v\|_2$, and the first display of Lemma 2 bounds $\|\Pi_J g_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2$ by $C_g^2(1 - 2^{-2s'})^{-1} 2^{-2J_n s'}$. By $(x+y)^2 \le 2x^2 +$

$2y^2$,

$$\Delta^2 \;\le\; (\bar{\Delta}_n^{\mathcal{G}})^2 \;=\; \frac{512(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{G}_n)}{\gamma^2} + \frac{64\,\mathcal{B}_{J_n}^2}{\alpha'\gamma} + \frac{2C_g^2}{1-2^{-2s'}}\,2^{-2J_n s'}, \qquad \text{(S7.2)}$$

with $(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{G}_n) \le \varrho^2(\lambda_{n,1}^{\mathcal{G}})^2|\mathcal{G}_n|$ by Lemma S5.2(iii). The residual probability $\mathrm{P}(\mathcal{A}_n^c)$ tends to zero by Lemma S6.1 and does not depend on the components; it is the ω_n of Corollary 2(iii), and by Lemma S7.1(iii) this paragraph is the proof of that part. The bound $\bar{\Delta}_n^{\mathcal{G}}$ depends on the components only through the constants of Assumption 5, so everything above holds uniformly over the class, which is what allows the target to vary with n in Assumption 6.

The order of $\bar{\Delta}_n^{\mathcal{G}}$. By the proof of Theorem 3, $(\lambda_n^{\mathcal{G}})^2\mathcal{W}(\mathcal{G}_n) \le \bar{\varrho}^2 C_\lambda \sigma^2 pq(b_n + 2^{J_n})/n \asymp n^{-2s'/(2s'+1)}$, and $2^{-2J_n s'} \le n^{-2s'/(2s'+1)}$ by the definition of J_n. Hence $\bar{\Delta}_n^{\mathcal{G}} \le c(\alpha,\alpha')\rho_n^{\mathcal{G}}$ with $c(\alpha,\alpha') < \infty$. Given $\epsilon > 0$, take $\alpha = \alpha' = \epsilon/4$ and n large enough that $\mathrm{P}(\mathcal{A}_n^c) < \epsilon/2$; then $\mathrm{P}\{\Delta > c(\epsilon/4,\epsilon/4)\rho_n^{\mathcal{G}}\} < \epsilon$, that is $\Delta = O_p(\rho_n^{\mathcal{G}})$, uniformly over the class.

Part (i). If $t_n/\rho_n^{\mathcal{G}} \to \infty$, then $\mathrm{P}(\Delta \le t_n) \to 1$, and on that event Lemma S7.1(i) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}$. No property of $\nu_{\min,n}$ has been used.

Part (ii). If $\mathcal{S}$ is empty, part (ii) is part (i). Otherwise, the condition $\limsup_n t_n/\nu_{\min,n} < 1$ gives $\eta > 0$ and n_0 with $\nu_{\min,n} - t_n \ge \eta\,\nu_{\min,n}$ for $n \ge n_0$, and Assumption 6 gives $(\nu_{\min,n} - t_n)/\rho_n^{\mathcal{G}} \ge \eta\,\nu_{\min,n}/\rho_n^{\mathcal{G}} \to \infty$.

Hence $\mathrm{P}(\Delta \leq t_n) \to 1$ and $\mathrm{P}(\Delta < \nu_{\min,n} - t_n) \to 1$, and on the intersection of the two events Lemma S7.1(iii) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$. For the existence, $t_n = (\rho_n^{\mathcal{G}} \nu_{\min,n})^{1/2}$ has $t_n/\rho_n^{\mathcal{G}} = (\nu_{\min,n}/\rho_n^{\mathcal{G}})^{1/2} \to \infty$ and $t_n/\nu_{\min,n} = (\rho_n^{\mathcal{G}}/\nu_{\min,n})^{1/2} \to 0$. □

At the resolution of Theorem 1. Theorem 1 gives $\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|^2_{L_2[0,1]} = O_p\{(\tilde{\rho}_n^{\mathcal{G}})^2\}$, with $\tilde{\rho}_n^{\mathcal{G}}$ as in Section 3.6 of the paper, hence $\Delta = O_p(\tilde{\rho}_n^{\mathcal{G}})$, and the bound in its proof depends on the components only through the constants of Assumption 5 and of the design, so that this holds uniformly over the class. The proofs of parts (i) and (ii) above use nothing about Δ beyond $\Delta = O_p(\rho_n^{\mathcal{G}})$ uniformly over the class, together with Lemma S7.1 and Assumption 6, and they go through verbatim with $\tilde{\rho}_n^{\mathcal{G}}$ in place of $\rho_n^{\mathcal{G}}$.

Proof of Corollary 3

Write r_n for the rate of the statement and $\Delta_{\ell m}$ as in Lemma S7.2. For the fit of Theorem 3, $\left\|\widehat{f} - f\right\|_n^2 = O_p(r_n^2)$ and $\sum_{\ell,m} \Delta_{\ell m}^2 = O_p(r_n^2)$ by Theorem 3 and Corollary 1; for the lasso, by Theorem S8.2 and Corollary S8.1. In both cases $\mathrm{P}(\mathcal{E}_\Sigma) \to 1$, by the proofs of Theorems 3 and S8.2, and $\|\mathbf{b}\|_n^2 = O_p(\mathcal{B}_{J_n}^2) = O_p(r_n^2)$, by the choice of J_n.

Part (i). The bound for the components is Lemma S7.2(i) with $|\mathcal{S}| \leq pq$. For prediction, $\left\|\widehat{f}^{(t)} - f\right\|_n^2 \leq 2\left\|\widehat{f}^{(t)} - f_{J_n}\right\|_n^2 + 2\|\mathbf{b}\|_n^2$, and on $\mathcal{E}_\Sigma$,

$\lambda_{\max}(\widehat{\Sigma}) < \Lambda$ and $\lambda_{\min}(\widehat{\Sigma}) > \gamma/2$, so that Lemma S7.2(iii) gives, for every t,

$$\left\|\widehat{f}^{(t)} - f\right\|_n^2 \le \frac{8\Lambda}{\gamma}\left\|\widehat{f} - f_{J_n}\right\|_n^2 + 4\Lambda t^2 |\mathcal{S}_{J_n} \setminus \widehat{\mathcal{S}}(t)| + 2\|\mathbf{b}\|_n^2,$$

with $\left\|\widehat{f} - f_{J_n}\right\|_n^2 \le 2\left\|\widehat{f} - f\right\|_n^2 + 2\|\mathbf{b}\|_n^2 = O_p(r_n^2)$.

Part (ii). By Lemma S7.1(ii), $\mathcal{S} \subseteq \widehat{\mathcal{S}}(t_n)$ whenever $t_n < \nu_{\min,n} - \Delta$. With $\limsup_n t_n/\nu_{\min,n} < 1$ there are $\eta > 0$ and n_0 with $\nu_{\min,n} - t_n \ge \eta\nu_{\min,n}$ for $n \ge n_0$, and since $\Delta^2 \le \sum_{\ell,m}\Delta_{\ell m}^2 = O_p(r_n^2)$ and $\nu_{\min,n}/r_n \to \infty$, $\mathrm{P}(\Delta < \nu_{\min,n} - t_n) \to 1$. On that event Lemma S7.2(ii) holds and, since $\mathcal{S}_{J_n} \subseteq \mathcal{S}$, so does the second case of Lemma S7.2(iii), which on $\mathcal{E}_\Sigma$ gives $\left\|\widehat{f}^{(t_n)} - f\right\|_n^2 \le (4\Lambda/\gamma)\left\|\widehat{f} - f_{J_n}\right\|_n^2 + 2\|\mathbf{b}\|_n^2$. If moreover $t_n/r_n \to \infty$, part (ii) of Corollary 2, or of Corollary S8.2 for the lasso, gives $\mathrm{P}\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \to 1$, and Lemma S7.2(ii) concludes. □

S8 The coordinatewise lasso

[k = 4: the statements of this section stood in Sections 3.2 to 3.6 of the paper, and their proofs in Sections S5 to S7 of this supplement.]

This section gives the theory of the estimator obtained by replacing $\|\boldsymbol{\theta}\|_{\mathcal{G},w}$ by $\|\boldsymbol{\theta}\|_1$ in equation (2.5) of the paper, the lasso of Tibshirani (1996) with the levels left unpenalized, which the software offers as an option. Assumptions 1 to 5, Lemmas 1 and 2, Proposition 2 and Lemmas S5.1,

S5.3 and S5.4 are used as they stand, with the notation S_0, s_0 and v of Section S1. The penalty level is

$$\lambda_n \;=\; 2\sigma B_X\sqrt{2C_U}\,\sqrt{\frac{2\log(2d_n/\alpha)}{n}} \;\asymp\; \sigma B_X\sqrt{C_U}\,\sqrt{\frac{J_n}{n}}, \tag{S8.1}$$

with d_n the number of penalized columns at resolution J_n.

Statements

Theorem S8.1 (Oracle inequality for the lasso)**.** *Let Assumption 1 hold, suppose* $\mathbf{A}$ *has rank* p*, and take* $\lambda \geq 2\lambda_0$ *with* $\lambda_0 = \sigma\widehat{\sigma}_{\max}\{2\log(2d/\alpha)/n\}^{1/2}$*. On the event* $\mathcal{T} = \{\left\|\widetilde{\mathbf{B}}^\top\boldsymbol{\varepsilon}/n\right\|_\infty \leq \lambda/2\}$*, which has probability at least* $1-\alpha$*:*

(i) with no condition on the design, $\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \leq 6\lambda\left\|\boldsymbol{\theta}^*\right\|_1 + 4\left\|\mathbf{b}\right\|_n^2$*;*

(ii) if $\widetilde{\gamma} > 0$*,*

$$\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}} + 16\left\|\mathbf{b}\right\|_n^2, \qquad \left\|v\right\|_2^2 \leq \frac{64\lambda^2 s_0}{\widetilde{\gamma}^2} + \frac{16\left\|\mathbf{b}\right\|_n^2}{\widetilde{\gamma}};$$

(iii) in either regime, $\left\|\widehat{f} - f\right\|_n \leq \left\|\widetilde{\mathbf{B}}v\right\|_n + 2\left\|\mathbf{b}\right\|_n + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n$*, and* $\mathrm{E}(\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2 \mid \mathbf{X}, \mathbf{U}) \leq \sigma^2 p/n$*.*

Theorem S8.2 (Rate of linear approximation for the lasso)**.** *Let Assumptions 1 to 5 hold with* p*,* q *and the constants fixed, and take*

$$J_n = \min\bigl\{J \in \mathbb{N} :\; 2^J \geq (n/\log n)^{1/(2s'+1)}\bigr\}, \tag{S8.2}$$

so that $J_n = \log_2 n/(2s'+1) + O(\log\log n)$. *Then the requirement of Proposition 2(ii) holds along* J_n, *being itself of the exact order* $(\log n/n)^{2s'/(2s'+1)}$, *and, with* λ_n *as in* (S8.1),

$$\left\|\widehat{f} - f\right\|_n^2 \;=\; O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s'/(2s'+1)}\Big\}.$$

At this resolution the estimation term and the bias term are of the same order, and the term $\sigma^2 p/n$ *of Theorem S8.1(iii) is of strictly smaller order.*

Corollary S8.1 (Components and coefficients for the lasso)**.** *In the setting of Theorem S8.2, and with* r_n *denoting the rate of that theorem,*

$$\sum_{\ell,m}\left\|\widehat{g}_{\ell m} - g_{\ell m}\right\|_{L_2[0,1]}^2 = O_p(r_n), \qquad \left\|\widehat{\beta}_\ell - \beta_\ell\right\|_{L_2([0,1]^q)} = O_p(r_n^{1/2})$$

for $\ell = 1, \ldots, p$, *and the same statements hold with the rate of Theorem S8.3 in place of* r_n *under the conditions of that theorem.*

The level λ_n is admissible only while $2^{J_n}\log d_n/n \to 0$, which Lemma S5.3 needs for $\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U$; above that ceiling $\widehat{\sigma}^2_{\max}$ grows with $2^J/n$. With $c_\psi = 7\left\|\psi\right\|_\infty^2/(12C_U)$ as in Section S6, write

$$\mu_J = 1 + c_\psi\frac{2^J\log d}{n}, \qquad \lambda_n^+ = \mu_{J_n}^{1/2}\lambda_n; \tag{S8.3}$$

when $2^{J_n}\log d_n/n \to 0$, $\lambda_n^+ = \lambda_n\{1+o(1)\}$.

Proposition S8.1 (Slow rate for the lasso, no design condition)**.** *Let Assumptions 1, 4 and 5 hold, together with the upper bounds in Assump-*

tions 2 and 3, with $\mathbf{A}$ *of rank* p *and* λ_n *as in* (S8.1). *Then* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{\lambda_n \|\boldsymbol{\theta}^*\|_1 + \mathcal{B}_{J_n}^2 + p/n\}$, *where* $\|\boldsymbol{\theta}^*\|_1 \le pqC_g \sum_{j<J_n} 2^{j(1/2-s)}$ *does not depend on* π, *whenever* $2^{J_n}\log d_n/n \to 0$; *with* λ_n^+ *of* (S8.3) *in place of* λ_n, *in the estimator and in the bound, the same holds for every sequence* J_n. *In particular, (i) if* $s > 1/2$ *and* $s' > 1/4$ *then* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{(J_n/n)^{1/2}\}$ *for every* $J_n \ge \log_2 n/(4s')$; *and (ii) if* $s < 1/2$ *and* $s' > s/2$, *the choice* $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ *gives* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{(J_n/n)^{2s'/(1-2s+4s')}\}$, *an exponent that equals* $2s/(2s+1)$ *when* $\pi \ge 2$. *With* λ_n^+:

(iii) (the pairs that (i) and (ii) exclude, all with $\pi < 2$*) (a) if* $s > 1/2$ *and* $s' \le 1/4$, *the choice* $2^{J_n} \asymp (n/\log n)^{2/(1+4s')}$ *gives* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{(\log n/n)^{4s'/(1+4s')}\}$; *(b) if* $s < 1/2$ *and* $s' \le s/2$, *the choice* $2^{J_n} \asymp (n/\log n)^{1/(1-s+2s')}$ *gives* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{(\log n/n)^{2s'/(1-s+2s')}\}$; *in both,* $2^{J_n} \gtrsim n/\log n$, *above the ceiling of Lemma S5.3, and the rates join those of (i) and (ii) at* $s' = 1/4$ *and* $s' = s/2$;

(iv) (the boundary $s = 1/2$, *with* $\pi > 1$*)* $\|\boldsymbol{\theta}^*\|_1 \le pqC_gJ_n$, *and: if* $s' \ge 1/4$, *that is* $\pi \ge 4/3$, *the choice* $2^{J_n} \asymp (n/\log n)^{1/(4s')}$ *gives* $\left\|\widehat{f}-f\right\|_n^2 = O_p\{n^{-1/2}(\log n)^{3/2}\}$; *if* $s' < 1/4$, *that is* $1 < \pi < 4/3$, *the choice* $2^{J_n} \asymp \{n/(\log n)^2\}^{2/(1+4s')}$ *gives* $\left\|\widehat{f}-f\right\|_n^2 = O_p[\{(\log n)^2/n\}^{4s'/(1+4s')}]$.

Proposition S8.1 reads Theorem S8.1(i) at $\boldsymbol{\theta}^*$ and pays $\lambda|\theta_a^*|$ also for

the coefficients smaller than λ, which would cost less as bias. Lemma S8.3 below holds for every comparator on the same event, and the comparator truncated at $t > 0$, $\bar{\boldsymbol{\theta}}_a = \theta^*_a \mathbf{1}\{|\theta^*_a| > t\}$ for $a = 1, \dots, d$, charges the penalty only for the coefficients above t. What remains is

$$\mathcal{R}_1(\boldsymbol{\theta}; t) \;=\; \sum_{a=1}^{d} \min\bigl(t|\theta_a|,\, \theta_a^2\bigr) \;\le\; t\,\|\boldsymbol{\theta}\|_1\,, \tag{S8.4}$$

the counterpart in the ℓ_1 norm of the ideal risk $\sum_a \min(\theta_a^2, t^2)$.

Proposition S8.2 (Slow rate for the lasso through the truncated comparator)**.** *Let the assumptions of Proposition S8.1 hold.*

(i) On the event $\mathcal{T}$, for every $\lambda \ge 2\lambda_0$ and every $t > 0$, with $\bar{\boldsymbol{\theta}}$ truncated at t,

$$\bigl\|\widehat{f} - f\bigr\|_n^2 \;\le\; 18\lambda\,\bigl\|\bar{\boldsymbol{\theta}}\bigr\|_1 + 48\,\bigl\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\bigr\|_n^2 + 48\,\|\mathbf{b}\|_n^2 + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2\,.$$

If t is deterministic, $\mathrm{E}\,\bigl\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\bigr\|_n^2 \le \kappa_2 C_U\,\bigl\|\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}\bigr\|_2^2$. With $t = \lambda$ deterministic, $18\lambda\,\bigl\|\bar{\boldsymbol{\theta}}\bigr\|_1 + 48\kappa_2 C_U\,\bigl\|\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}\bigr\|_2^2 \le \max(18, 48\kappa_2 C_U)\mathcal{R}_1(\boldsymbol{\theta}^*; \lambda)$, and

$$\bigl\|\widehat{f} - f\bigr\|_n^2 \;=\; O_p\bigl\{\mathcal{R}_1(\boldsymbol{\theta}^*; \lambda_n) + \mathcal{B}_{J_n}^2 + p/n\bigr\}$$

whenever $2^{J_n} \log d_n / n \to 0$; with λ_n^+ of (S8.3) in place of λ_n, in the estimator and in the bound, the same holds for every sequence J_n with $2^{J_n} \le n^2$.

(ii) For every J and every $t > 0$, with $\tau = (s + 1/2)^{-1}$: if $s < 1/2$,

$$\mathcal{R}_1(\boldsymbol{\theta}^*; t) \;\le\; pq\, C_g^{\tau}\, t^{4s/(2s+1)} \Big\{ \frac{1}{1 - 2^{s-1/2}} + \frac{1}{1 - 2^{-\min(\pi,2)s'}} \Big\}; \quad \text{(S8.5)}$$

if $s \ge 1/2$, nothing is gained in order: $\mathcal{R}_1(\boldsymbol{\theta}^*; t) \le t \, \|\boldsymbol{\theta}^*\|_1$, and for $t \le C_g$ the class of Assumption 5 contains a $\boldsymbol{\theta}^*$ with $\mathcal{R}_1(\boldsymbol{\theta}^*; t) \ge t C_g$ if $s > 1/2$, and one with $\mathcal{R}_1(\boldsymbol{\theta}^*; t) \ge t C_g \min\{J, \lfloor \log_2(C_g/t) \rfloor + 1\}$ if $s = 1/2$, the order of $t \, \|\boldsymbol{\theta}^*\|_1$ in Proposition S8.1.

(iii) If $s < 1/2$ and $s' > s/(2s + 1)$, then with $J_n = \lceil c \log_2 n \rceil$ and c in the window (S8.6),

$$\big\| \widehat{f} - f \big\|_n^2 \;=\; O_p\Big\{ \Big(\frac{\log n}{n} \Big)^{2s/(2s+1)} \Big\},$$

the prediction rate of Theorem S8.3.

(iv) If $s < 1/2$ and $s' \le s/(2s + 1)$, which forces $\pi < 2$, then with λ_n^+ and the choice $2^{J_n} \asymp (n/\log n)^{2s/\{s+s'(2s+1)\}}$, which has $2^{J_n} \gtrsim n/\log n$,

$$\big\| \widehat{f} - f \big\|_n^2 \;=\; O_p\Big\{ \Big(\frac{\log n}{n} \Big)^{4ss'/\{s+s'(2s+1)\}} \Big\};$$

at $s' = s/(2s + 1)$ the exponent is $2s/(2s + 1)$, that of (iii).

Proposition S8.2 is a statement on prediction: it uses the design as Proposition S8.1 does, and, in the new bias, the upper bound $\lambda_{\max}(\Sigma) \le \kappa_2 C_U$; the components still need the design condition (Corollary S8.1). When $s < 1/2$ and $\pi < 2$ it improves on Proposition S8.1 in every pair:

(iii) gives the rate of Theorem S8.3 in the window of that theorem, with a gain of $2(s-s')(1-2s)/\{(2s+1)(1-2s+4s')\}$ in the exponent over Proposition S8.1(ii), and (iv) replaces Proposition S8.1(iii)(b) and the part of Proposition S8.1(ii) with $s/2 < s' \le s/(2s+1)$. When $\pi \ge 2$ the rate is that of Proposition S8.1(ii), now for every $c \in [1/(2s+1), 1)$. When $s \ge 1/2$ nothing changes, for the reason given after Proposition S6.2.

For a vector $\boldsymbol{\theta}$ let $\theta_{(1)} \ge \theta_{(2)} \ge \cdots$ be the nonincreasing rearrangement of the absolute values of its entries and let $\boldsymbol{\theta}^{(M)}$ retain the M largest of them.

Lemma S8.1 (Besov regularity implies compressibility)**.** *Under Assumptions 4 and 5 with $\pi(s+1/2) > 1$, the oracle satisfies, for every J and every $k \ge 1$,*

$$\theta^*_{(k)} \;\le\; C_\tau\, k^{-1/\tau}, \qquad \tau = \frac{1}{s+1/2}, \qquad C_\tau = C_g (pq\, A_{s,\pi})^{s+1/2}, \qquad A_{s,\pi} = 1 + \frac{1}{1-2^{1-\pi(s+1/2)}},$$

with constants that do not depend on J and with $\tau \in (0,2)$ for every $s > 0$. Consequently $\sum_{k>M} \theta^{*2}_{(k)} \le \tau(2-\tau)^{-1} C_\tau^2 M^{1-2/\tau}$ for every $M \ge 1$.

The inclusion of Besov balls in weak ℓ_τ balls is classical (Johnstone, 2019, Proposition 9.15); what the lemma adds is the constant C_τ, uniform in J, for the oracle of the product design, which is the form in which the proof of Theorem S8.3 uses it.

Theorem S8.3 (Rate of nonlinear approximation for the lasso)**.** *Let Assumptions 1 to 5 hold with p, q and all the constants fixed as n grows, and set $\tau = (s + 1/2)^{-1}$. Suppose $s' > s/(2s+1)$ and take $J_n = \lceil c \log_2 n \rceil$ with*

$$\max\Big\{\frac{\tau}{2},\ \frac{2-\tau}{4s'}\Big\} \ \le\ c\ <\ 1, \tag{S8.6}$$

a window that is nonempty precisely under that condition. Then, with λ_n as in (S8.1)*,*

$$\big\|\widehat{f} - f\big\|_n^2 \ =\ O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s/(2s+1)}\Big\}, \qquad \sum_{\ell,m}\|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 \ =\ O_p\Big\{\Big(\frac{\log n}{n}\Big)^{2s/(2s+1)}\Big\}.$$

The logarithmic factor is the price of the single penalty level (S8.1), which takes a union bound over the d_n columns; Theorem 1 of the paper removes the factor $(\log n)^{2s'/(2s+1)}$ of it. Write $\rho_n = (\log n/n)^{s'/(2s'+1)}$ for the square root of the rate of Theorem S8.2.

Corollary S8.2 (Block structure by thresholding the lasso)**.** *Let the assumptions of Theorem S8.2 hold, with J_n and λ_n as there.*

(i) (Screening.) If the thresholds satisfy $t_n/\rho_n \to \infty$, then $\mathrm{P}\{\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}\} \to 1$.

(ii) (Recovery.) If, in addition, Assumption 6 holds with ρ_n in place of $\rho_n^{\mathcal{G}}$ and $\limsup_n t_n/\nu_{\min,n} < 1$, then $\mathrm{P}\{\widehat{\mathcal{S}}(t_n) = \mathcal{S}\} \to 1$. Such thresholds always exist; one is $t_n = (\rho_n \nu_{\min,n})^{1/2}$.

(iii) (A window at fixed n.) With α the level in (S8.1)*, for every n and every $\alpha' \in (0,1)$ there is an event, of probability at least $1-\alpha-\alpha'-\omega_n$ with $\omega_n \to 0$ not depending on the components, on which $\widehat{\mathcal{S}}(t) = \mathcal{S}$ for every t with $\bar{\Delta}_n \le t < \nu_{\min,n} - \bar{\Delta}_n$; here $\bar{\Delta}_n \le c(\alpha,\alpha')\rho_n$ is the explicit bound* (S8.12) *on $\max_{(\ell,m)} \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$.*

Parts (i) and (ii) hold also at the resolution of Theorem S8.3, with ρ_n replaced by $\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$, the square root of the rate of that theorem, both in the condition on t_n and in Assumption 6. Since $\tilde{\rho}_n \le \rho_n$, with a gap in the exponent when $\pi < 2$, the separation required is weaker exactly where the components are spatially inhomogeneous.

Proofs

Lemma S8.2 (Calibration of the penalty)**.** *Under Assumption 1, for every $\alpha \in (0,1)$, $\mathrm{P}(\left\|\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\right\|_\infty > \lambda_0) \le \alpha$ with $\lambda_0 = \sigma\widehat{\sigma}_{\max}\{2\log(2d/\alpha)/n\}^{1/2}$.*

Proof. Condition on the covariates, so that $\widetilde{\mathbf{B}}$ and $\widehat{\sigma}_{\max}$ are fixed and the errors are independent and sub-Gaussian with parameter σ. For a column a, $(\widetilde{\mathbf{B}}^\top\boldsymbol{\varepsilon})_a/n$ is sub-Gaussian with parameter $\sigma\|\widetilde{\mathbf{B}}^{(a)}\|_2/n$, and, $M_{\mathbf{A}}$ being an orthogonal projection, $\|\widetilde{\mathbf{B}}^{(a)}\|_2 \le \|\mathbf{B}^{(a)}\|_2 = \sqrt{n}\,(\widehat{\Sigma})_{aa}^{1/2} \le \sqrt{n}\,\widehat{\sigma}_{\max}$. With $t = \lambda_0$ the sub-Gaussian tail bound is at most α/d per column; a union bound over the d columns and the tower property finish the proof. □

Proof of Theorem S8.1

Step 1, the basic inequality. By Lemma S5.1, $\widehat{\boldsymbol{\theta}}$ minimizes the residualized objective; comparing its value at $\widehat{\boldsymbol{\theta}}$ with its value at $\boldsymbol{\theta}^*$, and using $M_{\mathbf{A}}\mathbf{A} = 0$, so that $M_{\mathbf{A}}\mathbf{Y} = \widetilde{\mathbf{B}}\boldsymbol{\theta}^* + M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$, gives, with $r = M_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon})$,

$$\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + 2\lambda \left\|\widehat{\boldsymbol{\theta}}\right\|_1 \;\leq\; 2\langle r, \widetilde{\mathbf{B}}v\rangle_n + 2\lambda \left\|\boldsymbol{\theta}^*\right\|_1 . \tag{S8.7}$$

Step 2, the two noises. Since $M_{\mathbf{A}}\widetilde{\mathbf{B}} = \widetilde{\mathbf{B}}$ and $M_{\mathbf{A}}$ is symmetric, $\langle r, \widetilde{\mathbf{B}}v\rangle_n = \langle \mathbf{b}+\boldsymbol{\varepsilon}, \widetilde{\mathbf{B}}v\rangle_n$. The stochastic part is handled by duality: on $\mathcal{T}$, $2|\langle \boldsymbol{\varepsilon}, \widetilde{\mathbf{B}}v\rangle_n| \leq 2\left\|\widetilde{\mathbf{B}}^\top \boldsymbol{\varepsilon}/n\right\|_\infty \|v\|_1 \;\leq\; \lambda \|v\|_1$. The deterministic part is handled by the Cauchy-Schwarz and Young inequalities, and not by duality, because $\mathbf{b}$ is not small coordinatewise: $2|\langle \mathbf{b}, \widetilde{\mathbf{B}}v\rangle_n| \;\leq\; 2\|\mathbf{b}\|_n \left\|\widetilde{\mathbf{B}}v\right\|_n \;\leq\; \left\|\widetilde{\mathbf{B}}v\right\|_n^2/2 + 2\|\mathbf{b}\|_n^2$. Substituting both in (S8.7),

$$\tfrac{1}{2}\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + 2\lambda \left\|\widehat{\boldsymbol{\theta}}\right\|_1 \;\leq\; \lambda \|v\|_1 + 2\lambda \left\|\boldsymbol{\theta}^*\right\|_1 + 2\|\mathbf{b}\|_n^2 . \tag{S8.8}$$

Step 3, the slow rate. Bounding $\|v\|_1 \leq \left\|\widehat{\boldsymbol{\theta}}\right\|_1 + \left\|\boldsymbol{\theta}^*\right\|_1$ in (S8.8) gives $\tfrac{1}{2}\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + \lambda \left\|\widehat{\boldsymbol{\theta}}\right\|_1 \leq 3\lambda \left\|\boldsymbol{\theta}^*\right\|_1 + 2\|\mathbf{b}\|_n^2$, which is part (i). No property of $\widehat{\Sigma}$ was used.

Step 4, the cone. Since $\boldsymbol{\theta}^*$ vanishes off S_0, $\left\|\widehat{\boldsymbol{\theta}}\right\|_1 \geq \left\|\boldsymbol{\theta}^*\right\|_1 - \left\|v_{S_0}\right\|_1 + \left\|v_{S_0^c}\right\|_1$ and $\|v\|_1 = \|v_{S_0}\|_1 + \left\|v_{S_0^c}\right\|_1$; substituting in (S8.8) and cancelling $2\lambda \left\|\boldsymbol{\theta}^*\right\|_1$,

$$\tfrac{1}{2}\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + \lambda \left\|v_{S_0^c}\right\|_1 \;\leq\; 3\lambda \left\|v_{S_0}\right\|_1 + 2\|\mathbf{b}\|_n^2 . \tag{S8.9}$$

Step 5, two cases. If $2\|\mathbf{b}\|_n^2 \le \lambda\|v_{S_0}\|_1$, then (S8.9) gives $\frac{1}{2}\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + \lambda\left\|v_{S_0^c}\right\|_1 \le 4\lambda\|v_{S_0}\|_1$; by the Cauchy-Schwarz inequality and (S5.2), $\|v_{S_0}\|_1 \le \sqrt{s_0}\|v\|_2 \le \sqrt{s_0}\left\|\widetilde{\mathbf{B}}v\right\|_n/\sqrt{\widetilde{\gamma}}$, so $\left\|\widetilde{\mathbf{B}}v\right\|_n \le 8\lambda\sqrt{s_0/\widetilde{\gamma}}$ and $\|v\|_2^2 \le \left\|\widetilde{\mathbf{B}}v\right\|_n^2/\widetilde{\gamma} \le 64\lambda^2 s_0/\widetilde{\gamma}^2$; the same display gives $\left\|v_{S_0^c}\right\|_1 \le 4\|v_{S_0}\|_1$ and hence $\|v\|_1 \le 5\|v_{S_0}\|_1 \le 40\lambda s_0/\widetilde{\gamma}$. If instead $2\|\mathbf{b}\|_n^2 > \lambda\|v_{S_0}\|_1$, then (S8.9) gives $\frac{1}{2}\left\|\widetilde{\mathbf{B}}v\right\|_n^2 < 8\|\mathbf{b}\|_n^2$, that is $\left\|\widetilde{\mathbf{B}}v\right\|_n^2 < 16\|\mathbf{b}\|_n^2$, together with $\lambda\|v\|_1 < 10\|\mathbf{b}\|_n^2$ and $\|v\|_2^2 < 16\|\mathbf{b}\|_n^2/\widetilde{\gamma}$. Each quantity is bounded by the maximum of the two cases, hence by their sum, which is part (ii).

Step 6, prediction. By (S5.1), $\widehat{f} - f = \widetilde{\mathbf{B}}v + \Pi_{\mathbf{A}}(\mathbf{b}+\boldsymbol{\varepsilon}) - \mathbf{b}$, and $\Pi_{\mathbf{A}}$ being a projection, $\|\Pi_{\mathbf{A}}\mathbf{b}\|_n \le \|\mathbf{b}\|_n$, which gives the triangle inequality of part (iii). Conditionally on the covariates the errors are independent, centered and of variance at most σ^2, so

$$\mathrm{E}(\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2 \mid \mathbf{X}, \mathbf{U}) = \frac{1}{n}\operatorname{tr}\{\Pi_{\mathbf{A}}\operatorname{Cov}(\boldsymbol{\varepsilon} \mid \mathbf{X}, \mathbf{U})\} \le \frac{\sigma^2}{n}\operatorname{tr}(\Pi_{\mathbf{A}}) \le \frac{\sigma^2 p}{n},$$

since the diagonal entries of $\Pi_{\mathbf{A}}$ are nonnegative and its trace is the rank of $\mathbf{A}$. □

Remark S8.1 (Why there is no cone condition)**.** The usual treatment of unpenalized coordinates (Bühlmann and van de Geer, 2011, Section 6.9) places them inside the support and requires compatibility on the cone $\|\delta_{S^c}\|_1 \le 3\|\delta_S\|_1$. Profiling removes them from the problem, and (S5.2)

transfers the full smallest eigenvalue of $\widehat{\Sigma}$ to the residualized block, so that Proposition 2(ii) is used as it stands and the constant does not depend on S_0. If only the restricted version of the design condition were available, the cone would return, and with the constant 4 rather than 3, since Step 5 produces $\left\|v_{S_0^c}\right\|_1 \leq 4\left\|v_{S_0}\right\|_1$: the bias consumes part of the slack.

Corollary S8.3 (The two rates with the design condition instantiated)**.** *Add Assumptions 2 to 5 and suppose* $2^J \log d/n \to 0$*, which is implied by the requirement of Proposition 2(ii). Take* $\lambda = 2\sigma B_X\sqrt{2C_U}\{2\log(2d/\alpha)/n\}^{1/2}$*. Then, with probability at least* $1-\alpha-\alpha'-\alpha''$ *up to terms that tend to zero,*

$$\left\|\widehat{f}-f\right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 c_U}\,\frac{s_0\log(pqN_J)}{n} + \frac{\mathcal{B}_J^2}{\alpha'} + \frac{\sigma^2 p}{n\alpha''},$$

and, with no condition on the design, the same bound with $\lambda\left\|\boldsymbol{\theta}^*\right\|_1$ *of Lemma S5.4 in place of the first term.*

Proof. Intersect four events: $\mathcal{T}$, of probability at least $1-\alpha$, by Lemma S8.2; $\{\widehat{\sigma}_{\max}^2 \leq 2B_X^2 C_U\}$, by Lemma S5.3, which makes the stated λ admissible, that is $\lambda \geq 2\lambda_0$; $\{\lambda_{\min}(\widehat{\Sigma}) \geq \gamma/2\}$, by Proposition 2(ii), which with (S5.2) gives $\widetilde{\gamma} \geq \gamma/2$; and $\{\|\mathbf{b}\|_n^2 \leq \mathcal{B}_J^2/\alpha'\}$, which is Markov's inequality applied to $\mathrm{E}\,\|\mathbf{b}\|_n^2 = \|f - f_J\|_{L_2(P)}^2 \leq \mathcal{B}_J^2$, the identity holding because the observations are identically distributed. Substituting in parts (ii) and (iii) of Theorem S8.1 gives the statement, and part (i) with Lemma S5.4 gives the

unconditional version. □

The first lemma is Theorem S8.1 read at an arbitrary comparator. It consumes no new step: the proof of that theorem uses the oracle only through the decomposition of $\mathbf{Y}$ and through the size of its support.

Lemma S8.3 (Theorem S8.1 with an arbitrary comparator)**.** *Under Assumption 1, with* $\mathbf{A}$ *of rank* p *and* $\lambda \geq 2\lambda_0$*, on the event* $\mathcal{T}$ *the following hold simultaneously for every* $\bar{\boldsymbol{\beta}} = (\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}})$*, with* $\bar{\mathbf{b}} = f - \mathbf{Z}\bar{\boldsymbol{\beta}}$*,* $\bar{M} = |\operatorname{supp}(\bar{\boldsymbol{\theta}})|$ *and* $\bar{v} = \widehat{\boldsymbol{\theta}} - \bar{\boldsymbol{\theta}}$*:* $\frac{1}{2}\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 + \lambda\left\|\widehat{\boldsymbol{\theta}}\right\|_1 \leq 3\lambda\left\|\bar{\boldsymbol{\theta}}\right\|_1 + 2\left\|\bar{\mathbf{b}}\right\|_n^2$*; if* $\widetilde{\gamma} > 0$*,* $\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \leq 64\lambda^2\bar{M}/\widetilde{\gamma} + 16\left\|\bar{\mathbf{b}}\right\|_n^2$ *and* $\|\bar{v}\|_2^2 \leq 64\lambda^2\bar{M}/\widetilde{\gamma}^2 + 16\left\|\bar{\mathbf{b}}\right\|_n^2/\widetilde{\gamma}$*; and*

$$\left\|\widehat{f} - f\right\|_n^2 \;\leq\; 3\min_{\bar{\boldsymbol{\beta}}}\left\{\frac{64\lambda^2\bar{M}}{\widetilde{\gamma}} + 20\left\|\bar{\mathbf{b}}\right\|_n^2\right\} + 3\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2. \tag{S8.10}$$

Proof. Follow the proof of Theorem S8.1 with $(\bar{\mathbf{c}}, \bar{\boldsymbol{\theta}}, \bar{\mathbf{b}}, \operatorname{supp}(\bar{\boldsymbol{\theta}}), \bar{M}, \bar{v})$ in place of $(\mathbf{c}, \boldsymbol{\theta}^*, \mathbf{b}, S_0, s_0, v)$. Step 1 uses only $\mathbf{Y} = \mathbf{A}\bar{\mathbf{c}} + \mathbf{B}\bar{\boldsymbol{\theta}} + \bar{\mathbf{b}} + \boldsymbol{\varepsilon}$, which is the definition of $\bar{\mathbf{b}}$, and $M_{\mathbf{A}}\mathbf{A} = 0$. Step 2 uses $\mathcal{T}$, which does not depend on the comparator, and $\left\|M_{\mathbf{A}}\bar{\mathbf{b}}\right\|_n \leq \left\|\bar{\mathbf{b}}\right\|_n$. Steps 3 to 5 are algebra with $\operatorname{supp}(\bar{\boldsymbol{\theta}})$ in place of S_0, using $\bar{\boldsymbol{\theta}}_{\operatorname{supp}(\bar{\boldsymbol{\theta}})^c} = 0$, the Cauchy-Schwarz inequality and (S5.2). Step 6 gives $\widehat{f} - \mathbf{Z}\bar{\boldsymbol{\beta}} = \widetilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \boldsymbol{\varepsilon})$ and hence $\widehat{f} - f = \widetilde{\mathbf{B}}\bar{v} + \Pi_{\mathbf{A}}(\bar{\mathbf{b}} + \boldsymbol{\varepsilon}) - \bar{\mathbf{b}}$. Since $\mathcal{T}$, $\widetilde{\gamma}$ and the condition on λ do not involve $\bar{\boldsymbol{\beta}}$, the inequalities hold simultaneously on the same event and the minimum may be taken; (S8.10) follows from $(x + y + z)^2 \leq 3(x^2 + y^2 + z^2)$. □

Proof of Lemma S8.1

The envelope. Fix $\epsilon > 0$ and count $N(\epsilon) = \#\{a : |\theta^*_a| > \epsilon\}$. Inside a block and a level, Markov's inequality in the ℓ_π norm gives $\#\{k : |\theta^*_{\ell m,jk}| > \epsilon\} \le \epsilon^{-\pi} \left\|\theta^*_{\ell m,j\cdot}\right\|_\pi^\pi$, and Assumption 5 gives $\left\|\theta^*_{\ell m,j\cdot}\right\|_\pi^\pi \le C_g^\pi 2^{-j\pi(s+1/2-1/\pi)} = C_g^\pi 2^{j(1-a)}$ with $a = \pi(s+1/2) > 1$. Since a level has 2^j coordinates,

$$N_{\ell m}(\epsilon) \;\le\; \sum_{j\ge 0} \min\{2^j,\, C_g^\pi \epsilon^{-\pi} 2^{j(1-a)}\}.$$

The two arguments agree at $2^{j_\epsilon a} = (C_g/\epsilon)^\pi$, that is at $2^{j_\epsilon} = (C_g/\epsilon)^\tau$, since $\pi/a = (s+1/2)^{-1} = \tau$. Below that level the minimum is 2^j and the sum is at most 2^{j_ϵ}; above it the minimum is the second term, which decreases geometrically with ratio $2^{1-a} < 1$, and the sum is $2^{j_\epsilon}/(1-2^{1-a})$. Adding the two parts and the pq blocks gives $N(\epsilon) \le pqA_{s,\pi}(C_g/\epsilon)^\tau$; truncating at $j < J$ only decreases the count, so the bound holds for every J. Finally, $\epsilon < \theta^*_{(k)}$ implies $N(\epsilon) \ge k$, hence $\epsilon \le C_g(pqA_{s,\pi}/k)^{1/\tau}$, and letting ϵ increase to $\theta^*_{(k)}$ gives the envelope; and $\tau = (s+1/2)^{-1} < 2$ for every $s > 0$.

The tail. With $2/\tau > 1$, $\sum_{k>M} \theta^2_{(k)} \le C_\tau^2 \sum_{k>M} k^{-2/\tau} \le C_\tau^2 \int_M^\infty x^{-2/\tau} dx = C_\tau^2 M^{1-2/\tau}\tau/(2-\tau)$. □

Proof of Proposition S8.1

By parts (i) and (iii) of Theorem S8.1, on $\mathcal{T}$, $\left\|\widetilde{\mathbf{B}}v\right\|_n^2 \le 6\lambda\left\|\boldsymbol{\theta}^*\right\|_1 + 4\left\|\mathbf{b}\right\|_n^2$ and $\left\|\widehat{f}-f\right\|_n^2 \le 3(\left\|\widetilde{\mathbf{B}}v\right\|_n^2 + 4\left\|\mathbf{b}\right\|_n^2 + \left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2)$, and no property of $\widehat{\Sigma}$ enters. Lemma S5.3 makes λ_n admissible with probability tending to one, since $2^{J_n}\log d_n/n \to 0$ in both regimes, and it is for this that the conditions $s' > 1/4$ in (i) and $s' > s/2$ in (ii) are needed: without them the choice in (i) gives $2^{J_n} \ge n^{1/(4s')} \ge n$, and the balance in (ii) gives $2^{J_n} \gtrsim n/\log n$, and parts (iii) and (iv) cover those pairs with λ_n^+; Markov's inequality applied to $\mathrm{E}\left\|\mathbf{b}\right\|_n^2 \le \mathcal{B}_{J_n}^2$, from Lemma 2, and to $\mathrm{E}(\left\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\right\|_n^2 \mid \mathbf{X},\mathbf{U}) \le \sigma^2 p/n$ gives the general statement. If $s > 1/2$, Lemma S5.4 gives $\left\|\boldsymbol{\theta}^*\right\|_1 = O(1)$ and $\lambda_n\left\|\boldsymbol{\theta}^*\right\|_1 \asymp (J_n/n)^{1/2}$, and it suffices that the bias not dominate, that is $2^{-2J_n s'} \le (J_n/n)^{1/2}$, which holds for $J_n \ge \log_2(n/J_n)/(4s')$. If $s < 1/2$, then $\left\|\boldsymbol{\theta}^*\right\|_1 \asymp 2^{J_n(1/2-s)}$, the two terms are $(J_n/n)^{1/2}2^{J_n(1/2-s)}$ and $2^{-2J_n s'}$, and equating them gives $2^{J_n} \asymp (n/J_n)^{1/(1-2s+4s')}$ with common value $(J_n/n)^{2s'/(1-2s+4s')}$; when $s' = s$ the denominator is $1+2s$.

The level above the ceiling, for (iii) and (iv). By (S6.6), $\mathrm{P}(\widehat{\sigma}^2_{\max} > 2B_X^2 C_U \mu_J) \le 1/d$ for every J. Off that event,

$$2\lambda_0 \;\le\; 2\sigma(2B_X^2 C_U \mu_J)^{1/2}\Big\{\frac{2\log(2d/\alpha)}{n}\Big\}^{1/2} \;=\; \lambda_n^+,$$

and the argument of the general statement goes through with λ_n^+ and no

condition on J_n: on $\mathcal{T}$, $\left\|\widehat{f}-f\right\|_n^2 \le 18\lambda_n^+ \|\boldsymbol{\theta}^*\|_1 + 24\,\|\mathbf{b}\|_n^2 + 3\,\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$, and Markov's inequality concludes. In every choice of (iii) and (iv), $2^{J_n} \le n^2$, so that $\log d_n \asymp J_n \asymp \log n$ and

$$\lambda_n^+ \;\asymp\; \Big(\frac{\log n}{n}\Big)^{1/2}\Big(1+\frac{2^{J_n}\log n}{n}\Big)^{1/2}, \tag{S8.11}$$

in particular $\lambda_n^+ \asymp 2^{J_n/2}\log n/n$ if $2^{J_n} \gtrsim n/\log n$.

Part (iii)(a). With $s > 1/2$, $\|\boldsymbol{\theta}^*\|_1 = O(1)$ as in (i). With $2^{J_n} \asymp (n/\log n)^{2/(1+4s')} \gtrsim n/\log n$, because $s' \le 1/4$, (S8.11) gives $\lambda_n^+\|\boldsymbol{\theta}^*\|_1 \asymp (\log n/n)(n/\log n)^{1/(1+4s')} = (\log n/n)^{4s'/(1+4s')}$, and the bias is $2^{-2J_n s'} \asymp (\log n/n)^{4s'/(1+4s')}$; p/n is of smaller order.

Part (iii)(b). With $s < 1/2$, $\|\boldsymbol{\theta}^*\|_1 \asymp 2^{J_n(1/2-s)}$ as in (ii). With $2^{J_n} \asymp (n/\log n)^{1/(1-s+2s')} \gtrsim n/\log n$, because $s' \le s/2$, (S8.11) gives $\lambda_n^+\|\boldsymbol{\theta}^*\|_1 \asymp 2^{J_n(1-s)}\log n/n \asymp (\log n/n)^{1-(1-s)/(1-s+2s')} = (\log n/n)^{2s'/(1-s+2s')}$, which is also the order of the bias $2^{-2J_n s'}$. At the boundaries, $s' = 1/4$ in (a) and $s' = s/2$ in (b), $2^{J_n} \asymp n/\log n$, μ_{J_n} is bounded, and the rates are $(\log n/n)^{1/2}$ and $(\log n/n)^s$, those of (i) and (ii) at these points.

Part (iv). With $s = 1/2$, Lemma S5.4 gives $\|\boldsymbol{\theta}^*\|_1 \le pqC_gJ_n$, and $J_n \asymp \log n$ in both choices. If $s' \ge 1/4$, $2^{J_n} \asymp (n/\log n)^{1/(4s')} \lesssim n/\log n$, μ_{J_n} is bounded, $\lambda_n^+\|\boldsymbol{\theta}^*\|_1 \asymp (\log n/n)^{1/2}\log n$, and the bias, $2^{-2J_n s'} \asymp (\log n/n)^{1/2}$, is of smaller order. If $s' < 1/4$, $2^{J_n} \asymp \{n/(\log n)^2\}^{2/(1+4s')}$, which exceeds $n/\log n$ for n large, and $\lambda_n^+\|\boldsymbol{\theta}^*\|_1 \asymp 2^{J_n/2}(\log n)^2/n \asymp \{(\log n)^2/n\}^{4s'/(1+4s')}$,

the order of the bias. □

Remark S8.2 (Why the resolution passes the ceiling)**.** In the pairs of (iii) the bound $\lambda_n \|\boldsymbol{\theta}^*\|_1 + \mathcal{B}_J^2$ is dominated by the bias at every J below the ceiling, and the best that can be done there is to stop at it: $2^{J_n} = n/(K \log n)$ with $K > 4\|\psi\|_\infty^2/(3C_U)$ fixed keeps λ_n admissible, because the exponent of Lemma S5.3 at $t = B_X^2 C_U$ is then $-\{3C_U K/(4\|\psi\|_\infty^2)\}\log n$ and $d_n \le pqn$, and gives $O_p\{(\log n/n)^{2s'}\}$. Above the ceiling λ_n^+ pays for the growth of $\widehat{\sigma}_{\max}$, and the estimation term, $2^{J/2}(\log n/n)\|\boldsymbol{\theta}^*\|_1$, grows more slowly than the bias decreases until J reaches the J_n of (iii). The exponents of (iii), $4s'/(1+4s')$ and $2s'/(1-s+2s')$, are strictly larger than $2s'$ away from $s' = 1/4$ and $s' = s/2$, and the J_n of (iii) minimizes the bound over J up to a constant, so that this is the best bound that the argument gives. In (iv) with $s' \ge 1/4$, the factor $(\log n)^{3/2}$ comes from $\|\boldsymbol{\theta}^*\|_1 \asymp J_n \asymp \log n$, which no choice avoids.

Proof of Proposition S8.2

Part (i). Lemma S8.3 with $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \bar{\boldsymbol{\theta}})$ gives, discarding $\lambda\left\|\widehat{\boldsymbol{\theta}}\right\|_1 \ge 0$, $\left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n^2 \le 6\lambda\|\bar{\boldsymbol{\theta}}\|_1 + 4\|\bar{\mathbf{b}}\|_n^2$, and Step 6 in its proof gives $\left\|\widehat{f} - f\right\|_n \le \left\|\widetilde{\mathbf{B}}\bar{v}\right\|_n + 2\|\bar{\mathbf{b}}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n$; with $(x+y+z)^2 \le 3(x^2+y^2+z^2)$, $\left\|\widehat{f} - f\right\|_n^2 \le 18\lambda\|\bar{\boldsymbol{\theta}}\|_1 + 24\|\bar{\mathbf{b}}\|_n^2 + 3\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$. Since $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})$, $\|\bar{\mathbf{b}}\|_n^2 \le 2\|\mathbf{b}\|_n^2 +$

$2\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2$. If t is deterministic, so is $u = \boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}}$, and $\mathrm{E}\,\|\mathbf{B}u\|_n^2 \le \lambda_{\max}(\Sigma)\,\|u\|_2^2 \le \kappa_2 C_U\,\|u\|_2^2$, as in the proof of Proposition S6.2. With $t = \lambda$, on the coordinates with $|\theta_a^*| > \lambda$ the minimum in (S8.4) is $\lambda|\theta_a^*|$, and on the others it is θ_a^{*2}; summing gives the bound by $\max(18, 48\kappa_2 C_U)\mathcal{R}_1(\boldsymbol{\theta}^*; \lambda)$. The form in O_p is that of the proof of Proposition S8.1: the event $\mathcal{T}$ with $\lambda_n \ge 2\lambda_0$, by Lemma S5.3 below the ceiling, or with $\lambda_n^+ \ge 2\lambda_0$, by (S6.6) for every J, and Markov's inequality for $\|\mathbf{b}\|_n^2$, $\left\|\mathbf{B}(\boldsymbol{\theta}^* - \bar{\boldsymbol{\theta}})\right\|_n^2$ and $\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2$.

Part (ii), $s < 1/2$. Fix a block and a level j, write $\theta_j \in \mathbb{R}^{2^j}$ for its coefficients, with $\|\theta_j\|_\pi \le C_g 2^{-j(s+1/2-1/\pi)}$ by Assumption 5, and let $v_j = \sum_k \min(t|\theta_{jk}|, \theta_{jk}^2)$. The first bound is in the ℓ_1 norm: by Hölder's inequality in $\mathbb{R}^{2^j}$, $v_j \le t\,\|\theta_j\|_1 \le t 2^{j(1-1/\pi)}\,\|\theta_j\|_\pi \le t C_g 2^{j(1/2-s)} =: A_j$. The second is the tail, as in the proof of Proposition S6.2(ii) with coordinates in place of chunks: $v_j \le t^{2-\pi}\,\|\theta_j\|_\pi^\pi \le t^{2-\pi} C_g^\pi 2^{-j\pi s'}$ if $\pi \le 2$, and $v_j \le \|\theta_j\|_2^2 \le C_g^2 2^{-2js}$ if $\pi \ge 2$; call it B_j. With

$$2^x = (C_g/t)^\tau, \qquad h = C_g^\tau t^{2-\tau} = C_g^\tau t^{4s/(2s+1)},$$

one has $A_j = h 2^{(j-x)(1/2-s)}$, because $1 + \tau(1/2-s) = \tau$, and $B_j = h 2^{-(j-x)\min(\pi,2)s'}$, because $\tau\pi s' = \pi - \tau$ and $2s\tau = 2 - \tau$. Summing A_j over $j \le x$ and B_j over $j > x$, two geometric series of ratios $2^{-(1/2-s)}$ and $2^{-\min(\pi,2)s'}$ that start from at most h, gives at most $h\{(1 - 2^{s-1/2})^{-1} + (1 - 2^{-\min(\pi,2)s'})^{-1}\}$ per block; truncating at $j < J$ only removes terms, and summing over the pq

blocks gives (S8.5). This is the computation of the proof of Lemma 3, with $\min(ty, y^2)$ in place of $\min(y^2, \eta)$.

Part (ii), $s \geq 1/2$. The first statement is $\min(t|y|, y^2) \leq t|y|$. For the second, the block with $\theta^*_{\ell m,00} = C_g$ and all its other coefficients zero satisfies Assumption 5, and gives $\mathcal{R}_1(\boldsymbol{\theta}^*; t) \geq \min(tC_g, C_g^2) = tC_g$. If $s = 1/2$, set, at each level $j < J$ with $2^j \leq C_g/t$, all the 2^j coefficients equal to $C_g 2^{-j} \geq t$; then $\|\theta_j\|_\pi = C_g 2^{-j(1-1/\pi)}$, the bound of Assumption 5 at $s = 1/2$, and each such level contributes $2^j t C_g 2^{-j} = tC_g$.

Part (iii). With $J_n = \lceil c \log_2 n \rceil$ and $c < 1$, $2^{J_n} \log d_n / n \asymp n^{c-1} \log n \to 0$, and (i) holds with $\lambda_n \asymp (\log n/n)^{1/2}$. By (S8.5), $\mathcal{R}_1(\boldsymbol{\theta}^*; \lambda_n) \lesssim (\log n/n)^{2s/(2s+1)}$. The lower end of the window (S8.6) is $s/\{(2s+1)s'\}$, because $(2-\tau)/(4s') = s/\{(2s+1)s'\}$ and $s' \leq s$, so that the bias is $\mathcal{B}^2_{J_n} \lesssim 2^{-2J_n s'} \leq n^{-2cs'} \leq n^{-2s/(2s+1)}$; and p/n is of smaller order.

Part (iv). Write $\eta = 2s/\{s + s'(2s+1)\}$, which lies in $[1, 2)$ exactly when $s' \leq s/(2s+1)$; then $2^{J_n} \asymp (n/\log n)^\eta \gtrsim n/\log n$, $2^{J_n} \leq n^2$ and, by (S8.11), $\lambda_n^+ \asymp 2^{J_n/2} \log n/n \asymp (\log n/n)^{1-\eta/2}$. By (S8.5), $\mathcal{R}_1(\boldsymbol{\theta}^*; \lambda_n^+) \lesssim (\log n/n)^{(1-\eta/2)4s/(2s+1)}$, and $(1-\eta/2)4s/(2s+1) = 4ss'/\{s+s'(2s+1)\}$; the bias is $2^{-2J_n s'} \asymp (\log n/n)^{2s'\eta}$, with the same exponent. At $s' = s/(2s+1)$, $\eta = 1$ and the exponent is $2s/(2s+1)$. Outside the window of (iii), $\pi < 2$, because $\pi \geq 2$ gives $s' = s > s/(2s+1)$. □

Proof of Theorem S8.2

By the definition of J_n, $(n/\log n)^{1/(2s'+1)} \le 2^{J_n} < 2(n/\log n)^{1/(2s'+1)}$ and $J_n \asymp \log n$, so

$$\frac{pq2^{J_n}\log(pq2^{J_n})}{n} \asymp \frac{(n/\log n)^{1/(2s'+1)}\log n}{n} = \Big(\frac{\log n}{n}\Big)^{2s'/(2s'+1)} \longrightarrow 0,$$

which is the requirement of Proposition 2(ii); (S5.2) then gives $\widetilde{\gamma} \ge \gamma/2$ with probability tending to one. On that event Corollary S8.3 with $s_0 \le d_n$ gives

$$\left\|\widehat{f} - f\right\|_n^2 \lesssim \frac{\sigma^2 B_X^2 C_U}{\kappa_1 c_U}\frac{d_n L_n}{n} + \frac{\mathcal{B}_{J_n}^2}{\alpha'} + \frac{\sigma^2 p}{n\alpha''}.$$

With $d_n \asymp pq2^{J_n}$ and $L_n \asymp \log n$ the first term is the display above; the second is $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \le (\log n/n)^{2s'/(2s'+1)}$ by the lower bound on 2^{J_n}; and the third is $O(1/n)$, of strictly smaller order because $2s'/(2s'+1) < 1$. □

Proof of Corollary S8.1

Inside a block the wavelets of level below J are orthonormal in $L_2[0,1]$, so $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]}^2 = \sum_{jk} v_{\ell m,jk}^2$, and summing over the pq blocks gives $\|v\|_2^2$; adding the bias of Lemma 2 in each block,

$$\sum_{\ell,m}\|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}^2 \le 2\|v\|_2^2 + \frac{2pqC_g^2}{1-2^{-2s'}}2^{-2Js'}.$$

Theorem S8.1(ii) bounds $\|v\|_2^2$ by $64\lambda^2 s_0/\widetilde{\gamma}^2 + 16\|\mathbf{b}\|_n^2/\widetilde{\gamma}$, and on the event of Theorem S8.2 the quantity $\widetilde{\gamma}$ is bounded below by the constant $\gamma/2$,

so the two terms are of the same order as those of Theorem S8.2, the only difference being a constant with γ^{-2} in place of γ^{-1}. This is where the design condition is used a second time. For the levels, Lemma S5.1 gives $\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c}) = \Pi_{\mathbf{A}}(\mathbf{b} + \boldsymbol{\varepsilon}) - \Pi_{\mathbf{A}}\mathbf{B}v$, so that $\|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n \leq \|\mathbf{b}\|_n + \|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n + \lambda_{\max}(\widehat{\Sigma})^{1/2}\|v\|_2$, while $\mathbf{A}^\top\mathbf{A}/n$ is a principal submatrix of $\widehat{\Sigma}$, whence $\|\widehat{\mathbf{c}} - \mathbf{c}\|_2^2 \leq \|\mathbf{A}(\widehat{\mathbf{c}} - \mathbf{c})\|_n^2/(\gamma/2)$. Finally the components of a functional coefficient depend on distinct coordinates of u and the basis is orthonormal, so $\|\widehat{\beta}_\ell - \beta_\ell\|_{L_2([0,1]^q)} \leq |\widehat{c}_\ell - c_\ell| + \sum_m \|\widehat{g}_{\ell m} - g_{\ell m}\|_{L_2[0,1]}$, with q fixed. The same argument applies verbatim under the conditions of Theorem S8.3. □

Proof of Theorem S8.3

A nonasymptotic bound. Apply Lemma S8.3 with $\bar{\boldsymbol{\beta}} = (\mathbf{c}, \boldsymbol{\theta}^{*,(M)})$, the best M term approximation of the oracle. Then $\bar{M} \leq M$ and $\bar{\mathbf{b}} = \mathbf{b} + \mathbf{B}(\boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)})$, so that, by the Cauchy-Schwarz inequality, Lemma S8.1 and (S6.2),

$$\|\bar{\mathbf{b}}\|_n^2 \leq 2\|\mathbf{b}\|_n^2 + 2\lambda_{\max}(\widehat{\Sigma})\left\|\boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)}\right\|_2^2 \leq 2\|\mathbf{b}\|_n^2 + \frac{2\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

Substituting in (S8.10), on the intersection of $\mathcal{T}$ with $\{\lambda_{\min}(\widehat{\Sigma}) \geq \gamma/2\}$,

$$\left\|\widehat{f} - f\right\|_n^2 \leq 3 \min_{1\leq M\leq d} h(M) + 120\|\mathbf{b}\|_n^2 + 3\|\Pi_{\mathbf{A}}\boldsymbol{\varepsilon}\|_n^2, \qquad h(M) = \frac{64\lambda^2}{\widetilde{\gamma}} M + \frac{40\Lambda C_\tau^2 \tau}{2-\tau} M^{1-2/\tau}.$$

The function h is strictly convex on the positive half line, since $1 - 2/\tau < 0$, and h' vanishes at $M^\star = \{5\Lambda C_\tau^2 \widetilde{\gamma}/(8\lambda^2)\}^{\tau/2}$, where $h(M^\star) = 2(2 -$

$\tau)^{-1}(64\lambda^2/\widetilde{\gamma})M^\star$; moreover $h(\lceil M^\star \rceil) \le h(M^\star) + 64\lambda^2/\widetilde{\gamma} \le 2h(M^\star)$ whenever $M^\star \ge 1$, because $(2-\tau)/2 < 1$.

The resolution. With $\lambda_n^2 \asymp \sigma^2 B_X^2 C_U J_n/n$, $\widetilde{\gamma} \in [\gamma/2, \Lambda]$ and the other constants fixed, $M_n^\star \asymp \lambda_n^{-\tau} \asymp (n/J_n)^{\tau/2}$ and $h(M_n^\star) \asymp \lambda_n^2 M_n^\star \asymp (J_n/n)^{(2-\tau)/2}$. Writing $J_n = \lceil c \log_2 n \rceil$, so that $2^{J_n} \asymp n^c$, the three inequalities of the window (S8.6) are, in order: $M_n^\star \le d_n$, which asks $(n/J_n)^{\tau/2} \lesssim pq\, n^c$, that is $c \ge \tau/2$ up to logarithms; that the bias not dominate, $n^{-2cs'} \lesssim (J_n/n)^{(2-\tau)/2}$, that is $c \ge (2-\tau)/(4s')$; and the requirement of Proposition 2(ii), $n^{c-1}\log n \to 0$, that is $c < 1$. The term $\sigma^2 p/n$ is of smaller order because $(2-\tau)/2 < 1$. The window is nonempty if and only if $\tau/2 < 1$, which always holds, and $(2-\tau)/(4s') < 1$, that is $s' > (2-\tau)/4$.

Under Assumption 5 the hypothesis is automatic. By Lemma S8.1 the envelope holds with $\tau = (s+1/2)^{-1}$, whence $2-\tau = 4s/(2s+1)$, $(2-\tau)/2 = 2s/(2s+1)$ and $(2-\tau)/4 = s/(2s+1)$, which is the condition $s' > s/(2s+1)$ of the statement; for $\pi \ge 2$ it holds for every $s > 0$, and for $\pi < 2$ it reads $2s^2/(2s+1) > 1/\pi - 1/2$. The rate for the components is the argument of Corollary S8.1 with $\left\|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}^*\right\|_2 \le \|\bar{v}\|_2 + \left\|\boldsymbol{\theta}^* - \boldsymbol{\theta}^{*,(M)}\right\|_2$ and Lemma S8.3, the only difference being an extra factor $\widetilde{\gamma}^{-1}$ in the first term, which does not change the order. Finally $2s/(2s+1) \ge 2s'/(2s'+1)$, with equality only when $s' = s$, by monotonicity of $x \mapsto 2x/(2x+1)$ and $s' \le s$, so the

rate is never worse than that of Theorem S8.2 and is strictly better when $\pi < 2$. □

Remark S8.3 (What compressibility buys)**.** Read at the oracle, Theorem S8.1(ii) has $s_0 = d$ and delivers the rate of linear approximation; read at a sparse comparator, through Lemma S8.3, it delivers the rate of Theorem S8.3. It is the same inequality in both cases, and neither says anything about $\mathrm{supp}(\widehat{\boldsymbol{\theta}})$: no minimal signal and no irrepresentability condition is used, and what is gained is a rate governed by the effective dimension $M_n^\star$ rather than by d_n.

Proof of Corollary S8.2

A nonasymptotic bound on Δ. Work at $J = J_n$ and $\lambda = \lambda_n$, and intersect the four events of the proof of Corollary S8.3: $\mathcal{T}$, of probability at least $1-\alpha$, by Lemma S8.2; $\{\widehat{\sigma}^2_{\max} \le 2B_X^2 C_U\}$, by Lemma S5.3, which makes λ_n admissible; $\{\lambda_{\min}(\widehat{\Sigma}) \ge \gamma/2\}$, by Proposition 2(ii), which with (S5.2) gives $\widetilde{\gamma} \ge \gamma/2$; and $\{\|\mathbf{b}\|_n^2 \le \mathcal{B}_{J_n}^2/\alpha'\}$, by Markov's inequality and Lemma 2. On this event Theorem S8.1(ii) with $s_0 \le d_n$ gives

$$\|v\|_2^2 \;\le\; \frac{64\lambda_n^2 d_n}{\widetilde{\gamma}^2} + \frac{16\,\|\mathbf{b}\|_n^2}{\widetilde{\gamma}} \;\le\; \frac{256\lambda_n^2 d_n}{\gamma^2} + \frac{32\mathcal{B}_{J_n}^2}{\alpha'\gamma}.$$

For a fixed block, the orthonormality of the wavelets of level below J_n gives $\|\widehat{g}_{\ell m} - \Pi_J g_{\ell m}\|_{L_2[0,1]} = \|v_{\ell m,\cdot}\|_2 \le \|v\|_2$, and the first display of Lemma

2 bounds $\|\Pi_J g_{\ell m} - g_{\ell m}\|^2_{L_2[0,1]}$ by $C_g^2(1-2^{-2s'})^{-1}2^{-2J_n s'}$. By $(x+y)^2 \le 2x^2+2y^2$,

$$\Delta^2 \;\le\; \bar{\Delta}_n^2 \;=\; \frac{512\,\lambda_n^2 d_n}{\gamma^2} + \frac{64\,\mathcal{B}_{J_n}^2}{\alpha'\gamma} + \frac{2C_g^2}{1-2^{-2s'}}\,2^{-2J_n s'} \tag{S8.12}$$

with probability at least $1-\alpha-\alpha'-\mathrm{P}\{\lambda_{\min}(\widehat{\Sigma}) < \gamma/2\}-\mathrm{P}\{\widehat{\sigma}^2_{\max} > 2B_X^2 C_U\}$, the last two terms tending to zero by Proposition 2(ii) and Lemma S5.3; their sum is the ω_n of Corollary S8.2(iii), and this paragraph is the proof of that part. By Lemma S7.1(iii), on that event $\widehat{\mathcal{S}}(t) = \mathcal{S}$ for every t with $\bar{\Delta}_n \le t < \nu_{\min} - \bar{\Delta}_n$, a window that is nonempty whenever $\nu_{\min} > 2\bar{\Delta}_n$. The bound $\bar{\Delta}_n$ depends on the components only through the constants of Assumption 5, and the two residual probabilities do not depend on them at all, so everything above holds uniformly over the class, which is what allows the target to vary with n in Assumption 6.

The order of $\bar{\Delta}_n$. At the resolution of Theorem S8.2 the three terms of (S8.12) are at most of order ρ_n^2: the first because $\lambda_n^2 d_n \asymp pq2^{J_n}J_n/n$, which is the quantity shown to be of order $(\log n/n)^{2s'/(2s'+1)}$ in the proof of Theorem S8.2, and the other two because $\mathcal{B}_{J_n}^2 \asymp 2^{-2J_n s'} \le (\log n/n)^{2s'/(2s'+1)}$ by the definition of J_n. Hence $\bar{\Delta}_n \le c(\alpha,\alpha')\rho_n$ with $c(\alpha,\alpha') < \infty$. Given $\epsilon > 0$, take $\alpha = \alpha' = \epsilon/4$ and n large enough that the two residual probabilities add up to less than $\epsilon/2$; then $\mathrm{P}\{\Delta > c(\epsilon/4,\epsilon/4)\rho_n\} < \epsilon$, that is $\Delta = O_p(\rho_n)$.

Part (i). If $t_n/\rho_n \to \infty$, then $\mathrm{P}(\Delta \le t_n) \to 1$, and on that event Lemma S7.1(i) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) \subseteq \mathcal{S}$. No property of $\nu_{\min,n}$ has been used.

Part (ii). If $\mathcal{S}$ is empty, part (ii) is part (i). Otherwise, the condition $\limsup_n t_n/\nu_{\min,n} < 1$ gives $\eta > 0$ and n_0 with $\nu_{\min,n} - t_n \ge \eta\,\nu_{\min,n}$ for $n \ge n_0$, and Assumption 6 with ρ_n in place of $\rho_n^{\mathcal{G}}$ gives $(\nu_{\min,n} - t_n)/\rho_n \ge \eta\,\nu_{\min,n}/\rho_n \to \infty$. Hence $\mathrm{P}(\Delta \le t_n) \to 1$ and $\mathrm{P}(\Delta < \nu_{\min,n} - t_n) \to 1$, and on the intersection of the two events Lemma S7.1(iii) with $\bar{\Delta} = \Delta$ gives $\widehat{\mathcal{S}}(t_n) = \mathcal{S}$. For the existence, $t_n = (\rho_n \nu_{\min,n})^{1/2}$ has $t_n/\rho_n = (\nu_{\min,n}/\rho_n)^{1/2} \to \infty$ and $t_n/\nu_{\min,n} = (\rho_n/\nu_{\min,n})^{1/2} \to 0$. □

At the resolution of Theorem S8.3. Theorem S8.3 gives $\sum_{\ell,m} \|\widehat{g}_{\ell m} - g_{\ell m}\|^2_{L_2[0,1]} = O_p(\tilde{\rho}_n^2)$ with $\tilde{\rho}_n = (\log n/n)^{s/(2s+1)}$, hence $\Delta = O_p(\tilde{\rho}_n)$, and the bound in its proof depends on the components only through the constants of Assumption 5, through C_τ of Lemma S8.1 and the design constants, so that this holds uniformly over the class. The proofs of parts (i) and (ii) above use nothing about Δ beyond $\Delta = O_p(\rho_n)$ uniformly over the class, together with Lemma S7.1 and Assumption 6, and they go through verbatim with $\tilde{\rho}_n$ in place of ρ_n.

S9 The simulation study: details and further results

This section completes Section 5 of the paper.

Test functions

Before they are centered and scaled to unit norm in $L_2[0,1]$, the two components of the uneven cell whose scale changes along the interval are

$$w(u)\Big[\exp\Big\{-\frac{(u-0.40)^2}{2(0.10)^2}\Big\}-0.8\exp\Big\{-\frac{(u-0.72)^2}{2(0.018)^2}\Big\}\Big] \quad \text{and} \quad w(u)\sin\{2\pi(u+3.5u^2)\},$$

with the window $w(u)=\eta(u/0.08)\,\eta\{(1-u)/0.08\}$, where $\eta(t)=e^{-1/t}/(e^{-1/t}+e^{-1/(1-t)})$ for $0<t<1$, $\eta(t)=0$ for $t\le 0$ and $\eta(t)=1$ for $t\ge 1$. The window equals one on $[0.08,0.92]$ and vanishes with all its derivatives at 0 and 1, so that the periodic extensions of the two components are infinitely differentiable and the cell measures uneven curvature and not a seam at $0\equiv 1$. The bumps, blocks and heavisine functions are those of Donoho and Johnstone (1994), with their constants, on $[0,1]$. The centering and the scaling constants are computed on 2^{16} midpoints of $[0,1]$.

The larger model

With $(p,q)=(6,4)$ the levels are $(1,2,-1.5,0.5,1,2)$, and the nonzero components are g_{11} (bumps), g_{12} (blocks) and g_{21} (heavisine), as in the inhomogeneous cell, and g_{33} (bumps), g_{34} (blocks) and g_{43} (heavisine); the coefficients β_5 and β_6 are constant. The grid of J stops at 7, since at $J=8$ the design has $6\times 4\times 255=6120$ penalized columns for 1000 observations.

The grids of the splines

In either search of `mgcv` a dimension above 80 whose smooths would have more than $2n$ coefficients is not fitted, which stops the grid at 120 in the mixed cell at $n = 1000$ and at 80 in the larger model, and a fit by REML whose smoothing parameters do not converge within 80 outer iterations leaves the search. At $n = 1000$ the grid of GAM (REML) stops at 160 because the fit at 240 did not converge in about a third of the replicates of the inhomogeneous cell, and in two of them did not finish in 15 hours; where it was fitted, it would lower the error of GAM (REML) by 0.3% in that cell and by 1.2% in the uneven one.

Further results

~~Table S1 gives the errors of every row at every sample size, together with the coordinatewise lasso of Section 4 of the paper, fitted at the pair chosen by cross-validation (Lasso) and thresholded by the one standard error rule (Lasso + threshold); Table S6 gives the recovery of the structure; Table S8 the arms at $n = 1000$ and the larger model; and Table S9 the resolution or dimension chosen, the fraction of replicates at the top of the grid, the time, and the slope of the logarithm of the mean squared errors against $\log n$.~~ Tables S1 to S5 give, one cell each, the errors and the frequency of

the exact structure of every row at every sample size, together with the coordinatewise lasso of Section 4 of the paper, fitted at the pair chosen by cross-validation (Lasso) and thresholded by the one standard error rule (Lasso + threshold); Table S6 gives the null blocks kept at $n = 1000$ and Table S7 the paired comparisons; Table S8 the arms at $n = 1000$ and the larger model; Tables S9 and S10 the resolution or dimension chosen, the fraction of replicates at the top of the grid and the time; and Table S11 the slope of the logarithm of the mean squared error against $\log n$.

The coordinatewise lasso with the threshold recovers the same structure as the WAFC from $n = 1000$, and less often at $n = 500$ in the mixed cell (69% against 84%). Its prediction error is 3% to 11% above that of the WAFC in the inhomogeneous and mixed cells at every sample size and 2% to 5% above it in the uneven cell, within 1% in the null cell, and in the smooth cell 7% below it at $n = 250$ and within 4% at the other sizes; the blocks of the penalty pay where the components are inhomogeneous. The WAFC reaches the top of its grid of J in at most 24% of the replicates of a cell. GAM (GCV), whose grid is not extended because of the cost of its fits, chooses its top in at least 99% of the replicates of the inhomogeneous cell from $n = 500$. GAM (REML) chooses the top of its grid in 71% and 65% of the replicates of the inhomogeneous and mixed cells at $n = 500$, where

the grid stops at 80, in 16% and 97% at $n = 1000$, where the size rule stops it at 120 in the mixed cell, and in every replicate of both at $n = 2000$. The time of the WAFC grows from 4 to 10 seconds at $n = 250$ to 130 to 250 seconds at $n = 2000$; that of GAM (REML) is comparable except in the mixed cell, where it reaches 3800 seconds at $n = 2000$, and that of GAM (GCV) reaches 340 to 1000 seconds. The slopes of the WAFC, -0.71 in the inhomogeneous and mixed cells and -0.95 to -1.04 in the others, are within 0.06 of those of the oracle support. Figure S1 adds to Figure 1 of the paper the pointwise 5% and 95% quantiles over the replicates; the bands of the two methods have about the same width, 0.82 and 0.77 on average on the bumps and 0.73 for both on the blocks.

Table S1: Simulation, the smooth cell, 100 replicates per sample size: the root mean squared error of prediction, the integrated squared error of the components times 100, and the frequency of the exact structure. The constant coefficients have a prediction error of 1.73 in the cells with varying coefficients, and coincide with the oracle support in the null cell; GAM (GCV) is not run in the mixed cell, and the lasso rows are the coordinatewise lasso of Section 4 of the paper.

	Prediction error				Components (×100)				Exact structure			
n	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
WAFC	0.317	0.228	0.165	0.118	9.49	4.89	2.57	1.34	1.00	1.00	1.00	1.00
WAFC (cv)	0.327	0.235	0.169	0.121	10.18	5.30	2.72	1.40	0.65	0.70	0.71	0.77
Lasso	0.332	0.248	0.183	0.137	10.35	5.92	3.23	1.81	0.00	0.00	0.00	0.00
Lasso + threshold	0.296	0.225	0.167	0.123	8.16	4.80	2.65	1.46	1.00	1.00	1.00	1.00
GAM (REML)	0.257	0.183	0.131	0.101	5.96	3.05	1.60	0.94	0.34	0.30	0.21	0.28
GAM (GCV)	0.299	0.203	0.144	0.107	8.53	3.89	2.01	1.08	0.16	0.16	0.11	0.18
B-spline group lasso	0.357	0.267	0.208	0.145	12.08	6.89	4.25	2.06	0.05	0.00	0.00	0.01
Adaptive spline	0.358	0.224	0.159	0.111	12.39	4.82	2.48	1.18	0.00	0.00	0.00	0.00
Block lasso (KP)	0.322	0.236	0.175	0.124	9.58	5.11	2.86	1.45	0.00	0.00	0.00	0.00
Oracle support	0.287	0.212	0.157	0.114	7.73	4.25	2.32	1.23	1.00	1.00	1.00	1.00

Table S2: Simulation, the uneven cell, 100 replicates per sample size: the root mean squared error of prediction, the integrated squared error of the components times 100, and the frequency of the exact structure.

	Prediction error				Components (×100)				Exact structure			
n	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
WAFC	0.487	0.343	0.247	0.179	22.75	11.39	5.92	3.15	1.00	1.00	1.00	1.00
WAFC (cv)	0.497	0.348	0.251	0.181	23.71	11.73	6.11	3.23	0.71	0.79	0.72	0.79
Lasso	0.562	0.392	0.278	0.200	30.36	14.94	7.52	3.95	0.00	0.00	0.00	0.00
Lasso + threshold	0.513	0.361	0.254	0.182	25.24	12.63	6.26	3.26	1.00	1.00	1.00	1.00
GAM (REML)	0.418	0.310	0.224	0.158	16.50	9.11	4.82	2.44	0.30	0.33	0.36	0.21
GAM (GCV)	0.440	0.307	0.222	0.158	18.44	9.01	4.76	2.47	0.13	0.16	0.09	0.09
B-spline group lasso	0.628	0.444	0.308	0.214	37.87	19.28	9.47	4.63	0.02	0.00	0.00	0.00
Adaptive spline	0.626	0.337	0.250	0.195	83.91	11.23	6.42	3.96	0.00	0.00	0.00	0.00
Block lasso (KP)	0.535	0.364	0.263	0.189	26.84	12.60	6.65	3.46	0.00	0.00	0.00	0.00
Oracle support	0.466	0.324	0.234	0.170	20.87	10.12	5.28	2.85	1.00	1.00	1.00	1.00

Table S3: Simulation, the inhomogeneous cell, 100 replicates per sample size: the root mean squared error of prediction, the integrated squared error of the components times 100, and the frequency of the exact structure.

	Prediction error				Components (×100)				Exact structure			
n	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
WAFC	1.240	0.986	0.768	0.592	148.57	97.72	61.84	36.02	0.36	0.91	1.00	1.00
WAFC (cv)	1.202	0.977	0.771	0.592	138.87	95.98	62.29	36.09	0.55	0.83	0.88	0.96
Lasso	1.324	1.081	0.831	0.640	168.99	117.83	71.91	42.53	0.00	0.00	0.00	0.00
Lasso + threshold	1.331	1.048	0.795	0.609	170.83	110.86	65.69	38.50	0.36	0.88	1.00	1.00
GAM (REML)	1.250	1.064	0.836	0.583	149.89	114.23	73.35	35.33	0.22	0.33	0.28	0.32
GAM (GCV)	1.171	0.932	0.785	0.609	132.27	87.02	64.63	38.53	0.10	0.14	0.09	0.08
B-spline group lasso	1.268	1.108	0.941	0.756	153.56	123.91	92.27	59.75	0.04	0.01	0.00	0.00
Adaptive spline	1.366	1.190	1.047	0.997	178.91	142.85	113.78	103.19	0.00	0.00	0.00	0.00
Block lasso (KP)	1.251	1.019	0.796	0.609	148.39	103.94	65.67	38.07	0.00	0.00	0.00	0.00
Oracle support	1.159	0.950	0.748	0.573	129.24	90.63	58.65	33.62	1.00	1.00	1.00	1.00

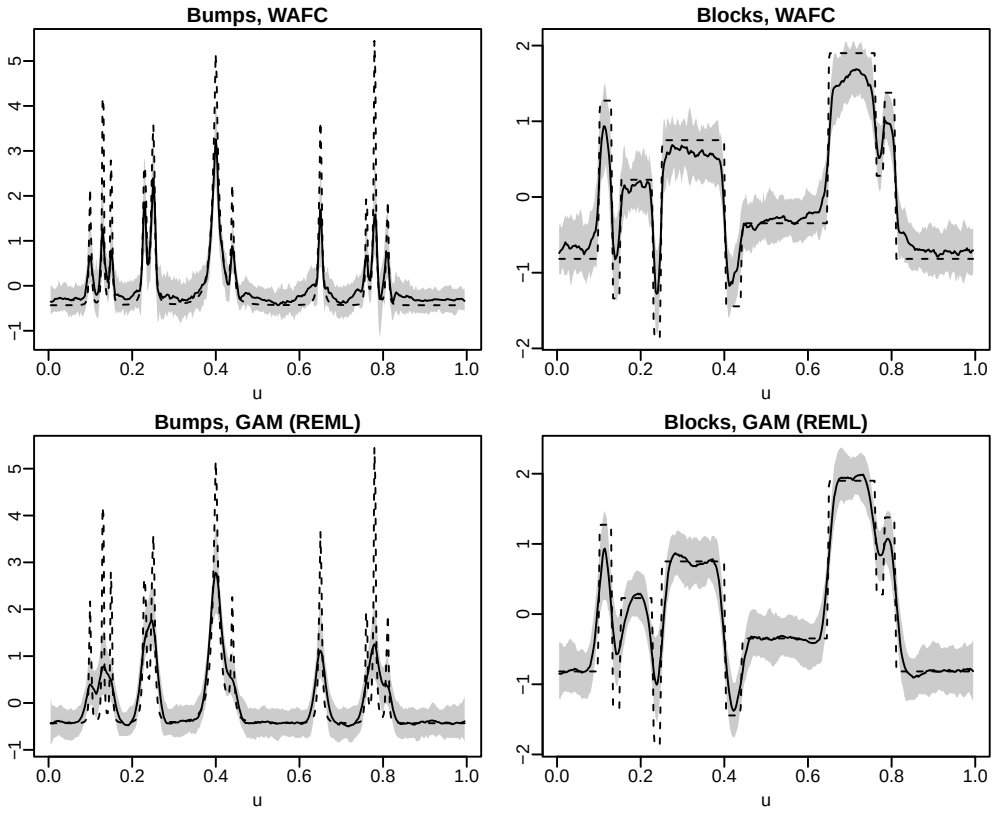


Figure S1: Simulation, inhomogeneous cell, $n = 1000$, the components of Figure 1 of the paper: the pointwise median (solid line) and the 5% and 95% quantiles (band) over the 100 replicates of the WAFC (top) and of GAM (REML) (bottom), with the truth (dashed line).

Table S4: Simulation, the mixed cell, 100 replicates per sample size: the root mean squared error of prediction, the integrated squared error of the components times 100, and the frequency of the exact structure.

	Prediction error				Components (×100)				Exact structure			
n	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
WAFC	1.301	1.052	0.820	0.625	166.97	109.92	69.77	40.13	0.25	0.84	1.00	1.00
WAFC (cv)	1.258	1.036	0.822	0.627	155.58	106.21	70.24	40.35	0.39	0.75	0.84	0.85
Lasso	1.426	1.207	0.937	0.724	199.81	144.01	90.61	53.76	0.00	0.00	0.00	0.00
Lasso + threshold	1.411	1.169	0.872	0.672	195.89	135.60	78.47	46.69	0.20	0.69	1.00	1.00
GAM (REML)	1.289	1.084	0.854	0.590	162.63	116.72	75.52	35.95	0.00	0.00	0.02	0.00
B-spline group lasso	1.309	1.149	0.975	0.794	167.52	130.81	98.37	65.80	0.00	0.00	0.00	0.00
Adaptive spline	1.630	1.284	1.085	1.022	259.03	163.84	121.24	107.59	0.00	0.00	0.00	0.00
Block lasso (KP)	1.343	1.140	1.015	0.810	173.10	126.53	105.20	67.72	0.00	0.00	0.00	0.00
Oracle support	1.184	0.962	0.758	0.575	137.80	91.10	59.43	33.88	1.00	1.00	1.00	1.00

Table S5: Simulation, the null cell, 100 replicates per sample size: the root mean squared error of prediction, the integrated squared error of the components times 100, and the frequency of the exact structure.

	Prediction error				Components (×100)				Exact structure			
n	250	500	1000	2000	250	500	1000	2000	250	500	1000	2000
WAFC	0.064	0.045	0.030	0.022	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00
WAFC (cv)	0.091	0.068	0.044	0.031	0.48	0.28	0.12	0.05	0.28	0.23	0.21	0.20
Lasso	0.093	0.070	0.048	0.036	0.54	0.32	0.16	0.09	0.16	0.15	0.08	0.13
Lasso + threshold	0.064	0.046	0.030	0.022	0.01	0.00	0.00	0.00	0.99	1.00	1.00	1.00
GAM (REML)	0.096	0.069	0.048	0.033	0.54	0.28	0.16	0.06	0.06	0.07	0.07	0.08
GAM (GCV)	0.152	0.103	0.070	0.046	2.33	0.95	0.47	0.19	0.03	0.04	0.03	0.04
B-spline group lasso	0.083	0.061	0.045	0.029	0.31	0.18	0.13	0.04	0.35	0.34	0.19	0.32
Adaptive spline	0.219	0.142	0.100	0.066	4.29	1.85	0.94	0.39	0.00	0.00	0.00	0.00
Block lasso (KP)	0.111	0.088	0.065	0.043	0.58	0.36	0.20	0.08	0.08	0.07	0.05	0.07
Oracle support	0.064	0.045	0.030	0.022	0.00	0.00	0.00	0.00	1.00	1.00	1.00	1.00

Table S6: Simulation at $n = 1000$, the structure: the mean number of null blocks kept per replicate; no method drops an active block in any replicate at this sample size. The smooth, uneven and inhomogeneous cells have 3 active and 3 null blocks, the mixed cell 3 active and 13 null, and the null cell 6 null blocks. A smooth of `mgcv` counts as kept when its effective degrees of freedom exceed 0.1.

	Smooth	Uneven	Inhomogeneous	Mixed	Null
WAFC	0.00	0.00	0.00	0.00	0.00
WAFC (cv)	0.50	0.42	0.16	0.26	1.97
Lasso	3.00	3.00	3.00	13.00	3.45
Lasso + threshold	0.00	0.00	0.00	0.00	0.00
GAM (REML)	1.04	0.90	0.95	4.77	2.36
GAM (GCV)	1.45	1.57	1.64	–	2.73
B-spline group lasso	2.81	2.97	2.95	8.69	2.12
Adaptive spline	3.00	3.00	3.00	13.00	6.00
Block lasso (KP)	2.93	2.98	3.00	9.79	3.70

Table S7: Simulation at $n = 1000$, paired comparisons: the median over the replicates of the ratio of the prediction error of each method to that of the WAFC in the same replicate, and in parentheses the fraction of replicates in which the method has the smaller error.

	Smooth	Uneven	Inhomogeneous	Mixed	Null
WAFC (cv)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	1.29 (0.00)
Lasso	1.11 (0.13)	1.12 (0.04)	1.08 (0.01)	1.14 (0.00)	1.44 (0.04)
Lasso + threshold	1.00 (0.48)	1.02 (0.34)	1.03 (0.19)	1.06 (0.04)	1.00 (0.46)
GAM (REML)	0.80 (0.99)	0.90 (0.87)	1.09 (0.05)	1.04 (0.18)	1.42 (0.04)
GAM (GCV)	0.86 (0.83)	0.89 (0.90)	1.03 (0.26)	–	2.29 (0.01)
B-spline group lasso	1.27 (0.02)	1.24 (0.00)	1.23 (0.00)	1.19 (0.00)	1.38 (0.09)
Adaptive spline	0.96 (0.61)	1.01 (0.49)	1.36 (0.00)	1.34 (0.00)	3.45 (0.00)
Block lasso (KP)	1.06 (0.22)	1.06 (0.03)	1.03 (0.04)	1.24 (0.00)	2.27 (0.03)
Oracle support	0.96 (0.76)	0.95 (0.86)	0.97 (0.94)	0.92 (0.97)	1.00 (0.49)

Table S8: Simulation, the arms at $n = 1000$ and the larger model, 100 replicates each: (a) the root mean squared error of prediction; (b) the frequency of the exact structure. SNR 1: signal to noise ratio 1; copula: modulators coupled by a Gaussian copula; X on U: linear covariates dependent on the modulators; larger: $(p, q) = (6, 4)$ with 6 of 24 blocks active. GAM (GCV) is run only in the arms of the inhomogeneous cell.

	Inhomogeneous			Mixed			
	SNR 1	Copula	X on U	SNR 1	Copula	X on U	Larger
(a) Prediction error							
WAFC	1.322	0.769	0.765	1.353	0.817	0.818	1.458
WAFC (cv)	1.211	0.773	0.769	1.250	0.821	0.821	1.430
Lasso	1.313	0.836	0.828	1.390	0.940	0.937	1.652
Lasso + threshold	1.383	0.795	0.796	1.433	0.876	0.879	1.609
GAM (REML)	1.265	0.834	0.835	1.293	0.851	0.852	1.498
GAM (GCV)	1.186	0.788	0.784	–	–	–	–
B-spline group lasso	1.275	0.938	0.931	1.304	0.974	0.961	1.602
Adaptive spline	1.382	1.040	1.046	1.552	1.087	1.087	1.786
Block lasso (KP)	1.262	0.797	0.791	1.331	1.018	1.011	1.585
Oracle support	1.182	0.752	0.744	1.180	0.750	0.752	1.341
(b) Exact structure							
WAFC	0.06	1.00	1.00	0.07	1.00	1.00	0.65
WAFC (cv)	0.62	0.84	0.85	0.47	0.82	0.83	0.72
Lasso	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Lasso + threshold	0.08	1.00	1.00	0.07	1.00	1.00	0.47
GAM (REML)	0.18	0.26	0.31	0.00	0.00	0.01	0.00
GAM (GCV)	0.13	0.13	0.11	–	–	–	–
B-spline group lasso	0.03	0.00	0.00	0.00	0.00	0.00	0.00
Adaptive spline	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Block lasso (KP)	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Table S9: Simulation, tuning and time: the mean resolution J chosen by the WAFC and the mean dimension k chosen by the two splines, each followed by the fraction of replicates at the top of its grid, and the median time of a fit with its tuning, in seconds.

		WAFC		GAM (REML)		GAM (GCV)		Time (s)		
Cell	n	J	Top	k	Top	k	Top	WAFC	REML	GCV
Smooth	250	3.41	0.00	12	0.00	19	0.04	3.5	4.4	113.7
	500	3.49	0.00	15	0.02	15	0.01	12.1	4.9	125.8
	1000	3.83	0.00	14	0.00	24	0.16	40.5	46.3	128.3
	2000	4.15	0.00	14	0.00	26	0.09	131.3	123.9	552.9
Uneven	250	4.54	0.00	27	0.03	35	0.12	3.7	4.0	154.8
	500	5.00	0.00	28	0.05	42	0.18	12.2	4.6	165.9
	1000	5.12	0.00	43	0.00	50	0.27	40.9	40.7	178.0
	2000	5.20	0.00	50	0.00	59	0.12	130.2	108.7	708.6
Null	250	4.65	0.14	8	0.02	15	0.02	3.8	3.9	86.0
	500	5.29	0.19	11	0.03	15	0.02	15.8	4.7	78.7
	1000	4.56	0.11	15	0.02	14	0.03	52.0	44.7	77.6
	2000	5.56	0.16	19	0.02	20	0.05	174.5	115.8	339.8

Table S10: Simulation, tuning and time (continued from Table S9).

		WAFC		GAM (REML)		GAM (GCV)		Time (s)		
Cell	n	J	Top	k	Top	k	Top	WAFC	REML	GCV
Inhomogeneous	250	5.12	0.03	29	0.03	62	0.59	4.6	4.5	148.9
	500	6.31	0.01	68	0.71	80	0.99	15.8	5.1	210.5
	1000	7.23	0.24	125	0.16	80	1.00	56.3	46.7	239.7
	2000	8.04	0.06	240	1.00	120	1.00	176.4	139.4	1006.6
Mixed	250	4.80	0.04	27	0.02	–	–	10.1	85.1	–
	500	6.14	0.00	66	0.65	–	–	33.9	91.3	–
	1000	6.90	0.09	119	0.97	–	–	78.8	430.1	–
	2000	7.94	0.01	240	1.00	–	–	245.8	3806.4	–

Table S11: Simulation, the slope of the least squares line of the logarithm of the mean squared error of prediction against $\log n$, over $n = 250$, 500, 1000 and 2000.

	Smooth	Uneven	Inhomogeneous	Mixed	Null
WAFC	−0.95	−0.96	−0.71	−0.71	−1.04
WAFC (cv)	−0.96	−0.97	−0.68	−0.67	−1.05
Lasso	−0.85	−0.99	−0.70	−0.66	−0.93
Lasso + threshold	−0.84	−1.00	−0.76	−0.73	−1.02
GAM (REML)	−0.91	−0.94	−0.73	−0.75	−1.04
GAM (GCV)	−0.99	−0.98	−0.62	–	−1.16
B-spline group lasso	−0.85	−1.04	−0.50	−0.48	−1.00
Adaptive spline	−1.11	−1.10	−0.31	−0.45	−1.15
Block lasso (KP)	−0.91	−0.99	−0.69	−0.47	−0.90
Oracle support	−0.89	−0.97	−0.68	−0.69	−1.03

S10 The applications: further results

Prediction and structure

Table S12 gives, for the two applications, the root mean squared error on the held-out weeks, its mean over the 20 partitions with the standard error of Nadeau and Bengio (2003), which corrects for the overlap of the training samples, the ratio to the better of the two splines in each partition, and the paired difference to the spline with the smaller mean error, with the fraction of partitions in which it is negative. Table S13 gives, for every block, the frequency with which each method keeps it over the partitions, and Table S14 the choices of J, the penalty, the threshold and k.

At Marylebone Road the comparison is that of Section 6.3 of the paper; the constant coefficients have an error of 15.48 ppb, about a third more than the methods with varying coefficients. At Dongsi the WAFC is ahead of the spline fitted by REML, the better of the two splines in every partition, in all 20 partitions and with either threshold, with an error smaller by 8% on average and a paired difference of -0.070 (standard error 0.023). There both thresholds and both splines keep every block in the fit to the whole sample, and the one standard error rule drops the pressure along the day of the year in one partition and along the humidity in two.

Both splines choose $k = 80$, the top of the grid, in all 21 fits of each application, and `k.check` flags two of the four smooths at Marylebone Road in every fit and four to eight of the eight at Dongsi. The WAFC chooses $J = 8$, the top of its grid, in the fit to the whole sample at Marylebone Road and $J = 5$ or 6 in the partitions; at Dongsi it chooses $J = 4$ in the fit to the whole sample and in 19 of the 20 partitions, and every coefficient of a block it keeps is nonzero. At Marylebone Road, extending the grids does not change the comparison: with k up to 160 in the first three partitions, REML and GCV again choose the largest dimension and the error on the held-out weeks rises by about 3 to 4 percent, and in the fit to the whole sample the cross-validated error of the WAFC at $J = 9$ is larger than at $J = 8$. At Dongsi, where the splines also choose $k = 80$, the grids were not extended.

Beijing

The data are the hourly observations at the site Dongsi of the Beijing Multi-Site Air Quality data of the UCI Machine Learning Repository (Chen, 2017), from 1 March 2013 to 28 February 2017, under the attribution given in the Acknowledgements of the paper. Dongsi is one of the 12 nationally controlled sites of the data set, which are among the 36 sites whose PM2.5

Table S12: The two applications, prediction on the held-out weeks over 20 partitions: the root mean squared error (RMSE), mean and standard error with the correction of Nadeau and Bengio (2003); its ratio to that of the better of the two splines in each partition; the paired difference to the spline with the smaller mean error, GAM (REML) in both applications, with the corrected standard error, and the fraction of partitions in which it is negative; and R^2, one minus the ratio of the mean squared error to the variance of the held-out response. WAFC: the threshold of the one standard error rule; WAFC (cv threshold): the threshold of the smallest cross-validated error.

	RMSE	Ratio	Difference	Negative	R^2
Marylebone Road, total oxidant (ppb)					
WAFC	11.75 (0.28)	1.025	0.28 (0.22)	0.30	0.68
WAFC (cv threshold)	11.52 (0.32)	1.005	0.05 (0.11)	0.40	0.69
GAM (REML)	11.47 (0.32)	1.000	–	–	0.69
GAM (GCV)	11.49 (0.33)	1.002	0.02 (0.02)	0.10	0.69
Constant coefficients	15.48 (0.30)	1.352	4.02 (0.30)	0.00	0.44
Dongsi, logarithm of PM2.5					
WAFC	0.800 (0.017)	0.920	−0.070 (0.023)	1.00	0.53
WAFC (cv threshold)	0.797 (0.016)	0.916	−0.074 (0.021)	1.00	0.54
GAM (REML)	0.871 (0.027)	1.000	–	–	0.45
GAM (GCV)	0.892 (0.029)	1.024	0.021 (0.006)	0.00	0.42
Constant coefficients	1.059 (0.016)	1.218	0.188 (0.026)	0.00	0.18

Table S13: The two applications, structure: for every block, the number of the 20 partitions in which each method keeps it and, in parentheses, the range of the component in the fit to the whole sample over the points drawn in Figure 2 of the paper and Figure S2, which is zero when that fit drops the block; in ppb for the background and in percentage points for the slope at Marylebone Road, and in the logarithm of PM2.5 at Dongsi, per unit of the covariate (m/s, °C, hPa) for the coefficients. WAFC (cv): the threshold of the smallest cross-validated error.

Block	WAFC	WAFC (cv)	GAM (REML)	GAM (GCV)
Marylebone Road				
Background, date	20 (24.2)	20 (24.2)	20 (22.9)	20 (24.4)
Background, wind speed	12 (0)	20 (5.75)	20 (7.82)	20 (7.38)
Slope on NOx, date	20 (15.8)	20 (15.8)	20 (20.8)	20 (21.0)
Slope on NOx, wind speed	0 (0)	19 (0.795)	18 (1.15)	19 (1.78)
Dongsi				
Level, day of the year	20 (6.33)	20 (6.33)	20 (8.36)	20 (8.89)
Level, humidity	20 (2.26)	20 (2.26)	20 (2.15)	20 (2.10)
Wind speed, day of the year	20 (0.417)	20 (0.417)	20 (0.685)	20 (0.744)
Wind speed, humidity	20 (0.294)	20 (0.294)	20 (0.211)	20 (0.212)
Temperature, day of the year	20 (0.101)	20 (0.101)	20 (0.178)	20 (0.198)
Temperature, humidity	20 (0.0735)	20 (0.0735)	20 (0.0654)	20 (0.0663)
Pressure, day of the year	19 (0.109)	20 (0.109)	20 (0.186)	20 (0.221)
Pressure, humidity	18 (0.0501)	20 (0.0501)	20 (0.0350)	20 (0.0358)

Table S14: The two applications, tuning: the choices in the fit to the whole sample and their median (range) over the 20 partitions. J and λ are those of the WAFC, which the two thresholds share; the threshold is in percent of the largest block norm of the fit, followed by the number of nonzero coefficients after it; k is the dimension chosen by the two splines, at both modulators, and the ~~last row~~ row of smooths flagged the number of smooths that `k.check` of `mgcv` flags, of 4 at Marylebone Road and of 8 at Dongsi, which is the same for REML and GCV. The grids are $J \in \{2, \ldots, 8\}$ and $k \in \{5, 10, 20, 40, 80\}$. The time is that of one fit with its tuning, in seconds on one core of an Intel Core i7-10875H.

	Marylebone Road		Dongsi	
	Whole	Partitions	Whole	Partitions
J	8	5 (5–6)	4	4 (3–4)
$\lambda \times 10^3$	101.9	40.2 (29.7–73.2)	0.43	0.88 (0.56–1.60)
WAFC				
Threshold (%)	77	77 (56–93)	0.95	0.98 (0.55–1.13)
Nonzero coefficients	458	77 (62–152)	120	120 (56–120)
WAFC (cv threshold)				
Threshold (%)	5.7	10.6 (3.9–33.3)	0.46	0.46 (0.28–1.02)
Nonzero coefficients	488	108 (77–178)	120	120 (56–120)
GAM (REML) and GAM (GCV)				
k	80	80 (80–80)	80	80 (80–80)
Smooths flagged	2	2 (2–2)	4	4 (4–8)
Time per fit (s)				
WAFC	2163	2387 (1764–3546)	8830	7355 (6420–7909)
GAM (REML)	9.2	7.1 (6.9–7.4)	27.6	23.4 (22.5–25.9)
GAM (GCV)	202	150 (143–152)	474	424 (359–461)

series Zhang et al. (2017) analyze; the meteorological variables of a site are those of the nearest station of the China Meteorological Administration, which for Dongsi changes in February 2015 (Zhang et al., 2017, Table S1 of their Supplementary Material). The 34,287 hours with PM2.5, temperature, pressure, dew point and wind speed observed, a relative humidity in $(0, 100]$ and a positive PM2.5 are used. The response is the logarithm of PM2.5, the linear covariates are the constant, the wind speed (m/s), the temperature (°C) and the pressure minus 1000 hPa, and the modulators are the day of the year, from 0 on 1 January with the hour as a fraction of the day, and the relative humidity (%), computed from the temperature and the dew point by the Magnus formula; every coefficient is modulated by both, so that $p = 4$, $q = 2$ and the model has eight blocks. The day of the year is periodic, as the periodized basis is. The splits, the methods and their tuning are those of Section 6 of the paper.

The winter heating season of Beijing and the North China Plain usually runs from 15 November to 15 March, and the weather-adjusted PM2.5 level rises significantly at its start and falls at its end in every season from 2010 to 2014 (Liang et al., 2015, Section 7); the effective dates move with the temperature (for instance, the season started on 3 November in 2012), so the marks at 15 November and 15 March in Figure S2 are nominal. Along

the day of the year, the mean of the component of a coefficient inside the nominal season minus its mean outside it is, in the fit of the WAFC to the whole sample, -0.18 per m/s for the wind speed (-0.19 to -0.13 over the partitions), -0.056 per °C for the temperature (-0.058 to -0.031) and -0.028 per hPa for the pressure (-0.032 to 0), and the two splines give -0.17 to -0.18, -0.062 and -0.033. The three differences are negative in every fit of the WAFC and of the two splines, except that of the pressure in the partition in which the WAFC drops that block: inside the season a unit of wind speed goes with a larger decrease of PM2.5, and a degree of temperature with a smaller increase, than outside it.

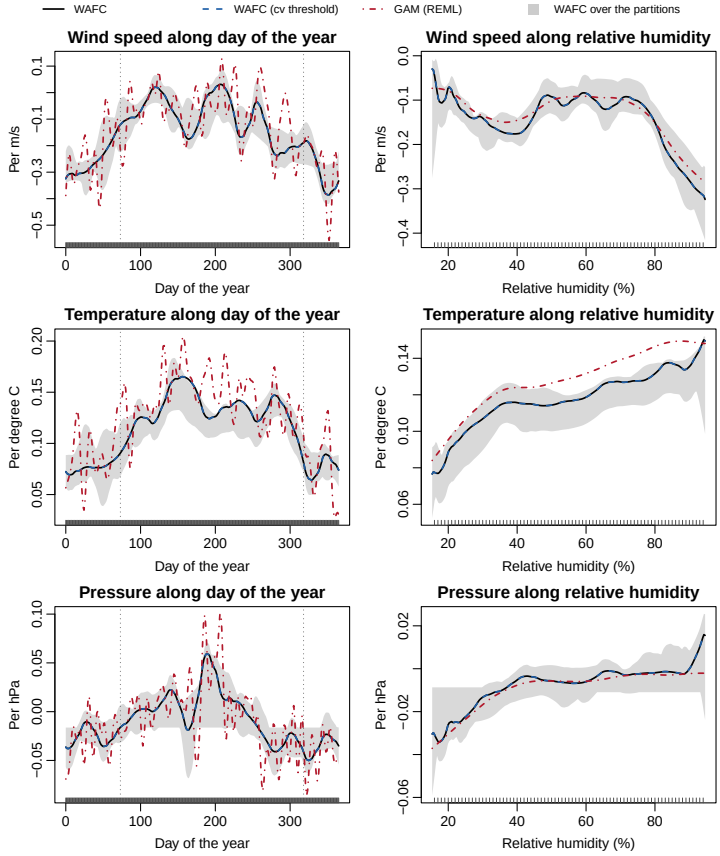


Figure S2: Beijing, Dongsi, hourly, March 2013 to February 2017: the coefficients of the wind speed (top, per m/s), the temperature (middle, per °C) and the pressure (bottom, per hPa) for the logarithm of PM2.5, along the day of the year (left) and the relative humidity (right), with the component of the other modulator at its sample mean; the band is the range of the WAFC over 20 fits on 70% of the weeks, and the vertical lines mark the nominal heating season, 15 November to 15 March. Not shown: the level, which moves along the year with the coefficient of the temperature, and the relative humidity below about 15% and above about 95%, where the periodized basis joins the two ends of its range; ticks on the axes mark the observed values.

References

Bühlmann, P. and S. van de Geer (2011). *Statistics for High-Dimensional Data: Methods, Theory and Applications*. Berlin: Springer.

Cai, T. T. (1999). Adaptive wavelet estimation: a block thresholding and oracle inequality approach. *The Annals of Statistics 27*(3), 898–924.

Chen, S. (2017). Beijing Multi-Site Air Quality. UCI Machine Learning Repository, `https://doi.org/10.24432/C5RK5G`, accessed 20 September 2026.

Cohen, A. (2003). *Numerical Analysis of Wavelet Methods*, Volume 32 of *Studies in Mathematics and its Applications*. Amsterdam: Elsevier.

Cohen, A., I. Daubechies, and P. Vial (1993). Wavelets on the interval and fast wavelet transforms. *Applied and Computational Harmonic Analysis 1*(1), 54–81.

Daubechies, I. (1992). *Ten Lectures on Wavelets*, Volume 61 of *CBMS-NSF Regional Conference Series in Applied Mathematics*. Philadelphia: Society for Industrial and Applied Mathematics.

Donoho, D. L. and I. M. Johnstone (1994). Ideal spatial adaptation by wavelet shrinkage. *Biometrika 81*(3), 425–455.

Donoho, D. L. and I. M. Johnstone (1998). Minimax estimation via wavelet shrinkage. *The Annals of Statistics 26*(3), 879–921.

Hsu, D., S. M. Kakade, and T. Zhang (2012). A tail inequality for quadratic forms of subgaussian

random vectors. *Electronic Communications in Probability 17*(52), 1–6.

Johnstone, I. M. (2019). Gaussian estimation: Sequence and wavelet models. Draft version of September 16, 2019.

Klopp, O. and M. Pensky (2015). Sparse high-dimensional varying coefficient model: nonasymptotic minimax study. *The Annals of Statistics 43*(3), 1273–1299.

Liang, X., T. Zou, B. Guo, S. Li, H. Zhang, S. Zhang, H. Huang, and S. X. Chen (2015). Assessing Beijing's $PM_{2.5}$ pollution: severity, weather impact, APEC and winter heating. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 471*(2182), 20150257.

Lounici, K., M. Pontil, S. van de Geer, and A. B. Tsybakov (2011). Oracle inequalities and optimal inference under group sparsity. *The Annals of Statistics 39*(4), 2164–2204.

Mallat, S. (2009). *A Wavelet Tour of Signal Processing: The Sparse Way* (Third ed.). Burlington, MA: Academic Press.

Nadeau, C. and Y. Bengio (2003). Inference for the generalization error. *Machine Learning 52*(3), 239–281.

Simon, N. and R. Tibshirani (2012). Standardization and the group lasso penalty. *Statistica Sinica 22*(3), 983–1001.

Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological) 58*(1), 267–288.

Triebel, H. (1983). *Theory of Function Spaces*, Volume 78 of *Monographs in Mathematics*. Basel: Birkhäuser. Reimpresso em 2010 na série Modern Birkhäuser Classics, DOI 10.1007/978-3-0346-0416-1.

Tropp, J. A. (2012). User-friendly tail bounds for sums of random matrices. *Foundations of Computational Mathematics 12*(4), 389–434.

Tsybakov, A. B. (2009). *Introduction to Nonparametric Estimation*. Springer Series in Statistics. New York: Springer.

Zhang, S., B. Guo, A. Dong, J. He, Z. Xu, and S. X. Chen (2017). Cautionary tales on air-quality improvement in Beijing. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences 473*(2205), 20170457.